Sewing: environment report

Construction, coupled physics, experiments and evidence

Gym Anything for Robotics

Environment `sewing` (aliases `sew`, `sewing-machine`, `lockstitch`, `sew-plain-seam`) · Newton 1.5 rigid world (MuJoCo CPU solver) coupled every 0.5 ms tick to a repository-owned implicit cloth solver in Warp (orthotropic woven plies with bending friction, the fabric's measured compression, Coulomb friction and the air under the cloth) and an industrial lockstitch machine whose needle, feed dog, presser foot, hook and threads act on the cloth only through that solve · default robot `dual-fr3` (two FR3 arms with Franka Hands), also `dual-panda`, `single-fr3` and `g1` (the Unitree G1 humanoid with Dex3 hands, standing at the table) · action group *sewing*.

Status. Built and checked as described below; nothing here is a policy, and every demonstration is a scripted program or a labelled phantom. What holds: the fabric's own KES instruments run on the discrete model return the fabric's values; feed, stitch and puncture match their design criteria and are converged at 600 sti/min (E17 passes); every start state rests on every deployment (20 soaks); legal abuse stays finite; a still hand presses the work steadily on all four deployments; a robot hand slides the stack, steers it while the machine sews (the needle intact), draws the sewn work through and pinches the hanging drape and holds it, by contact; the bench seam settles in one shape in six repeats; with labelled assistance the load passes every check, the plain seam is sewn end to end with both backtacks (16 of 17 checks) and the curve on a short stack is sewn flat with every stitch formed (its accuracy checks fail). What does not, stated plainly and kept open: the work turns under the running foot by itself, 0.55 deg a stitch at first, from the feed dog's unequal rows under the foot's unequal toes, whose widths are an estimate, and every guiding policy meets it; the baste is not sewn: its needle fails by the 25th-32nd stitch guided or not; the shear lock does not bound the plies' shear under a hand held against the feed; a Franka Hand squeezing the plies holds them only lightly (2 N a finger; 20 N throws the fingers open); the pivot's lines and the task's own curve are not demonstrated (§11); no robot program sews a whole seam; the dual-FR3 rig cannot put both hands at the foot (§16). Through the API every deployment's commands of 0.02 s and longer return inside 200x on a quiet host with all three default cameras (0.5 s sewing actions at 118-123x); on a shared host: 0.5 s sewing actions at 120-163x on every deployment (1-minute load 9-14), commands of 0.1 s and longer inside 200x on every deployment (0.02 s and longer on the FR3 pair and the G1, 0.05 s on the single FR3), 0.01 s not (234-292x) (§14).

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1 The real job

The action group (14 rows) is operating a sewing machine to join, hem and stitch cloth, leather, vinyl and paper. The flagship is a sewing machine operator's core cycle at an industrial single-needle lockstitch head: lay cut plies under the presser foot, guide them with both hands while the machine feeds them, keep the seam edge at the allowance, regulate tension against the feed without pulling the needle, pivot about a lowered needle, backtack, trim, and lay the piece off (GSD gives 10-20 s of basic time for a 42 cm straight seam with backtacks). An automated cell drives the head through its control box (pedal speed, knee lifter, touch-back reverse, needle-up/down stop, trimmer) the way the policy here does. The design proposed 16 tasks on four presets (sample room, home sewing, bindery paper, upholstery); this build implements preset A's sample room with five of its tasks (§10) and states the rest as work not built (§16).

2 Scene at real scale

2.1 Physical geometry

Frame: origin where the needle axis meets the throat-plate top, +x to the operator's right (toward the machine's arm and handwheel), +y away from the operator (the feed direction), +z up; the table top is z = 0, the floor -0.762 m.

Item	Size and placement	Source
Sewing table	1,200 x 533 x 30 mm top at 762 mm, K-leg stand	Juki DDL-9000C table drawing; Apparel Resources (standard 21 in depth, ~30 in height) [read]
Head	Juki DDL-9000C class: bed 517 x 178 mm flush with the table, needle-to-arm 303 mm, pillar, arm and head sections	manual [read]; arm clearance and sections from product photographs [estimate]
Needle bar / presser bar	30.7 mm stroke; presser bar behind the needle carrying the foot	manual [read]
Presser foot	sole 26 mm, toes 6.0 / 4.5 mm, 2 mm slot; 40 N; lift 5.0 / 8.5 mm	manual [read]; foot geometry [estimate]
Feed dog	two 3.5 mm rows either side of the needle, 10 mm long, 0.8 mm above the plate	manual [read: height]; footprint [estimate]
Robots	two FR3 + Franka Hand on one pedestal 0.55 m in front of the needle, 0.20 m above the table (a seated operator's shoulders), left at x = -0.30 m, right at x = +0.10 m, facing +y; or the Unitree G1 (Dex3-1 hands) standing 0.13 m behind the table's front edge on the needle line	E0b reach preflights (110 key poses each) [authored]
Plies	two 200 x 420 mm muslin plies, 125 g/m2, 0.61 mm free thickness	Ramaiah et al. fabric C [read]

The machine's moving parts (needle bar, presser foot, feed dog, take-up lever, handwheel) are separate USD assets authored in Blender (`assets/build_machine.py`) at these dimensions and driven every frame by the simulated state. The robots and the head collide in MuJoCo with the head's static boxes (pillar, arm, head, presser and needle bars, foot shank, handwheel); the cloth sees the bed, the plate, the feed dog, the foot's sole and toes, the pillar, the head's underside and each hand's finger boxes and housing.

2.2 The room, the machine's appearance and the cameras

The cell stands in a closed sample room (`room.py`, render-only: nothing in it collides, and all of it is outside every robot's sweep): 6.8 x 5.4 m, 3.0 m high, two windows with frames, mullions and sills in the left wall (daylight from the pinned sky HDRI enters only through them), a door with its frame, lever handle and kick plate in the right wall, and skirting. Two more lockstitch workstations (the cell's own table and head, unthreaded and idle) stand either side of the cell with stools and blue bundle bins holding cut parts; a 2.6 x 1.0 m cutting table at the back carries a green cutting mat, a length of navy fabric with three kraft-paper pattern pieces (front and back bodice, sleeve: grain lines, notches, labels), an aluminium L-square, shears, pattern weights and a tape measure; six more pattern pieces hang from a rail on the back wall; steel shelving holds nine fabric bolts; a dress form with a muslin toile, an ironing station, a drawer cabinet, a wall clock and two signs (sample-room safety, the seam specification) complete it. Pinned assets are CC0 (Poly Haven, hashes in `visual_assets.json`); the pattern paper, cutting mat and signs are drawn by `room.py`; no brand is shown. Six 4,000 K ceiling fixtures light the room; the machine's needle lamp lights the needle area.

The head's appearance (`assets/build_machine.py`, Blender) is modelled on the Juki DDL-9000C class within

the collision envelope of §2.1: rounded cast arm, pillar and head (a voxel-remeshed union of bevelled sections), the bed with its throat plate (needle hole, four dog slots, seam-guide marks), slide plate, face plate, two thread-tension assemblies (discs, spring, nut) with the check spring, thread guides, the take-up lever's slot, the touch-back switch beside the needle bar, the stitch-length dial and reverse-feed lever, an operation panel, the oil-level window, the bobbin winder on the arm, the presser lifter lever behind the head, the direct-drive motor housing and handwheel, and the thread stand with two cones of navy Tex 30 polyester; the sewing thread is drawn in the same navy (a contrasting thread, as a sample room uses to inspect a seam). The plies are drawn from the simulated mid-surface as unbleached muslin at true scale (`render.muslin_textures`): the CC0 rough-linen photograph's weave (its threads measure 1.09 x 1.47 mm by FFT over its 270.7 mm) scaled to muslin's ~0.4 mm thread pitch, with slub yarns (thick places along the warp and weft, 5-40 mm long, +-6 % brightness), seed-coat flecks (0.15-0.9 mm, brown, ~0.8 per cm2) and a +-2 % mottle [authored, from photographs], tinted to ecru (238, 226, 198). (The linen at 5 cm a tile and no flecks reads as card.)

Policy cameras: `operator` (above and between the robot's shoulders, 50 deg) and `needle` (a D405-class camera on the face plate, 62 deg); the G1 sees through its head camera (`head_rgb`). Diagnostic cameras: `room` (the room from its front-right corner at a standing person's eye height), `cell` (the whole cell from behind the table's left end at standing height: the work, the machine and both arms beyond it; from the robots' side the arms hid the work at any height), `work` (the work from the table's left side), `hero`, `machine`, `macro` (the needle area at 30 mm), `overhead` and `seam` (behind the foot, looking at the stitches as they leave the needle).

2.3 Screenshots

Every camera of each task's start state, rendered from the simulated state:



Figure 1: The sample room from its front-right corner: the cell (the FR3 pair on its pedestal at the lockstitch head), two more workstations, the cutting table with pattern pieces, shelving, the dress form



Figure 2: The cell from behind the table's left end: the plies under the foot, the head and its thread stand, both arms with their hands above the work



Figure 3: The plain-seam start state from the hero camera: the dual FR3 at the machine



Figure 4: The policy's operator camera: the stack under the foot, its near part over the table's front edge

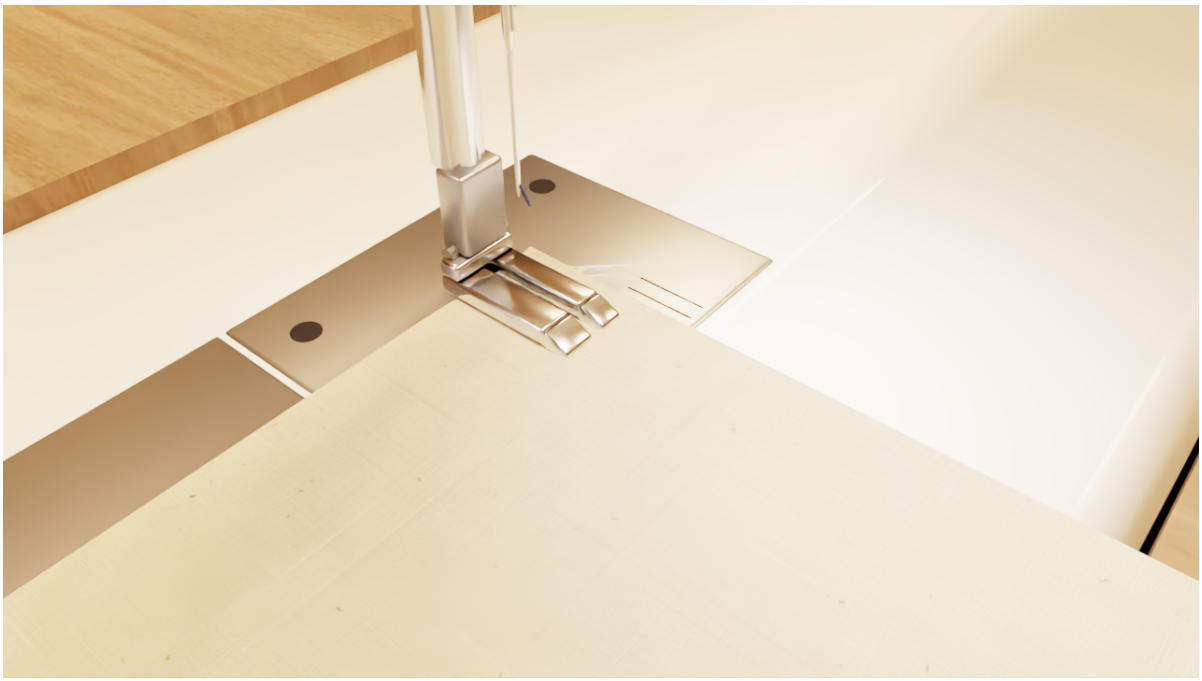


Figure 5: The face-plate needle camera (policy)

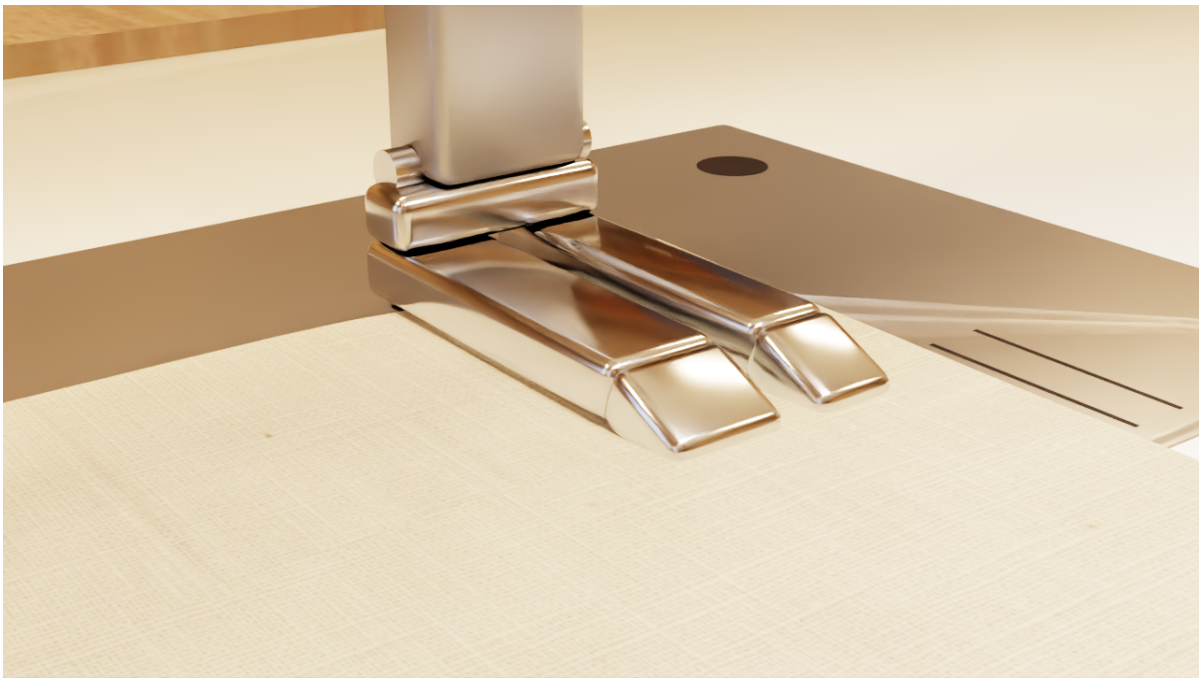


Figure 6: The presser foot on the plies (macro, diagnostic): the muslin's weave and a seed fleck at true scale



Figure 7: The work from the table's left side: the stack under the foot, the right hand's fingers above it



Figure 8: Task start states from overhead: load (the plies on the table, the foot down on the bare plate), pivot, curve



Figure 9: The pivot start state



Figure 10: The curve start state (the rounded corner)

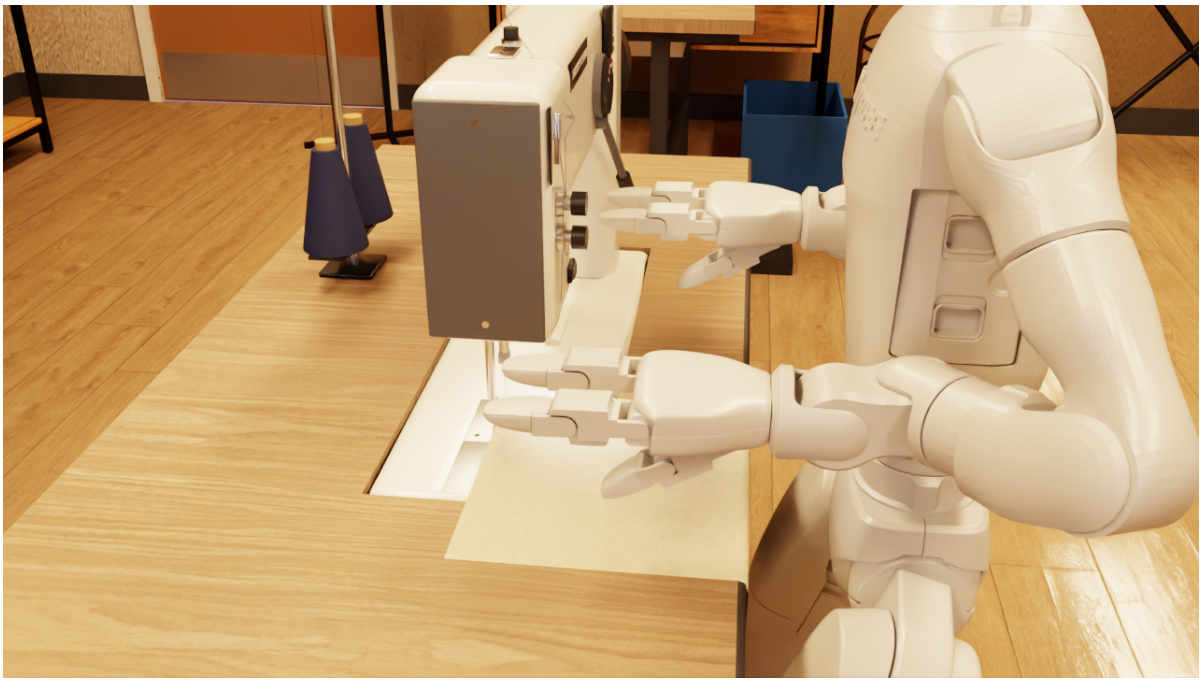


Figure 11: The humanoid configuration: the G1 standing at the table



Figure 12: The G1's head camera (policy): the face plate hides the needle

3 The cloth

Every ply is a triangle mesh in its cut-panel frame (material coordinates (u, v) in metres; the warp along v, the length of the ply, the weft along u). The mesher (`mesh.py`) puts 2.5 mm edges in a 22 mm band along each seam the task sews (the right edge and its seam line; for the pivot also the near edge and its seam line; for the curve the whole rounded path) and 7.5 mm edges in the bulk. One solver (`cloth.py`, a repository-owned implicit Newton solver in Warp, in place of Newton's VBD cloth) advances all plies, the presser foot and the sewing constraints together, one backward-Euler step per 0.5 ms tick, the minimiser of an incremental potential:

$$\begin{aligned}\mathbf{x}_{n+1} &= \arg\min_{\mathbf{x}} \sum_i \frac{m_i}{2\Delta t^2} |\mathbf{x}_i - \mathbf{y}_i|^2 + E_{\text{membrane}} + E_{\text{bend}} + E_{\text{contact}} + \Psi_{\text{friction}} + D_{\text{air}} + E_{\text{sewing}}, \\ \mathbf{y} &= \mathbf{x}_n + \Delta t\, \mathbf{v}_n + \Delta t^2\, \mathbf{g}\end{aligned} \tag{1}$$

with Newton iterations (two, then up to six while a step exceeds 10^{-4} m/s $\times\, \Delta t$, or tangentially 1/20 of the friction's pre-slip), a matrix-free block-Jacobi conjugate gradient (40 iterations) and a backtracking line search on the same potential, so a step never raises it. Positions are float64 (in float32 a resting vertex's per-tick motion rounds away).

Membrane. Each triangle's deformation gradient F gives the yarn stretches $|F\mathbf{a}_w|$ and $|F\mathbf{a}_f|$ ($\mathbf{a}_w$, $\mathbf{a}_f$ the warp and weft directions) and the shear angle γ between them. Each yarn family follows a toe-region bilinear tension law $n(e)$ (N/m) on its strain $e = |F\mathbf{a}| - 1$, through the KES point:

$$n(e) = \begin{cases} K_0\, e & 0 \le e \le e_t \\ K_0\, e_t + K_1 (e - e_t) & e > e_t \\ k_c K_0\, e & e < 0 \end{cases} \tag{2}$$

stiffness K_0 up to the toe strain e_t, K_1 beyond, both fitted to KES EMT (the strain at 490 N/m) and LT (linearity) with a toe carrying 10 % of the KES load (estimate); compression is resisted at $k_c = 0.08$ of the toe stiffness (woven cloth buckles rather than shortening); shear has the KES rigidity G (N/m per rad). Muslin: Ramaiah et al. fabric C (125.2 g/m2, 0.61 mm, LT 0.597, WT 9.91, EMT ~6.6 %, G 0.52 gf/cm deg).

Shear locking. The shear energy $\frac{1}{2} G A s^2$, with $s = (F\mathbf{a}_w) \cdot (F\mathbf{a}_f)$ ($\sin\gamma$ for straight yarns), is calibrated by KES-FB1 over ±8 deg; its force $\frac{1}{2} G \sin 2\gamma$ falls past 45 deg and vanishes at 90, where a woven fabric's rises steeply as the yarns of a set touch. Past the locking angle the energy gains

$$\tfrac{1}{2} k_{\text{lock}} A \left(|s| - \sin\gamma_{\text{lock}} \right)^2, \qquad |s| > \sin\gamma_{\text{lock}} \tag{3}$$

with $\gamma_{\text{lock}} = 50$ deg (derived: the yarns of a set touch when their spacing $p\cos\gamma$ reaches the yarn width w; a plain cotton of 137.8 g/m2 at 23.6 x 23.2 threads/cm, the muslin's own count not being in Ramaiah's table, gives 45-62 deg for round to flattened yarns) and k_{lock} the fabric's tensile toe stiffness K_0 (estimate). Below 50 deg the law is unchanged (a test checks both). Without the lock, the plies under a still hand beside the foot shear to 72-87 deg.

Membrane friction (tensile and shear hysteresis). Woven cloth is not elastic in its plane either: the yarns slide at their crossings as the crimp interchanges and as the net shears. KES-FB1 of the same fabric returns RT 49.94 % of the tensile work on unloading, and its shear loop is 2HG 1.24 gf/cm wide at 0.5 deg and 2HG5 1.79 gf/cm at 5 deg (read). Each triangle carries, per yarn direction, four elastic-perfectly-plastic Iwan elements on the yarn strain e (the bending friction's Huber energy, with stick strains (2i+1)/8 of e_s = 0.3 %, estimate) whose slip forces sum to $\varphi\, n_{\text{el}}(e)$, the elastic tension times

$$\varphi = c_{\text{TF}}\,\frac{1-RT}{1+RT}, \tag{4}$$

Coulomb friction under the crossings' pressure, which grows with the tension, lagged by one tick so each tick's step is still a minimisation. The elastic law is the KES law divided by $1+\varphi$, so loading follows the KES curve and unloading runs at $(1-\varphi)/(1+\varphi)$ of it; $c_{\text{TF}} = 1.125$ (`TF_COEF`) accounts for the work the micro-slip reversal returns. On the shear angle four more elements slip at a force per unit width $H_G(\gamma) = H_{G0} + H_{G1}|\gamma|$ through 2HG/2 at 0.5 deg and 2HG5/2 at 5 deg (the crossings jam as the net shears; 0.58 to 0.88 N/m), stick angles over 0.5 deg (estimate), and the elastic G is the KES slope less H_{G1} (the friction's growth steepens both branches of the loop). The return mapping runs at the end of every tick with the bending friction's. The virtual KES-FB1 loops (§8.3) return the fabric's values. Without this the membrane dissipates nothing. It does not cure the runaway of a short stack at the machine's top speed, which remains open (§8.6).

Bending. Discrete hinges across every interior edge e, with the dihedral angle θ_e:

$$E_e = \tfrac{1}{2}k_e\,\theta_e^2, \qquad k_e = s_e\,B_{\mathbf{d}}\,\frac{|e|^2}{A_1+A_2}, \qquad B_{\mathbf{d}} = B_w(\mathbf{d}\cdot\mathbf{a}_w)^2 + B_f(\mathbf{d}\cdot\mathbf{a}_f)^2 \tag{5}$$

for the curvature direction $\mathbf{d}$ across the edge and the areas A_1, A_2 of its two triangles. The coefficients s_e are calibrated per ply, per mesh class (seam band, bulk) and per direction so that exact isometric wraps of the ply's own mesh onto cylinders give the continuum energy $\frac{1}{2}B\kappa^2 A$ along the warp and the weft (a 2x2 solve; with one global coefficient the mesher's strips give B 0.83x along the warp and 1.23x along the weft). Rest angles are zero (no pressed folds in the built tasks).

Bending friction (hysteresis). Woven cloth bends with friction at the yarn crossings: KES-FB2's moment-curvature loop has width 2HB (Ramaiah fabric C: 2HB 0.0277 gf cm/cm, i.e. HB = 1.36e-4 N m/m, the elastic moment at 0.51 /cm). Each hinge carries an Iwan assembly of four elastic-perfectly-plastic elements in parallel, element i with slip torque $T_e/4$ ($T_e = t_e\,H_B\,|e|$, t_e calibrated like s_e) and stick range $(2i+1)/8$ of $\kappa_s h_e$ (κ_s = 0.25 /cm, read off the rounded reversal of the paper's KES loop; any $\kappa_s < 0.75$ /cm reproduces 2HB at ±1 /cm). Within a tick each element is the Huber energy of its angle past its plastic angle p_i:

$$\psi_i(d) = \begin{cases} \frac{1}{2}k_i\,d^2 & |d| \le T_i/k_i \\ T_i|d| - \dfrac{T_i^2}{2k_i} & |d| > T_i/k_i \end{cases}, \qquad d = \theta - p_i, \tag{6}$$

the exact backward-Euler incremental potential of perfect plasticity; after the tick the return mapping moves p_i to the edge of the stick range and books $T_i|\Delta p_i|$ as dissipated. The virtual KES-FB2 loop (§8.1) returns B 0.98/0.96 and 2HB 0.98/0.97 of the inputs along warp/weft, 0.91 and 0.97 along the bias (not calibrated).

Contact and compression. Cloth vertices carry the ply's free half-thickness as a contact radius against the machine's and the hands' boxes and against the other ply (vertex-face pairs, the closest face per ply, persistent across ticks unless another face is 1 um closer). The overlap d is resisted by the fabric's measured compression, on a surface

$$P(d) = a\left(e^{2bd} - 1\right), \tag{7}$$

two compressible surfaces in series sharing the overlap, fitted through the KES-FB3 points of the same fabric (T0 0.610 mm, Tm 0.311 mm at 50 gf/cm2 from WC 0.181 gf cm/cm2 and LC 0.242: $2/x - 2/(e^x - 1) = LC$ with $x = b(T_0 - T_m) = 8.25$, so a = 1.29 Pa, b = 27,600 /m). Under the presser's ~154 kPa a ply is 0.186 mm. One deviation from the KES curve is built into this fit: the law's zero pressure sits at T0, whereas KES measures T0 at its 0.5 gf/cm2 (49 Pa) reference, so at 49 Pa the model's ply is 0.477 mm thick, not 0.610 (the virtual KES-FB3 test, §8.2). A law that also passes 49 Pa at T0 and keeps LC 0.242 needs a third parameter and a

zero-pressure ("free") thickness, which KES does not measure. Contact pressures under the foot (~10^5 Pa) and the feed dog lie far above 49 Pa, where the fit holds; a ply resting under its own weight (~1 Pa) measures 0.589 mm in the virtual test, and between 1 and 49 Pa the model is up to 0.13 mm thinner than KES's curve. Where the law's stiffness reaches kappa m/dt^2 (kappa = 1e5, ~0.9 MPa) it continues linearly (a fibre- packing stop). The ply-ply contact direction is the face's vertex normals interpolated at the contact point, fixed for the tick. The compression's hysteresis is the fabric's measured resilience (KES RC 37.8 %): each vertex surface keeps a compression set $s = (1 - RC)$ of its largest compression and the pressure follows the loading law on the elastic remainder, $P = P_l\big((d - s)/RC\big)$, which reproduces the KES loading curve exactly while loading and returns exactly RC of the work on unloading; the set recovers (1 s, estimate) once the surface is unloaded below 0.5 gf/cm2 (without the set, cloth leaving the presser foot springs up at up to 0.67 m/s).

Friction. Coulomb friction with static stick. Each cloth contact, a cloth vertex on a shape or a ply on another ply, stores its elastic tangential displacement $\mathbf{d}$ from tick to tick (by shape, or by vertex for ply-ply contacts). Within a tick the contact pays the Huber potential of $\mathbf{d} + \mathbf{u}_t$, $\mathbf{u}_t$ this tick's tangential slip:

$$\psi(\mathbf{d}) = \begin{cases} \dfrac{\mu N}{2\delta}\,|\mathbf{d}|^2 & |\mathbf{d}| \le \delta \\ \mu N\,(|\mathbf{d}| - \delta/2) & |\mathbf{d}| > \delta \end{cases} \tag{8}$$

a tangential spring of stiffness $\mu N/\delta$ up to the Coulomb limit μN and sliding at μN beyond it. At the end of the tick the return mapping cuts a slid contact's displacement back to δ along its slip. This is the elastic-perfectly-plastic element of the yarn crossings (Dahl's and Iwan's pre-sliding model), and it is the exact backward-Euler incremental potential of rate-independent Coulomb friction, so a step is still a minimisation. A load below the limit is held at a fixed displacement: a draped stack does not creep. The normal force N and the tangent plane are lagged from the tick start. The pre-sliding displacement is $\delta = 0.1$ mm (estimate: fibre bending at the asperities of fibrous contacts; no measured value for cotton on steel was found). Measured: coupons on an incline hold at 6 deg (32 um of elastic pre-slip, acceleration zero) and slide at 12 and 18 deg at 0.6006 and 1.6328 m/s^2 against the analytic 0.6003 and 1.6320 (§8.5); the plain-seam start state's draped stack stays at rest (§13).

How the stored displacement follows a changing load. The spring's stiffness $\mu N/\delta$ grows with the normal force, so a displacement stored under a light load would hold a force multiplied by a later rise of N: a ply that slid a fraction of δ on the feed dog's teeth under its own weight would hold up to its full Coulomb force once the presser foot's 20 N per toe came down on it, and the clamp at rest would lock 1.5-2 N of opposed sideways force between the foot's soles and the teeth. A real contact pressed harder at a fixed tangential displacement keeps its tangential force: the area the extra load adds carries no shear (Mindlin and Deresiewicz, as discrete-element models apply it; reference knowledge, not measured here). On the cloth's contacts with shapes each stored displacement therefore carries the peak normal force N_p it has been held at since its stick state was set; where $N > N_p$,

$$\mathbf{d} \leftarrow \mathbf{d}\,N_p/N, \qquad N_p \leftarrow N, \tag{9}$$

which keeps the force; below the peak $\mathbf{d}$ is kept and the force follows the load reversibly; a contact that slides sets its stick state again at its present load. (Rescaling against the previous tick's load instead loses force on every jitter of a contact's load and lets the draped stack creep off the table's edge at 15 um/s.) The ply-on-ply contacts keep the displacement-carried law: the same rule on them makes the top ply's free end flip up and heap behind the foot at the 4 mm feed in 10 of 11 runs, against none of 12 without it; why is not established (open, §16), and the bench clamp at rest holds 0.5-0.8 N of the plies' opposed shear.

IPC's smoothed friction (Li et al. 2020) is not used: it sticks only inside a zone of $\epsilon_v\,\Delta t$, 0.5 nm a tick at the cell's ϵ_v = 1e-6 m/s. The Newton loop cannot resolve it, so every tick runs to its iteration cap, and the remaining error moves a resting stack at 3.9 um/s without end. Raising ϵ_v only raises the creep.

The feed falls short of the dog's stroke: with the stroke equal to the dial the reference stack (two plies, 40 N, 600 sti/min) fed 2.180 mm at the 2.5 mm setting, 87 % (E2). A dial is graduated in the stitch the machine sews on the fabric it is set for (the E2 criterion: within ~3 % of the setting), so the stroke is fitted to the dial, 1.145 mm per mm of setting (`machine.feed_stroke_per_setting`, labelled fitted: the stitch length is calibrated, not predicted); it feeds the 1.5, 2.5 and 4 mm settings at -1.3, -0.2 and -0.4 %. The shortfall hardly depends on δ (2.198, 2.180 and 2.116 mm at δ = 0.05, 0.1 and 0.2 mm).

Coefficients: cotton-steel 0.15 (read 0.10-0.21), cloth-cloth 0.65 (read 0.59-0.82), cloth on the laminate table 0.30 (estimate), the feed dog's teeth 1.0 (effective, open), the hands' finger pads 0.80 (the robot catalog's generic link value, not a measured fingertip-on-cotton value), the hand housing and the Dex3 palm 0.4 and the machine's painted castings 0.3 (estimates). The three that decide whether a hand can guide the work (pad, table, dog) are unsourced.

Air. Under a ply near a support surface the air is squeezed or drawn in: a damping pressure $c(h)\,v_n$ with

$$c(h) = \left(\frac{1}{c_{\text{perm}}} + \frac{h^3}{12\,\mu_{\text{air}} L^2} \right)^{-1} \tag{10}$$

($c_{\text{perm}} = 125$ Pa over the fabric's air permeability at 125 Pa, ASTM D737; muslin 0.4 m/s, prior), plus a quadratic drag $\frac{1}{2}\rho C_d |v_n| v_n$ on every vertex (C_d 1.2, a flat plate). Not modelled: the added mass of air (for a free-hanging 200 mm flap about 1.5 times the cloth's own mass).

4 The machine

The head (`machine.py`) is a Juki DDL-9000C-class direct-drive single-needle lockstitch (stitch type 301), commanded only through its control box.

Shaft. The commanded speed is reached through the soft-start ramp (25,000 sti/min per s); 0 stops at the selected needle position (up, or down at the loop-pickup angle), creeping at 200 sti/min for the last part of the turn.

Needle bar. A slider-crank, stroke 30.7 mm, rod/crank 3.5: the point at

$$z(\theta) = z_{\text{BDC}} + r + r\cos\theta + \sqrt{l^2 - r^2 \sin^2\theta} - l, \tag{11}$$

12 mm below the plate at bottom dead centre.

Feed dog. A four-motion dog (two rows 3.5 mm wide either side of the needle, 10 mm long), rising 0.8 mm above the plate and sinking 0.5 mm below it, its horizontal stroke 1.145 times the stitch-length setting, the calibration that makes the fed length the dial's (§3, Friction; reversed while touch-back is held, switched at the next feed stroke). It is a kinematic shape: the plies move only through friction with it and the foot.

Presser foot. A body in the cloth solve (0.12 kg, free along z only) carrying the sole and toes (26 mm long, toes 6.0 and 4.5 mm with a 2 mm needle slot), pushed down by the presser spring (40 N) and gravity, resting on whatever is under it and stopped by the lifter (rate-limited to 0.2 m/s; lift1 5.0 mm, lift2 8.5 mm). Its load reaches the cloth only through contact; the foot rides up over thick places and bounces at speed by itself.

Needle puncture. When the point crosses a ply at a material point, a hole is created there and held to the needle axis by an implicit constraint whose axial force is capped by the penetration law per ply, $F_{\text{pen}} = \frac{1}{2} s_f \, (F_{\text{qs}} + c_v v_{\text{tip}})$ (F_{qs} 1.56 N, c_v 0.44 N s/m: a fit to Chmielowiec's two-ply measurements, 1.7 N at 200 rpm, 5.45 N at 5,500 rpm; s_f 0.8 for muslin with an Nm 80 needle, prior 0.6-1.0), rising over the point's 1.5 mm and falling to 0.6 of it on the shank, 0.3 on withdrawal (the ply is dragged up: flagging).

Needle deflection and damage. The holes' lateral pull deflects the needle as a cantilever from the bar (stiffness from E = 205 GPa, Nm, the free length to the plate and the factor alpha = 4.6 fitted to Tanaka et al.'s measured deflections); the engaged holes share the needle (series compliance 1/k_pin + n/k_n). The needle bends when its root moment exceeds yield (sigma_y 1.5 GPa, estimate), strikes the plate when deflected past the plate hole's edge (0.7 mm), strikes the hook beyond the hook clearance (0.07 mm), and breaks on a robot finger or the hand's housing in its path.

Stitch formation. At loop pickup (2 mm past bottom dead centre) a stitch forms only if the needle is intact, its deflection over the cycle stayed within the hook clearance and the loop's reach and timing, and no ply lifted more than 1 mm with the needle (flagging); otherwise a skipped stitch is recorded with its reason. A formed stitch binds the holes of every pierced ply (interlock: rest length the stack's thickness + 0.05 mm, stiffness the thread's EA over it) and adds needle- and bobbin-thread elements from the previous interlock, created slack and drawn to their set tension by the take-up once the needle has left the cloth. Set tensions at the take-up: the dial tension (needle 0.59 N, bobbin 0.25 N) times the speed factor $g(n) = 1 + 1.68\,(n - 200)/5300$ (Chmielowiec, n in sti/min). The seam's elements carry the pulls at the interlock: the bobbin's set tension, and the needle thread's set tension divided by its path loss 2.36. The loss is derived, not fitted: the default dial pair sews balanced stitches, and an interlock inside the stack is in balance when the two threads pull it equally, so the loss is the default ratio 0.59/0.25 (the capstan friction of the needle thread's path through the eye and the hole, $e^{\mu\theta}$ with $\theta \approx 205$ deg at $\mu = 0.24$). At the default pair a balanced seam therefore carries no tension couple across the stack; a tight top thread does, and curls and puckers the seam. Balance is a law: the ratio of the two pulls within the capstan band $e^{\pm\mu\pi}$ (μ 0.24), not an emergent interlock position. Threads break above $(1 - 0.40) \times 14.9$ N $\times\, 0.68$.

Thread to the needle and the bobbin; trimmer. Until trimmed, the needle thread runs from the last stitch to the needle's eye and the bobbin thread to the bobbin (tension-only elements capped at the static dial tensions). While the shaft turns, the take-up lever supplies and takes up this thread and sets each stitch, so these elements are slack; when the shaft stops, their rest length is their length at that moment, and pulling the

work away pays thread out through the tension discs at the static tension (a constant pull while sewing would curl the seam). The trimmer stops needle-up and cuts both threads, leaving 3 mm tails.

5 Robots and the hand-cloth coupling

The robots are the repository catalog's (`robots/`), selected by alias:

Alias	Rig	What stands at the machine
`dual-fr3` (default)	`sewing-fr3`	two Franka FR3 arms with Franka Hands on one pedestal in the operator's seat (bases 0.55 m in front of the needle, 0.20 m above the table, left at x = -0.30 m, right at x = +0.10 m), `sewing` profile: hands down 100 mm above the table in front of the work
`dual-panda`	`sewing-panda`	the same hands on Pandas
`single-fr3`	`sewing-fr3-single`	the right arm alone (one hand can guide a seam)
`g1`	`sewing-g1`	the Unitree G1 (29 body joints, two Dex3-1 hands) standing on the floor 0.13 m behind the table's front edge, centred on the needle line, facing the machine; the repository's standing G1 profile (`whiteboard`)

Rigid world. Newton 1.5 with its MuJoCo solver on the CPU, on the same 0.5 ms clock as the cloth. Joint targets pass a rate limiter at the joints' speed and acceleration limits (the motion generator copied from welding): the FR3's 10 rad/s^2 per joint and the Panda's 15/7.5/10/12.5/15/20/20 (libfranka's rate limiter, read for each robot); the G1, which has no motion generator, gets an environment ramp of 10 rad/s^2. An FR3 or Panda arm can also be made compliant: the raw operation `joint_impedance` sets its joints' stiffness and damping as libfranka's Robot::setJointImpedance does (§15). MuJoCo's midphase is disabled: its per-body bounding trees, built when the model is compiled, do not cover the collision geoms the shared robot adapter places afterwards (29 geoms outside their trees, the worst an FR3 link 171 mm out), so contacts between the arms, the table and the machine would be silently dropped. No shape carries a contact-detection gap: Newton's default 0.1 m gap widens detection under MuJoCo 3.9 without generating force, and with it the G1 carries ~1,000 force-free pairs of its own links up to 200 mm apart through every step (1,075 MuJoCo contacts a tick against 8 without; its rigid step then costs 25 wall seconds per simulated second on a shared host).

FR3 and Panda. The Franka Hand has a position mode and a force mode (up to 35 N per finger, 70 N per hand). The arms carry libfranka's collision reflexes (10/20 N m joint-torque thresholds); the joint torques that trigger them include the cloth's reaction on the hand.

G1. The humanoid stands on its position servos (legs kp 800, waist and arms 120, wrists 80, Dex3 fingers 1.8) with no weld to the floor, no pelvis support and no balance controller, as in barcode scanning and the whiteboard environment. The waist and arm servos add a gravity feed-forward torque computed from the model at the commanded configuration, as Unitree's G1 manipulation stack sends it (barcode scanning's measured 4.2 deg waist sag without it). It has no protective stop: the G1's low-level joint interface has no collision or velocity reflex. The Dex3 hands are commanded joint by joint. Cartesian commands go through a seven-joint hand IK on a private copy of the MuJoCo state (`left`, `right`), or ten joints with the waist (`left+waist`, `right+waist`: the robot leans).

Coupling. A hand touches the cloth through its collision shapes in the cloth solve: each Franka Hand's two finger boxes (the URDF's collision boxes) and its housing, and each Dex3 hand's six finger links and its palm (the boxes around their URDF collision meshes), with the pad friction 0.80 (fingers) and 0.4 (housing, palm). The import merges the FR3's `fr3_link8` and `fr3_hand` into `fr3_link7`, so the housing box is posed on link 7 from the URDF's mount (0.107 m along the link's axis, turned -45 deg about it).

The shapes do not ride rigidly on the arm. Each hand's shapes ride on one body in the cloth solve, free along the vertical (the direction a hand presses the work) and held to the arm's pose by a spring with a structural damper:

$$f = k\,(z_a - z_b) + c\,(\dot{z}_a - \dot{z}_b), \qquad k = 3 \times 10^4 \text{ N/m},\ c = 2\zeta\sqrt{k\,m},\ \zeta = 0.3,\ m = 10 \text{ g} \tag{12}$$

(z_a the arm's commanded height for the hand, z_b the body's; `coupling.hand_stiffness`, `hand_damping_ratio`, `hand_mass`, all estimates). The spring is implicit in the cloth's step: the damper is folded into it for the tick ($k + c/\Delta t$ towards the target that reproduces the damper's force), so the impact of a hand on the work loads the spring within the same backward-Euler solve as the fabric's compression. k is the order of a collaborative arm's structural stiffness at its tool (harmonic drives, joint torque sensors, the fingers); no measured FR3 or Dex3 value was found. The body's mass is the pads' and fingertip shells' only: the hand's own mass is in the arm's model, which drives the spring. Its weight is compensated, so a hand held in the air keeps its pose. Horizontally and in rotation the shapes follow the arm rigidly.

What the arm feels is the spring's force along the vertical and the cloth's contact and friction forces across it. These are low-pass filtered (1 ms: a numerical damper of this explicit, one-tick-lagged coupling, physical value zero; at 5 ms a light hand pressing the work chatters, §8.9) and applied to the hand bodies as external wrenches in the next rigid step. A hand presses, pinches and drags the cloth only through these contacts; nothing attaches cloth to a hand. The machine's needle strikes and breaks on any of these shapes in its path.

Why the compliance: with rigid pads, the FR3 lowered onto the stack at 0.15 m/s crushes the two plies from 0.86 to 0.21 mm in 4.5 ms under 245 N, about 3.8 kg of effective arm mass stopped by the cloth over steel. With the sprung hand the same approach peaks at 21.6 N, and it scales with the stiffness: 8.3, 21.6 and 56 N at 1e4, 3e4 and 1e5 N/m. The body's mass is the pads' own: with 0.1 kg (the fingers' mass) the G1's palm body rings on the stack's compressed stiffness (~2e5 N/m) at ~180 Hz, its contact force swinging between 0 and 11 N while the spring stays smooth, and the damper does not reach that mode. With the pads' mass the mode lies beyond 700 Hz, where the implicit step damps it (E18, §8.9).

Reach and clearance (FR3). E0b (reach preflight): with the pedestal 0.42 m in front of the needle at table height 105 of 110 hand-down key poses are reachable with joint-limit margin down to 0; at 0.55 m in front and 0.20 m above the table all 110 are reachable (>= 0.46 rad margin, median 0.72). E0c (static contacts at working points around the needle, 18 per hand and orientation): straight down, the right wrist meets the head near the needle (6-12 of 18 clear); tilted 30 deg forward the hand's housing meets the table top at the lowest points (right hand at yaw -90 deg: 7 of 18 clear); the left hand clears 12 of 18 at yaw 0. Robot motions near the machine must also be planned: a joint-space move from the ready pose to a point under the head swings the FR3's wrist through the machine's overarm.

Reach and posture (G1). E0b-G1: the same 110 key points with the hand flat (palm down, fingers toward the machine; the Dex3 palm faces the wrist link's +y on the right hand and -y on the left, measured by flexing a finger). Standing 0.08, 0.13, 0.18 or 0.23 m behind the table's edge the hands reach all 110 from each place, leaning on the waist for 17, 36, 54 and 71 of them (median joint-limit margin 0.62, 0.41, 0.31, 0.22 rad). The robot stands 0.13 m behind the edge. Its ready posture holds the palms flat 0.13 m over the table, the left 0.19 m left of the needle and 0.17 m in front of it, the right 0.15 m right and 0.20 m in front; it came from the hand IK and a contact check at the commanded joint positions of every robot link against the table, the machine and the robot's own links: from 0.13 m and further back it touches nothing (from 0.08 m the right fingers would stand 3.5 mm inside the machine). Standing in the final posture with the cloth for 15 s (E14), the pelvis stays 0.784 m above the floor, tilted 0.16 deg, the joints settling by at most 0.43 mrad. From the head camera the machine's face plate hides the needle when the robot stands square (a seated operator is centred on the needle line); the machine-mounted `needle` camera is the humanoid's second policy camera.

6 Numerics and the start state

Clock. One 0.5 ms tick for the robots and the cloth (2 kHz: at least 9 ticks of feed-dog contact per stroke at 5,000 sti/min). Each tick: the rigid step (MuJoCo, CPU), the hands' boxes posed in the cloth solve, the machine's pre-tick (shaft, dog, lifter), the cloth's implicit step, the machine's post-tick (punctures, deflection, loop pickup, threads), the hands' springs and contacts filtered onto the arms, the robots' protective stops.

Solver. Newton on the incremental potential: two iterations, then up to six in total while the accepted step exceeds 1e-4 m/s x dt (or, tangentially on a particle's dominant contact, 1/20 of the friction's pre-slip δ: an unconverged step leaves an error in the stored displacement, not a drift; a body on its stop has that axis in the active set); each a block-Jacobi PCG with 40 iterations and a backtracking line search on the same potential. The whole tick is a captured CUDA graph (`capture_while` for the convergence loop). Lagged per tick: the friction normal forces and tangent planes (the stored stick displacements are state, updated by the return mapping at the tick's end), the ply-ply contact pairs (closest face per ply, persistent) and their contact normals, the air layer's gaps and the drag's |v_n|. Everything else is implicit.

The mesh. Each ply is meshed in its material coordinates with 1.25 mm triangles within 22 mm of the seam edge and the seam line, growing to 7.5 mm over three coarse edges beyond (`mesh.py`; the plain-seam ply 12,000 vertices). The band is 1.25 mm because the baste task's 6 mm allowance puts the top ply's cut edge 0.2 mm outside the right toe's corner: on a 2.5 mm band one row of triangles straddles it, and at the 5 mm stitch the edge curls behind the foot and the work wads in every run; at 1.25 mm the same start bastes flat. The plain seam's turn under the foot is the same on both (5.5 against 5.8 deg after 1.0 s), and a tick costs 1.65 times the 2.5 mm mesh's while sewing.

The start state. The plies are laid analytically (flat, the part beyond the table's front edge draped concentric with its rounded edge) and brought to rest by dynamic relaxation: the whole cell steps normally for 1.0 s while the cloth's velocities are zeroed whenever its kinetic energy passes a peak, then 0.25 s settle undamped. The result is cached by configuration and by a hash of the physics source (the sewing modules that shape the physics,

the robot catalog, the robot models and the sewing rigs); a session loads it in seconds, and any change to that source prepares it anew. Measured on the plain-seam start: an analytic drape over a sharp table edge stores 0.32 mJ of chordal compression on the 7.5 mm mesh (released in 10 ms, it sets the overhang swinging for seconds); the relaxed start holds ~1e-11 J of cloth kinetic energy (section 13).

Rigid warnings. mujoco_warp's model conversion prints, for every configuration, that multi-contact collision is on but cylinder pairs get at most one contact. The dual-FR3 cell has no cylinder geoms; the G1 cell has four (the shoulder pitch and roll links, radius 30 mm), which touch nothing in the working poses. No other warning is printed. The scene passes gravity as the vector (0, 0, -9.81), as Newton 1.4 and later expect; Python's remaining deprecation warnings come from matplotlib's own use of pyparsing at import (third-party).

7 Parameter ledger

Every number the model uses is a ledger entry in `params.py` with its value, unit, label and source, and the fabric profiles carry a source per field; `python -m sewing.params` writes the ledger as `PARAMETERS.md`. 88 entries: 33 read from a manual, a datasheet or a paper; 8 derived from read values (among them the needle thread's path loss 2.36 and the muslin's compressed thickness); 1 literature; 2 from the repository; 25 estimates (mostly machine dimensions and needle-law thresholds that no source gives, the table edge's radius, the compression set's recovery time); 4 open (the needle's shank and withdrawal fractions, the cloth's friction on the feed dog and on the laminate); 2 priors (the air permeability per fabric and the squeezed gap's length); 11 authored numerics (the tick, the mesh, the contact penalty, the Newton and CG settings); 1 abstract-only (the thread's 40 % sewing strength loss); 1 read/open (the thread's capstan friction). Every named entry is read by the model or says why it is not: the servo's torque and the shaft's inertia (the shaft follows its speed command: a 6 N puncture on the 15.35 mm crank needs ~0.09 N m, 4 % of the servo's peak), the hand lifter (not built) and the rejected solver paths' substeps.

The fabric that the tasks use, `muslin_125`, is one fabric measured in one source (Ramaiah et al., KES-FB, fabric C: unfinished plain-weave cotton, 125.2 g/m2, 0.61 mm): areal density, thickness, tensile linearity and energy (the tension law through EMT and LT), shear rigidity, bending rigidity, bending hysteresis 2HB, and the compression curve (WC, LC, RC; Tm derived) are all read from that table; the warp/weft split of B uses Pal et al.'s ratio (estimate), the toe-load fraction of the tension law is an estimate, and its friction pairs are read ranges. Three more profiles exist for the presets not built (`shirting_126`, `denim_440`, `paper_75`); their missing KES values are derived from fabric C's ratios and labelled so.

Further published values for the least grounded friction coefficients and machine dimensions (values as published, not measured here; the ledger's entries in `params.py` do not cite them):

- the hand's pad on cloth (0.80, `repo`: the catalogue's value on every FR3 link): smooth ABS on cotton shirting 0.83 static, 0.80 kinetic (US patent 10458053 B2, table 4: an ASTM D1894 sled at 200 g, vendor data); a human forefinger sliding on plain-weave cotton 1.05 +- 0.32 (Nagatsu et al., Sen'i Gakkaishi 64(4), 2008: 15 subjects, ~1 N, 2 mm/s). The Franka Hand's fingertip material is not documented, so 0.80 stays a plastic-pad value, not a measured one;
- cloth on the laminate table (0.30) and the feed dog's effective grip (1.0) with its 0.1 mm pre-slip: no published value was found (Taylor and Pollet 2000 measured apparel fabrics on Formica at low load, but its numbers were not accessible; the only toothed-steel-on-cotton number found is a micro-pillar steel surface in the same patent, 2.69 static, 2.60 kinetic, not a feed dog): these stay open;
- the air permeability of muslin: no value at 100-125 Pa for a 110-130 g/m2 cotton was found; it stays a prior;
- the machine: Juki's DDL-9000C specification sheet confirms the needle bar's 30.7 mm stroke, the feed dog's 0.8 mm height, the 5.5 mm hand lift and the auto-lifter's 5.0 and 8.5 mm steps (and gives 15 mm for the knee lifter, 13.5 mm for the auto-lifter's maximum); no published drawing gives the presser foot's sole and toes, the feed dog's footprint and tooth pitch, the throat plate's size or the presser bar's moving mass, which stay estimates.

8 Component validation

Each experiment states its prediction before it runs and is discussed against it; the numbers below are copied from the result files by code.

8.1 Bending and its hysteresis: a virtual KES-FB2 loop (E1b)

60 x 60 mm strips of the mesher's own triangulation are wrapped isometrically onto cylinders through the KES-FB2 program (0 -> +2.5 -> -2.5 -> +2.5 /cm in 0.01 /cm steps); after each step the solver's own return mapping runs, and the moment per unit width is (1/A) dU/dK of the solver's hinge energy. Against the inputs (B 3.0e-6 warp, 2.3e-6 weft, the bias the mean; 2HB 0.0277 gf cm/cm), per unit of the area the hinges carry:

Mesh	Direction	B model / input	2HB model / input
2.5 mm	warp	0.980	0.978
2.5 mm	weft	0.963	0.974
2.5 mm	bias (not calibrated)	0.913	0.974
7.5 mm	warp	0.992	0.989
7.5 mm	weft	0.803	0.950
7.5 mm	bias (not calibrated)	0.877	1.006

The free-edge rows of a strip carry no hinge (3 % of the area at 2.5 mm, 8 % at 7.5 mm); against the full area the ratios are those times 0.97 and 0.92. The coarse mesh's weft deficit is the KES program's curvature (radius 4-20 mm) on 7.5 mm triangles. The loop's shape (rounded reversals over ~0.4 /cm, parallel branches 2HB apart) is in the figure below.

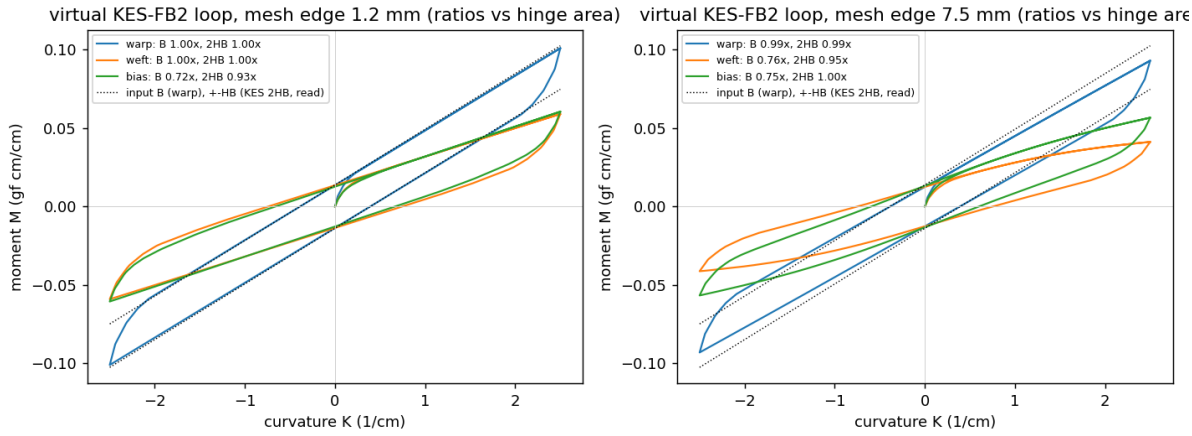


Figure 13: The virtual KES-FB2 loop on the discrete model (E1b): moment against curvature, warp, weft and bias, against the fabric's B and 2HB

8.2 Compression and its hysteresis: a virtual KES-FB3 test (E1c)

A 20 x 20 mm coupon of one ply on a static plate is squeezed by a kinematic platen moving at 0.1 mm/s, as the KES-FB3 head does, to 50 gf/cm2 and back (gravity off: the coupon's 1.2 Pa is below the test's resolution); the pressure is the platen's measured reaction over the coupon's area, the thickness the platen-plate gap. Against the same fabric's KES values (Ramaiah fabric C):

Quantity	Model	KES
Tm at 50 gf/cm2 (mm)	0.310	0.311
WC (gf cm/cm2)	0.184	0.181
RC (%)	37.8	37.8
T0 at 0.5 gf/cm2 (mm)	0.477	0.610
LC (from the model's own T0)	0.44	0.242

Tm, WC and RC hold to 2 %. T0 does not: the law is fitted with zero pressure at KES's T0, whereas KES reads T0 at 0.5 gf/cm2 (§3); LC computed against the model's own 0.477 mm inherits that. The loading branch passes 49 Pa at 0.477 mm and 1 Pa at 0.589 mm. On unloading the stack follows the set (thinner than on loading at every pressure) and shows a plateau at 0.5 gf/cm2 between 0.37 and 0.48 mm while the set recovers against the slowly lifting platen (1 s time constant, estimate), about 3 % of WC.

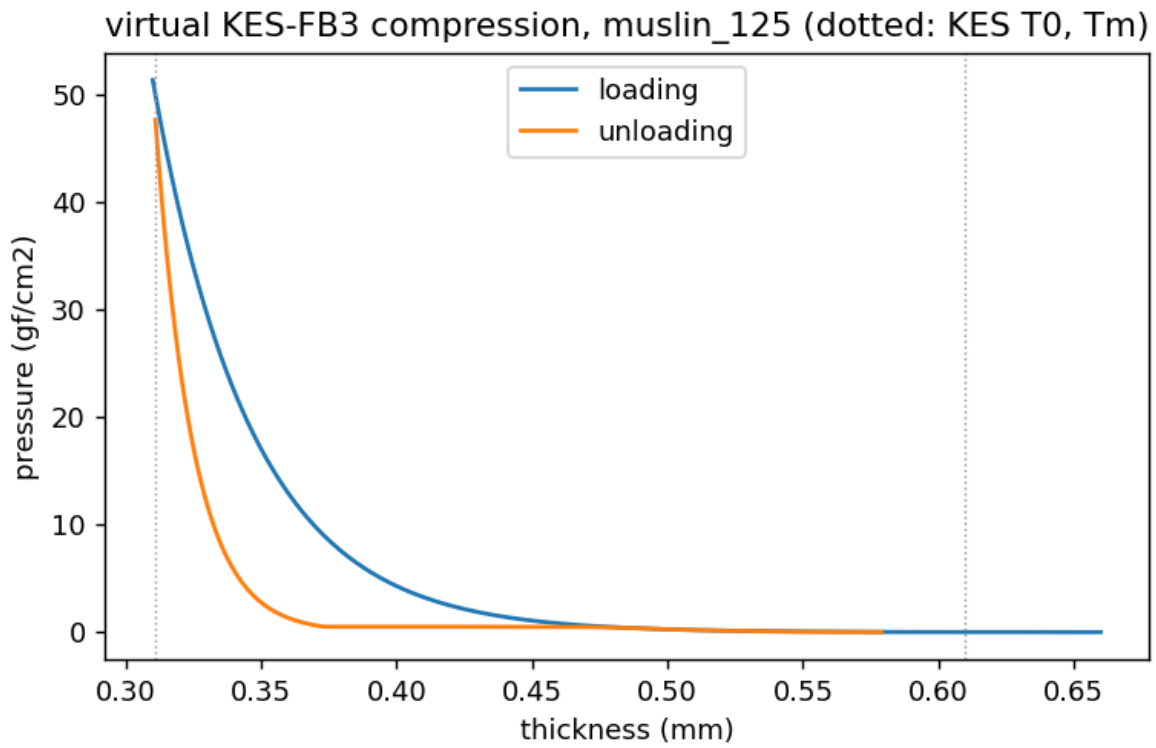


Figure 14: The virtual KES-FB3 compression test (E1c): pressure against thickness, loading and unloading, against the fabric's T0, Tm, WC, LC and RC

8.3 Tensile and shear hysteresis: virtual KES-FB1 loops (E1d)

A 40 x 40 mm square of the mesher's triangulation (2.5 and 7.5 mm) is deformed affinely through the KES-FB1 programs: pure stretch along one yarn family to 500 gf/cm and back to zero load (5e-5 strain steps), and a trellis shear (the yarns rotate without stretching, as a woven net shears) 0 -> +8 -> -8 -> +8 deg in 0.02 deg steps. After each step the solver's own return mapping runs; the force per unit width is (1/A) dU/dq of the solver's membrane energy at the current plastic state (central differences of 5e-5 strain and 2e-5 rad: the kernels form the yarn strain in float32, so steps of 1e-6 read the tension only to +-5 %). Against the same fabric's KES-FB1 values (Ramaiah fabric C):

Quantity	**Model (2.5 mm warp / weft, 7.5 mm warp / weft)**	**KES (read)**
RT (%)	49.93 / 49.93 / 49.93 / 49.93	49.94
LT	0.598 / 0.598 / 0.598 / 0.598	0.597
WT (gf cm/cm2)	9.89 / 9.89 / 9.89 / 9.89	9.91
EMT (%)	6.62 / 6.62 / 6.62 / 6.62	6.6 (law)
G (gf/cm deg), 2.5 / 7.5 mm	0.517 / 0.517	0.52
2HG (gf/cm)	1.236 / 1.238	1.24
2HG5 (gf/cm)	1.779 / 1.779	1.79

The affine deformation strains every triangle alike, so the mesh size does not enter. RT is calibrated (TF_COEF), LT, WT and EMT test that loading stays on the law the tension was fitted to, G, 2HG and 2HG5 test the shear friction's two-point fit.

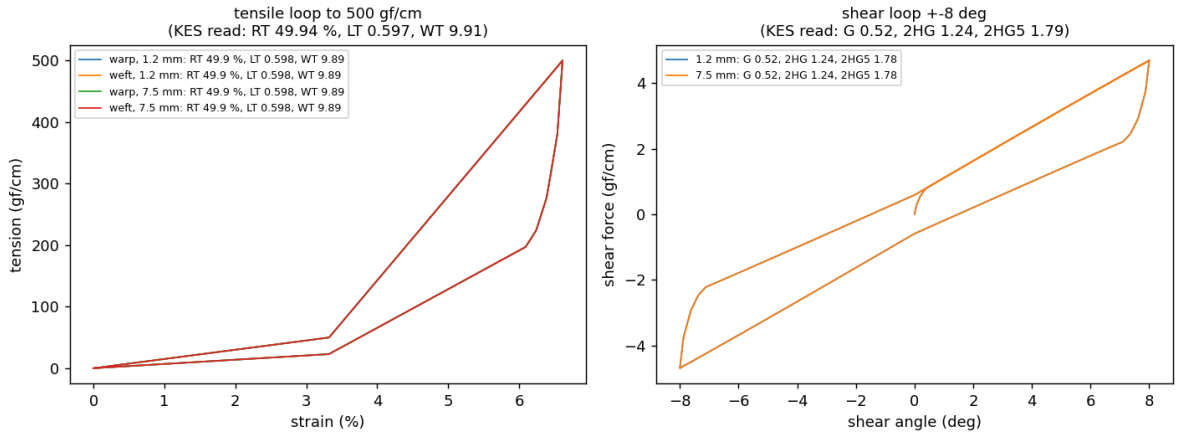


Figure 15: The virtual KES-FB1 tensile and shear loops (E1d) against the fabric's RT, LT, WT, EMT, G, 2HG and 2HG5

8.4 Reach and clearance (E0b, E0c, E0d)

See §5. E0b (FR3): from the chosen pedestal (0.55 m in front of the needle, 0.20 m above the table) all 110 hand-down key poses are reachable with at least 0.46 rad of joint margin (median 0.72); from 0.42 m in front at table height 105 of 110. E0b-G1: all 110 flat-hand key points from every stand 0.13-0.28 m behind the table's edge, the ready posture touching nothing at any of them. E0c (static contacts at 18 working points around the needle per hand and orientation, with complete contacts): straight down the right hand clears the head at 6-12 of 18, the left at yaw 0 at 12 of 18; tilted 30 deg forward the hand's housing meets the table top at the lowest points (the right hand at yaw -90 deg: 7 of 18 clear); at 45 and 60 deg of tilt almost every point touches the table or the pedestal.

E0b and E0c check each hand alone; E0d checks the pair working together, statically: both hands come from their ready poses along straight lines to 60 mm over their places on the work and down to their pressing heights, IK seeded from the previous point, and at every waypoint MuJoCo's contacts between any robot link and the table, the machine or the other arm are listed; then each hand's pressing height (the TCP height at which its lowest part meets the bed) and both hands travelling together with the work. The guided seam's first places (the right hand 120 mm in front of the needle at its seam edge, turned -90 deg and tilted 30 deg; the left 110 mm left of the needle) collide on the way in at 13 of 28 waypoints and at the places themselves: the left arm's wrist (links 6 and 7) sits up to 49 mm inside the right arm's link 5. No variant of those places was clear (orders, transit heights, the left hand moved to x -0.15 or -0.17 m or turned, the right turned or held straight down): with both FR3s on one pedestal 0.4 m apart, both hands working in front of the needle put the arms across each other. The places the guided seam uses, found by scanning: the right hand 170 mm in front of the needle, turned -60 deg and tilted 30 deg, the left 150 mm left of the needle and straight down; clear at every waypoint; pressing heights 26.5 and 10.5 mm (their lowest parts meet the bed at 25.8 and 9.4 mm); clear over 30 mm of travel with the work, the arms touching from 40 mm, so the hands regrip after 30 mm.

8.5 Cloth coupons (E1)

Three checks of the cloth model on its own (no machine, no robot), each at the 0.5 ms tick and at half of it.

Tension. A strip (warp, weft, bias) under a prescribed edge load (load control, test-only velocity damping that does no work at rest) against the tension law. Warp and weft strains are within 0.7 % of the law at 20 N/m, 1.5 % at 100 N/m and 5.0 % at 300 N/m; the tick does not change them. The bias strips are compared with what the model's own laws predict for a bias pull, each yarn at half the load on its law and the shear from the elastic G plus the shear friction's loading branch (KES 2HG, §3): at 1 and 3 N/m the shear stress is about the friction's slip force (0.58 N/m), so the net barely shears at 1 N/m and the elastic formula E45 (which would give 0.0089) overstates the strain fivefold; the model follows the friction-aware prediction.

Direction	Tension (N/m)	dt (ms)	Strain	Expected (law; bias: law and shear friction)	Error
warp	20	0.5	0.0136	0.0135	+0.4 %
warp	100	0.5	0.0370	0.0370	+0.1 %
warp	300	0.5	0.0520	0.0519	+0.3 %
weft	20	0.5	0.0136	0.0135	+0.4 %

Direction	Tension (N/m)	dt (ms)	Strain	Expected (law; bias: law and shear friction)	Error
weft	100	0.5	0.0370	0.0370	+0.1 %
weft	300	0.5	0.0520	0.0519	+0.2 %
bias	1	0.5	0.0019	0.0018	+3.6 %
bias	3	0.5	0.0167	0.0168	-0.6 %
warp	20	0.25	0.0136	0.0135	+0.3 %
warp	100	0.25	0.0370	0.0370	+0.0 %
warp	300	0.25	0.0519	0.0519	+0.1 %
weft	20	0.25	0.0136	0.0135	+0.3 %
weft	100	0.25	0.0370	0.0370	+0.0 %
weft	300	0.25	0.0519	0.0519	+0.1 %
bias	1	0.25	0.0019	0.0018	+3.6 %
bias	3	0.25	0.0167	0.0168	-0.6 %

Direction	Clamp	Mesh (mm)	dt (ms)	Bending friction	Chord angle (deg)	Elastica (deg)	Peirce B / input	Model error
warp	platform	1.25	0.5	off	43.0	41.5	0.997	-5.7 %
warp	platform	1.25	0.5	on	30.3	41.5	1.651	+56.1 %
warp	platform	0.625	0.5	off	46.3	41.5	0.879	-16.9 %
warp	platform	0.625	0.5	on	34.4	41.5	1.396	+32.0 %
warp	vertices	1.25	0.5	off	42.6	41.5	1.015	-4.0 %
warp	vertices	1.25	0.5	on	30.0	41.5	1.677	+58.5 %
warp	vertices	0.625	0.5	off	45.1	41.5	0.922	-12.8 %
warp	vertices	0.625	0.5	on	36.7	41.5	1.272	+20.3 %
warp	vertices	1.25	0.25	off	42.6	41.5	1.016	-4.0 %
weft	platform	1.25	0.5	off	57.7	56.4	1.059	-5.2 %
weft	platform	1.25	0.5	on	41.4	56.4	2.027	+81.3 %
weft	platform	0.625	0.5	off	66.0	56.4	0.715	-36.1 %
weft	platform	0.625	0.5	on	52.3	56.4	1.326	+18.6 %
weft	vertices	1.25	0.5	off	57.3	56.4	1.077	-3.6 %
weft	vertices	1.25	0.5	on	41.1	56.4	2.050	+83.5 %
weft	vertices	0.625	0.5	off	63.4	56.4	0.814	-27.2 %
weft	vertices	0.625	0.5	on	53.6	56.4	1.257	+12.5 %
weft	vertices	1.25	0.25	off	57.2	56.4	1.083	-3.1 %

Incline (deg)	dt (ms)	mu	Slide (mm)	Acceleration (m/s^2)	Expected
6	0.5	0.15	0.03	0.000	0.000
12	0.5	0.15	50.84	0.631	0.600
18	0.5	0.15	129.34	1.660	1.632
6	0.25	0.15	0.02	0.000	0.000
12	0.25	0.15	49.25	0.601	0.600
18	0.25	0.15	129.87	1.633	1.632

Peirce cantilever (ASTM D1388). The strip's chord angle against the exact heavy elastica for the same bending rigidity and overhang, so Peirce's own approximation (+5.8 % along the warp, +7.9 % along the weft on the exact curve) is separated from the model's error. The clamp is built as the boundary condition of the discrete strip: a straight mesh row on the clamp line, and the hinges across it stiffened to their free wing, since the clamp holds the other wing flat. With the bending friction off the bending length is 1.1-1.8 % short of the B-derived value on the seam band's 1.25 mm mesh, both clamps, warp and weft, at both ticks (B 3.1-5.3 % low; the E1 criterion: 5 %); every case settles. On a mesh half that (0.625 mm, a convergence check; no task uses it) it is 4.5-13.9 % short (B 13-36 % low, the weft on the platform clamp the worst), so the deficit, smaller at 1.25 mm than at 2.5 mm, grows again below it; why is not established (the hinges' calibration at that size and the solve's convergence there are not separated), and the acceptance check fails on it. A residual of 3.5-5.6 % in B on this mesher's strip is not isolated (a structured strip reads 1.3 %; the hinges' cylindrical calibration accounts for 1.5 and 0.5 %). With the bending friction on, the loaded cantilever reads 1.57-1.83 times the KES B on the 1.25 mm mesh (1.12-1.33 on the 0.625 mm one): the friction moment resists the bend on loading, as in a real Peirce test, whose B includes the hysteresis that KES B excludes. A quasi-static reference built on the model's own friction law predicts more (2.0-2.1 times); the probe releases the whole overhang at once where the standard slides it out slowly, which is not modelled (open). The tick does not change any of it.

Incline. A coupon on a steel plate tilted 6, 12 and 18 deg (cotton on steel mu 0.15, slip angle 8.5 deg): it sticks at 6 deg (0.02-0.03 mm of creep in 0.4 s) and slides at 12 and 18 deg. At half the tick it slides at g(sin - mu cos) to 0.1 %; at the default tick, on the seam band's 1.25 mm mesh the coupon is cut from, it slides 5.2 and 1.8 % fast (the sliding friction 2 % low, an effective 0.147 against 0.15), so the acceptance check (1 %) fails. The same

coupon meshed at 2.5 mm on the same physics slides to 0.05 % at both ticks: the error comes with the finer band at the 0.5 ms tick; why is not established.

8.6 The drop feed (E2)

The machine with the needle out (feed only), two plies 120 x 200 mm under the lowered foot, 0.6 s after the shaft has reached speed. The expectation: the advance is the dog's travel while above the plate less the slip under the foot's load, nearly the full setting for a light stack, with the top ply lagging (the reason walking feet exist).

Speed (sti/min)	Setting (mm)	Presser (N)	Bottom ply (mm/stitch)	Top ply	Shift (mm/stitch)	Highest cloth (mm)	Foot load (N)
600	2.5	40	2.522 (101 %)	2.499	-0.023	1.32	41.2
1500	2.5	40	2.498 (100 %)	2.480	-0.018	3.34	41.2
3000	2.5	40	2.367 (95 %)	2.372	+0.005	2.96	41.2
600	2.5	20	2.531 (101 %)	2.512	-0.019	1.25	21.2
600	2.5	60	2.504 (100 %)	2.475	-0.029	1.38	61.2
600	1.5	40	1.493 (100 %)	1.488	-0.005	1.38	41.2
600	4	40	4.009 (100 %)	3.980	-0.028	1.33	41.2
600	5	40	4.996 (100 %)	4.970	-0.025	1.38	41.2

The plies advance 95-101 % of the setting: 100.9 and 100.0 % (bottom and top ply) at the 2.5 mm, 600 sti/min, 40 N case the dog's stroke is fitted to (`fitted`, 1.145 of the dial; §7), within 1 % at 1.5 and 4 mm, at the baste's 5 mm (99.9 and 99.4 %) and at 1,500 sti/min, 95 % at 3,000 sti/min (the dog's stroke is shorter in time and the stack's inertia slips more), and 99-101 % at 20 and 60 N. The top ply lags the bottom by 0.019, 0.023 and 0.030 mm per stitch at 20, 40 and 60 N: 0.8, 0.9 and 1.2 % of the length, rising with the presser force as Carvalho measured on a drop feed (0.2-2.4 % over 50 cm, rising with force). No published feed efficiency for muslin on this head was found.

At top speed (E2b, UNASSISTED). Two plies 100 x 120 mm wholly on the bed, the machine alone at its 5,000 sti/min maximum for 0.42 s from rest. The model as built, six runs at the default tick and six at half the tick (runs are not bit-reproducible, §8.8):

Model	Repeat	Stitches formed / revolutions	Needle	Faults	Advance at the needle / 100 mm away (mm)	Cloth at the needle minus far end (mm)	Far end height, max (mm)	Foot unloaded (< 5 N)
as built (KES RT, 2HG; eps_v 1e-6)	0	27 / 26	ok	none	57.8 / 51.6	1.8 to 7.2	0.88	20 %
as built (KES RT, 2HG; eps_v 1e-6)	1	26 / 26	damaged	hook_struck_needle	57.6 / 51.7	1.8 to 7.9	0.88	20 %
as built (KES RT, 2HG; eps_v 1e-6)	2	27 / 26	ok	none	57.2 / 50.7	1.8 to 8.1	0.88	20 %
as built (KES RT, 2HG; eps_v 1e-6)	3	27 / 26	ok	none	58.8 / 52.8	2.1 to 7.4	0.89	20 %
as built (KES RT, 2HG; eps_v 1e-6)	4	27 / 26	ok	none	58.5 / 52.2	2.0 to 7.4	0.88	20 %
as built (KES RT, 2HG; eps_v 1e-6)	5	27 / 26	ok	none	58.1 / 51.6	1.5 to 7.6	0.88	20 %

Model	Repeat	Stitches formed / revolutions	Needle	Faults	Advance at the needle / 100 mm away (mm)	Cloth at the needle minus far end (mm)	Far end height, max (mm)	Foot unloaded (< 5 N)
as built, at half the tick (0.25 ms)	0	27 / 26	ok	none	57.7 / 52.4	1.4 to 6.2	0.89	24 %
as built, at half the tick (0.25 ms)	1	27 / 26	ok	none	58.1 / 52.9	1.2 to 6.2	0.89	24 %
as built, at half the tick (0.25 ms)	2	27 / 26	ok	none	57.8 / 52.1	1.3 to 6.3	0.89	24 %
as built, at half the tick (0.25 ms)	3	27 / 26	ok	none	57.1 / 51.6	1.3 to 6.2	0.89	24 %
as built, at half the tick (0.25 ms)	4	27 / 26	ok	none	56.9 / 51.6	1.3 to 5.9	0.89	24 %
as built, at half the tick (0.25 ms)	5	26 / 26	damaged	hook_struck_needle	58.0 / 53.7	1.3 to 5.6	0.90	24 %

Unguided at the maximum, a short stack can run away (open). In a clean run the strip between the needle and its far end stretches and relaxes every stitch (the cloth at the needle leads the far end by up to 8 mm); when the grip at the needle is lost the stretched strip springs back, the far end overruns the cloth at the needle, and the cloth drags the needle into the hook or the plate. The two ticks agree: at 0.5 ms and at 0.25 ms no run of six overran (the cloth at the needle stayed 1.5-8.1 and 1.2-6.3 mm ahead of the far end), and in one run of six at each the hook struck the deflected needle (26 stitches, the needle damaged); the rest ran clean. The outcome is sensitive to the cloth model near the needle (with a 2.5 mm seam band at half the tick one run of six overran by 12.3 mm and the hook struck the needle in all six), and six runs a tick cannot show that the dependence on the tick is gone. A missing loss is suspected (candidates: the presser bar's bushing friction and damping, not modelled; the foot's moving mass, an estimate; the dog's smooth box teeth). What real cloth does here is not measured: operators guide the work at speed. In every run the presser foot leaves the stack for 20-26 % of the time: at 5,000 sti/min the dog's teeth meet the cloth rising at 0.63 m/s and throw the foot (its moving mass, 0.12 kg, is an estimate). The stitch counter reads 27 against 26 shaft revolutions because the 27th loop was picked up in the revolution under way when the run ended.

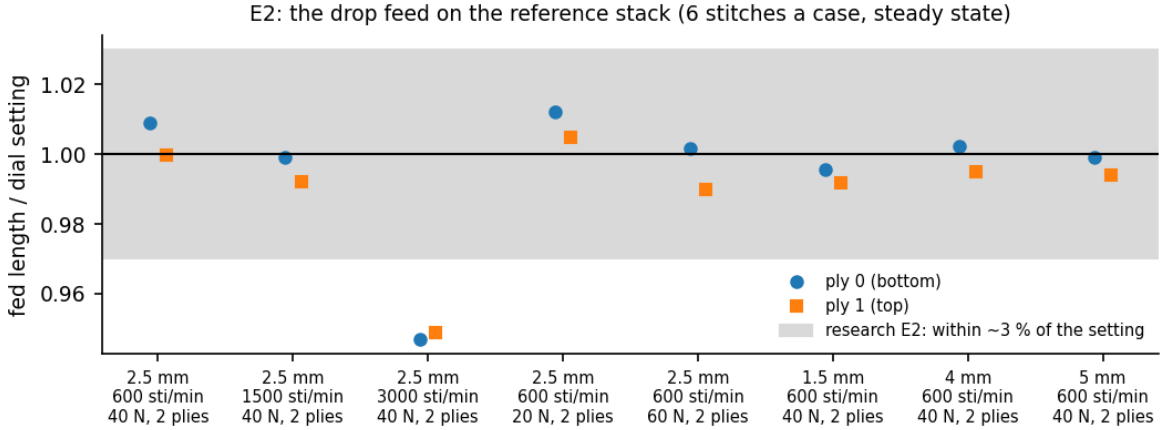


Figure 16: E2: the length fed per stitch against the dial, every case, both plies; grey: the ~3 % band of the E2 criterion

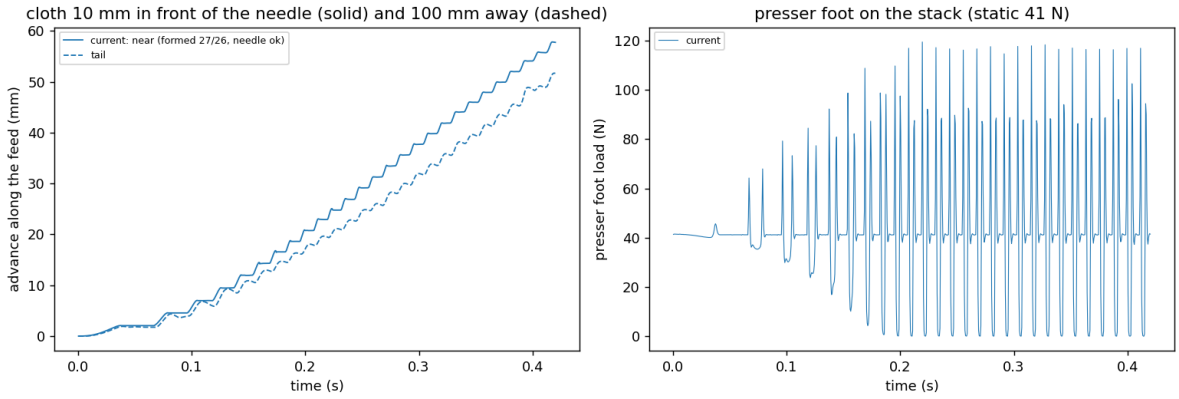


Figure 17: E2b: the machine alone at 5,000 sti/min on the bench

8.7 Needle deflection and the pivot (E6/E9, ASSISTED)

Both experiments stand in for an operator's hands and are labelled ASSISTED. They act through a phantom pad or couple, not through forces applied straight to the cloth's vertices (0.5 N on the loose 2.5 g stack flings it round the foot at ~200 m/s^2). E6 presses a phantom pad (30 x 20 mm, mu 0.8, 30 N, a firm press: estimate) onto the stack 30-50 mm in front of the needle and slides it across the seam at 10 mm/s; the pull is the pad's measured reaction, and every row carries the force balance across the seam (x) by shape. E9 turns the stack with a phantom couple (zero net force, as two hands apply) driven by a PD on the stack's measured angle toward 30 deg, with the foot raised.

Case	Needle	First damage: pad travel (mm)	pull (N)	needle lateral (N)	bed / plate / foot / needle across the seam (N)	stack at the needle moved (mm)
foot down	bent	18.895	15.5735	4.8702	5.5088 / 1.1593 / 7.938 / 0.9338	0.2074
foot raised	bent	14.005	6.7642	4.8735	4.4782 / 0.0377 / 0.0168 / 2.1607	0.5578
foot raised, alpha 1	struck_plate, bent	9.0	4.944	1.1036	4.4953 / 0.0186 / 0.009 / 0.421	0.2819

Case	Stack rotation (deg)	Centre from the needle axis (mm)	Bottom ply: rotation, centre (mm)	Top ply: rotation, centre (mm)
needle down	29.1	0.046	29.1, 0.04	29.1, 0.06
needle up	28.5	38.229	16.5, 9.35	40.5, 49.52
needle up, feed dog off	28.7	58.278	28.7, 58.26	28.8, 58.30
needle up, compression set off	28.7	53.391	28.3, 52.97	29.2, 53.80

Foot down. For the first ~12 mm of travel the pull goes into dragging the stack over the machine bed under the pad (mu 0.15 x 30 N = 4.5 N) and into the foot; the needle carries ~0.15 N. Beyond that the stack at the needle starts to move and the needle loads, bending after 18.9 mm of travel at 15.6 N of pull with 7.9 N on the foot, below a friction-only estimate for the foot (12-14 N at 40 N, mu 0.15 on two faces). E6b shows where the foot's sideways load comes from: at 20 mm of pad travel the foot carries 6.9 N across the seam (the needle already bent), through friction under its soles and through cloth that the pad drags toward the clamped stack and that buckles into a fold against the outer face of the left sole (8 contacts on that face, 4 at 10 mm of travel): normal forces on the foot's sides, as on a real foot. **Foot raised.** The needle carries the pull from the first millimetre and bends at 5.03 N (its yield moment at the 15 mm lever) after 13.8 mm of travel; with alpha = 1 (the free cantilever, 3,544 N/m instead of 16,303 N/m) it first strikes the throat-plate hole's edge at 1.1 N of lateral load and then bends. The needle's load points mostly toward the pad (the yarns between the pad and the needle turn diagonal); the force balance closes to under 0.001 N at the median row (0.08 N at worst).

Pivot. Needle down, the stack turns 29.1 deg about a centre 0.05 mm from the needle axis, both plies together (rigid fits to 0.02 mm). Needle up, the top ply turns 40.5 deg about a centre 50 mm from the axis while the bottom ply catches on the raised feed dog (at the needle-up stop the teeth stand 0.51 mm above the plate) and turns 16.5 deg about a centre 9 mm from the axis. With the teeth switched off both plies turn together, 28.7 deg about a centre 58 mm away; with the compression set switched off they turn together too (28.3 and 29.2 deg about 53 mm). The teeth are smooth boxes with an effective mu of 1.0 (open); how strongly real teeth hold an

unloaded ply is not measured here.

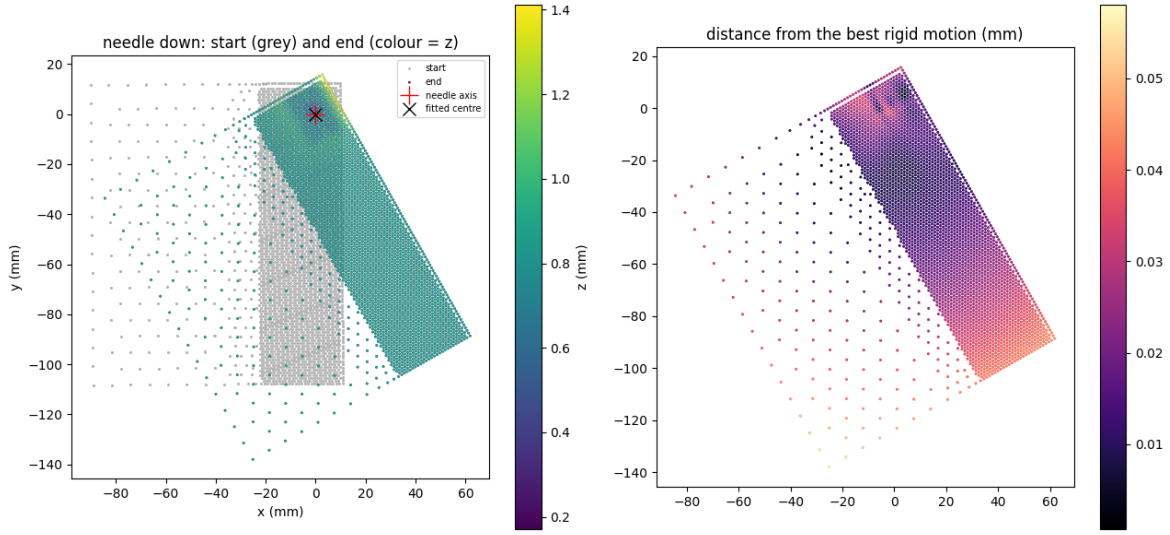


Figure 18: E9: the stack turned about the lowered needle (needle down)

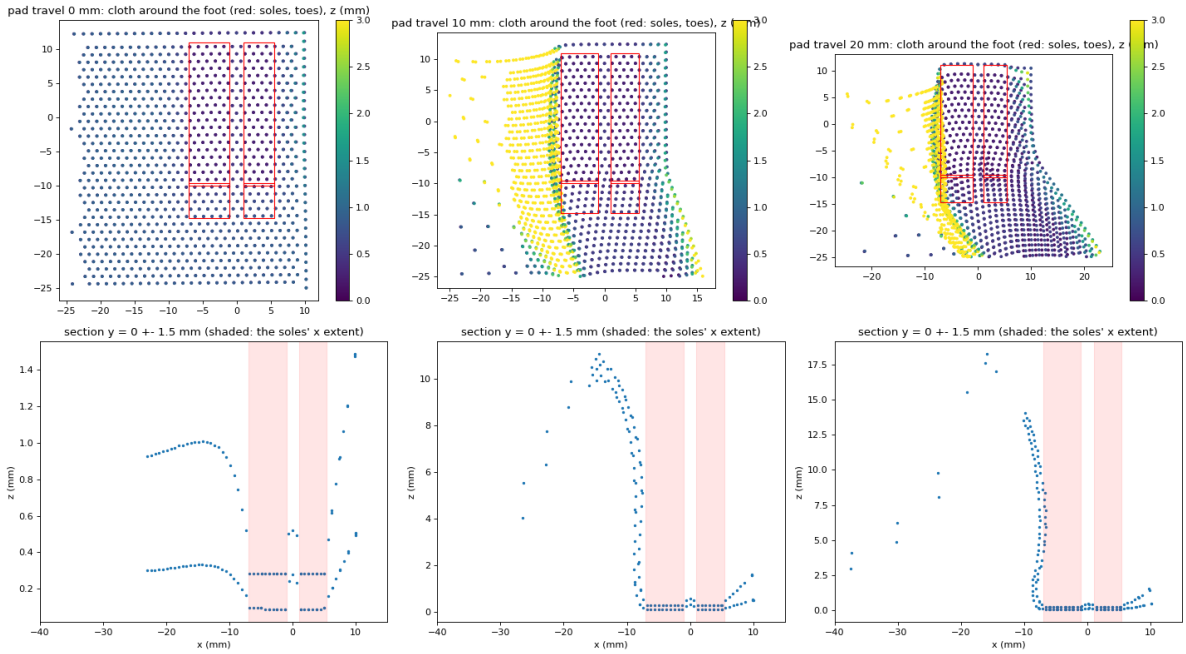


Figure 19: E6b: where the presser foot's sideways load comes from

8.8 Stitch formation, thread and puncture (E7 with E5)

Two plies (100 x 120 mm) on the bench, 16 stitches at a seam start, the stack left to settle 0.4 s.

Case	Formed / attempted	Skips	Balance	Stitch (mm)	Set at the take-up, needle / bobbin (N)	Pull at the interlock (N)	Thread tension after settling (N)	Puncture peak / law (N)	Hole offset max (mm)	Pucker RMS (mm)	Contraction
default tension, 600 sti/min	16 / 16	-	balanced 16	2.50	0.665 / 0.282	0.282 / 0.282	0.001 / 0.004	1.30 / 1.58	0.018	0.118	0.0019

Case	Formed / attempted	Skips	Balance	Stitch (mm)	Set at the take-up, needle / bobbin (N)	Pull at the interlock (N)	Thread tension after settling (N)	Puncture peak / law (N)	Hole offset max (mm)	Pucker RMS (mm)	Contraction
needle tension x3	16 / 16	-	top_tight 16	2.50	1.994 / 0.282	0.845 / 0.282	0.015 / 0.002	1.30 / 1.58	0.019	0.117	0.0045
needle tension x0.3	16 / 16	-	bottom_tight 16	2.50	0.199 / 0.282	0.085 / 0.282	0.001 / 0.004	1.30 / 1.58	0.018	0.118	0.0017
default tension, 3,000 sti/min	16 / 16	-	balanced 16	2.47	1.114 / 0.472	0.472 / 0.472	0.028 / 0.002	1.64 / 2.90	0.007	0.989	0.0483

Every stitch forms in all four cases; the balance flag follows the dial (balanced at the default pair, top-tight at three times the needle tension, bottom-tight at 0.3 times); at 3,000 sti/min the stitch is 2.47 mm against 2.50 mm. At 600 sti/min the seam lies flat by the authored gate (0.12 mm RMS against 0.5 mm in all three cases) and contracts 0.19 % at the default pair, 0.45 % at three times the needle tension and 0.17 % at 0.3 times: the tighter needle thread gathers the stitched line more, as expected for tension pucker. At 3,000 sti/min this run's seam came out lifted, 0.99 mm RMS and 4.8 % contraction, beyond both of the task's gates (0.5 mm, 2 %). The gathering stays in the plane (no waves: the seam band's 1.25 mm mesh barely resolves stitch-scale waves, and the in-plane compression, kc 8 % of the toe stiffness, an estimate, has no yarn jamming). The thread elements carry the pull at the interlock (the needle thread's set tension over its path loss 2.36, §4) and end nearly slack after settling (0.010-0.015 N): the stitched line gathers about as much as the thread's elastic stretch, the cloth's in-plane compression resisting almost nothing. Whether a real seam keeps more of its set tension, near the set values as expected, is not measured here.

Puncture (E5). The needle's peak axial force per stitch is 1.30 N at 600 sti/min against the penetration law's 1.58 N for two plies, and 1.64 N against 2.90 N at 3,000 sti/min: the law is a cap (the force at which the point parts the yarns), and at speed the plies yield into the plate's needle hole before the cap is reached. On a 2.5 mm band the peaks are 1.34 and 1.67 N; the 1.25 mm band does not raise them, so the yield is not the mesh's resolution around the 1.6 mm hole (what sets it is not established). The holes land within 0.02 mm of the needle's axis.

The seam's shape. The bench seam as built:

Variant	Seam band RMS (mm)	Highest point (mm)	where (x, y mm)
current	0.12	1.3	9, 31

A constant pull at the dial tension on the thread to the eye while sewing would curl the stitched stack up behind the foot by itself (§4). Runs of one configuration are not bit-reproducible. With the 1.25 mm band, E7's default case, E17's runs at both ticks and six repeats of the bench seam all settled flat (0.11-0.12 mm RMS; the repeats 0.106-0.107 mm after 2 s, §16), while E7's 3,000 sti/min case came out lifted (0.99 mm). On a 2.5 mm band the finished seam takes one of two shapes from run to run (flat 0.13 mm RMS, or lifted up to 1.75 mm). Which shape a real muslin seam would take is not measured.

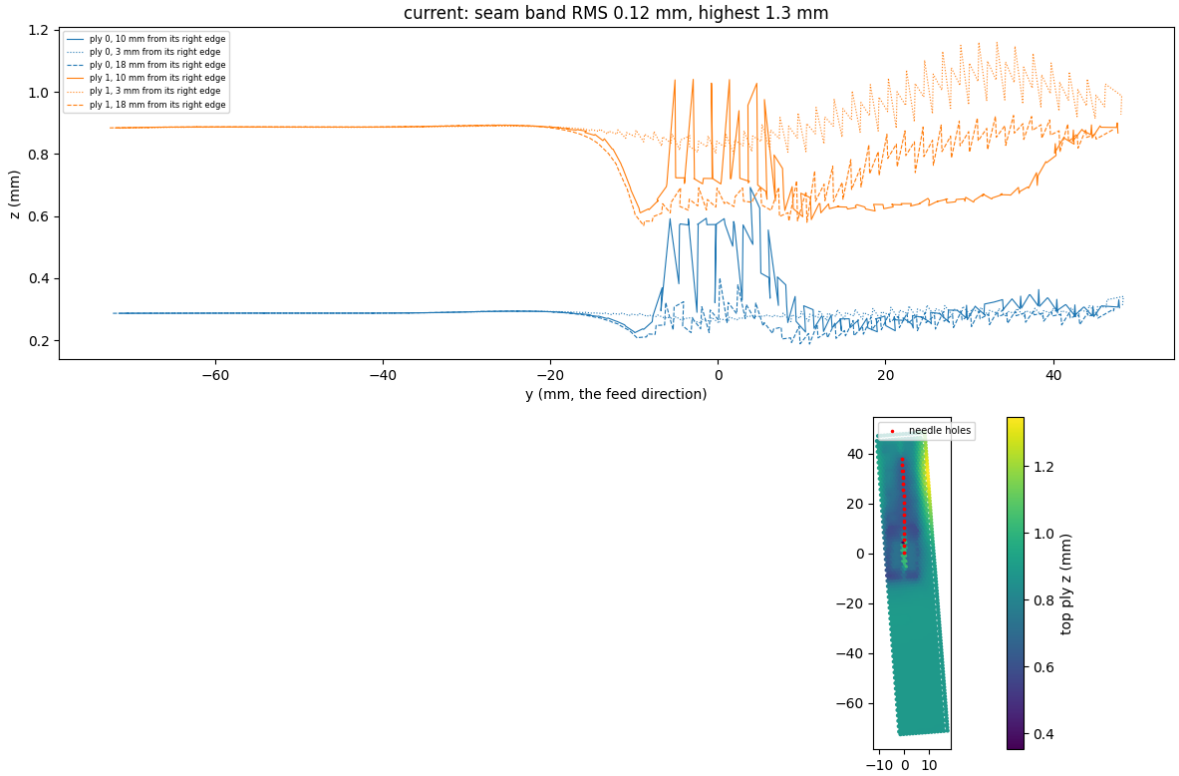


Figure 20: The seam's shape as built (E7's default case): the stitched stack's height along the seam

8.9 A hand pressing the work, held still (E18, SCRIPTED ROBOT)

From the plain-seam start state one hand comes down onto the stack and is held still: the FR3's or the Panda's left hand (fingers closed, straight down 0.11 m left of the needle and 0.13 m in front of it), the single FR3's right hand at the mirror of that place about its own base (0.09 m left of the needle; it comes round the head's front, because the right arm's one-shot joint-space move from its ready pose sweeps its wrist through the head), the G1's left Dex3 flat at its ready place. It lowers in 0.5 mm steps, waiting between steps while out of contact until the hand has arrived, then in 0.05 mm steps (the G1: 0.5 mm) until the cloth pushes back with a press that holds for 50 ms; then the commanded height is held for 0.5 s with the machine idle and 0.5 s while it sews at 600 sti/min under the still hand. The cloth's force on the hand is recorded every tick (the tick's average reaction, unfiltered). The hands are the compliant bodies of §5 and the coupling's reaction filter is 1 ms:

Robot	Reaction filter	Press reached (N)	Machine idle: mean +- sd (N)	min	max	below 0.2 N	Machine sewing: mean +- sd (N)	min	max	below 0.2 N	Dominant frequency (sewing)	Reflexes
FR3 pair	1 ms	3.7	3.54 +- 0.09	3.39	3.9	0 %	2.33 +- 0.73	1.51	3.7	0 %	none (slow drift)	0
Panda pair	1 ms	3.8	3.88 +- 0.05	3.75	4.1	0 %	2.44 +- 0.62	1.56	3.9	0 %	none (slow drift)	0
G1	1 ms	3.0	2.99 +- 0.03	2.93	3.0	0 %	2.80 +- 0.09	2.61	3.0	0 %	none (slow drift)	0
single FR3	1 ms	3.7	3.65 +- 0.08	3.54	3.9	0 %	2.57 +- 0.70	1.72	3.8	0 %	none (slow drift)	0

A still hand on the work should feel a steady press, and every deployment's does: the press never falls below 0.2 N and has no fluctuation peak above the stitch rate (the largest at 6 Hz, the Panda's idle press), and no arm stops on its reflex. Idle, the Franka hands hold 3.5-3.9 N with a slow relaxation (the plies' compression set); once the machine runs, the feed draws the cloth from under the still hand and the press falls to 1.5-1.7 N by the end of the 0.5 s (means 2.3-2.6 N). The G1's flat Dex3 hand holds 3.0 N idle and 2.6-3.0 N while sewing, its palm spread over more of the stack, the robot standing. The single FR3 presses with its right hand at the mirror of the pair's left-hand place, which it reaches round the head's front (a static check of the route's poses found a clear one).

Why 1 ms: with a 5 ms filter the G1's hand falls into a 116 Hz limit cycle on the work once the machine runs (0-14

N impacts, no contact a third of the time); with none the press alternates tick to tick by +-1.4 N (the explicit one-tick coupling's period-2 mode); 1 ms, two ticks, is the smallest value tested that holds a steady press on both robots. The filter has no physical counterpart (a pad's compliance is a spring in series, which the compliant hand is, not a delay). The approach matters too: stepping without waiting for the hand to arrive drives the FR3's fingers 4 mm into the stack at once (298 N); a direct joint-space move from the ready pose down to 15 mm meets the table and trips the FR3's reflex; and the single FR3's right arm, moved in one joint-space step from its ready pose (right of the needle) to its place left of the needle, sweeps its wrist through the machine's head and arm (joint 4's reflex at 341 and 520 N m), so its hand goes round the head's front.

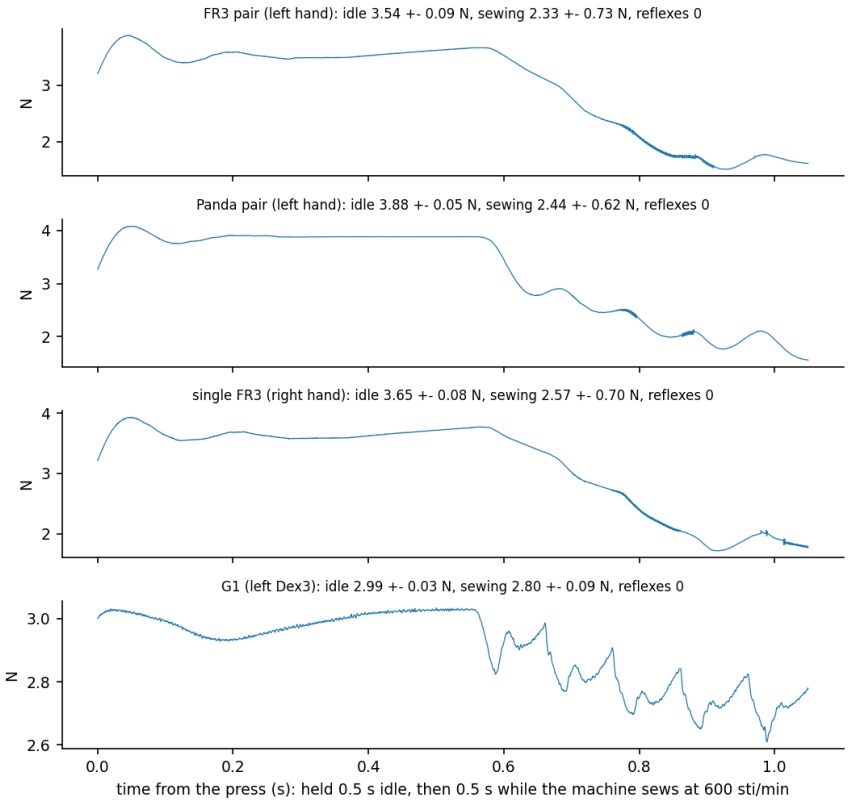


Figure 21: E18: the cloth's force on a hand pressing the work and held still, machine idle then sewing, every robot

9 Discretisation and convergence

In time (E17: E2's 600 sti/min case and E7's default case at half the tick):

Quantity	dt 0.5 ms	dt 0.25 ms	Change
bottom ply feed (mm/stitch)	2.523	2.525	0.1 %
top ply feed (mm/stitch)	2.499	2.501	0.1 %
stitch length (mm)	2.501	2.501	0.0 %
puncture peak (N)	1.299	1.386	6.8 %
pucker RMS (mm)	0.118	0.109	8.0 %

At half the tick the feed changes by 0.1 % and the stitch length by 0.01 %, the puncture peak by 6.8 % (1.30 against 1.39 N), and both runs' seams settle flat (0.118 and 0.109 mm RMS): the experiment's own check passes. With a 2.5 mm seam band it fails (the seam settles lifted, 1.75 mm RMS, at the default tick and flat, 0.26 mm, at half the tick); no run on the 1.25 mm band has shown the lifted shape at 600 sti/min (§8.8). E17b measures the seam allowance's raw edge:

Tick (ms)	Variant	Band RMS (mm)	Seam line mean (mm)	Raw edge mean / max (mm)	Raw edge in front of the needle at the last cycles (mm)
0.5	current	0.119	0.799	0.965 / 1.345	1.19, 1.21, 1.24, 1.28

Every stitch, the feed dog's teeth meet the resting bottom ply rising at ~80 mm/s (600 sti/min) and lift the clamped stack and the foot ~0.5 mm; the free allowance beside the foot is flung up and keeps part of the lift,

ratcheting in front of the needle. The raw edge stands 0.17 mm above the seam line on average at the default tick. Halving the tick halves the lift (0.41 against 0.20 mm, measured on an earlier cloth model; the half-tick run was not repeated on this one): a first-order discretisation error of the dog's impact on the cloth, resolved within one 0.5 ms tick. Neither friction makes it: with the bending friction or the membrane friction switched off the edge lifts about as much (2.6-3.0 mm at its highest). A converged edge needs a tick several times smaller, at several times the runtime (§14); the default tick keeps it, the pucker check leaves the allowance's outer 4 mm out, and the seam camera shows the lifted edge (§15). In space: the bending model at 2.5 and 7.5 mm (E1b, section 8.1) and the Peirce cantilever at 1.25 and 0.625 mm (section 8.5), whose deficit is smaller at 1.25 mm than at 2.5 mm and grows again at 0.625 mm (open). On the seam band's 1.25 mm mesh at the default tick a sliding coupon's friction reads 2 % low (§8.5; at half the tick, or on 2.5 mm, it is exact). The coarse bulk mesh also matters where the cloth wraps a sharp edge and where a seam runs outside the fine band, so every seam a task sews lies in the band.

At the machine's top speed (§8.6): unguided, a short stack on the bed runs the same at both ticks (no overrun in six runs at either, the hook striking the needle in one of six at each); the presser foot's landing peaks rise from 119-120 N to 187-188 N at half the tick (a near-rigid impact resolved within a tick; its peak force is not a converged quantity). The control box's speed output tells a policy so. At 600 sti/min (above) the feed, the stitch and the puncture are converged.

10 Tasks and evaluation

Five of the design's sixteen tasks are built, all in preset A (`sample-room`). Task ids: `sewing/v1/sample-room/(load,plain-seam,baste,pivot,curve)/seed-(0..9)`. The evaluator (`task.py`) reads only the simulated state: the holes' material coordinates on each ply, the stitch records (formed or skipped with the reason, reverse, the plies bound, balance), the plies' corners, the needle's state, the threads, the hands' cloth forces and the robots' reflex history. No check reads a scripted label.

Check	Rule	Source of the threshold
stitches_formed	at least 3 stitches binding both plies (pivot: 6)	authored
joined_far_end / joined_near_end	interlocks within 6 mm of each end of the plies (curve: far end and left edge, along the seam, on each ply, reverse stitches included)	authored
seam_line	every hole within 1/16 in (1.6 mm) of the allowance (10 mm; baste 6 mm)	training manual [read]
seam_mean	mean allowance within 0.8 mm	authored (half the line tolerance)
stitch_length	forward steps within 0.6-1.4 of the setting	authored
stitch_density	9-11 stitches in every inch window at the 2.5 mm setting (10 SPI)	US Army 11 +- 1, applied about the setting [read]
skips	at most one isolated skipped stitch, no run of two (curve: a cycle with no fabric under the needle, or with the needle bent or broken, is not a skipped stitch)	FED-STD-751a / trade [read]
backtack_start / backtack_end	at least 3 reverse stitches at each end (plain seam)	trade practice 3-4 [literature]
no_reverse	no reverse stitch (baste)	instruction
no_tucks	no stitch pierces a ply twice; no hole beyond 20 mm (curve: the first clause only, a guard the machine model cannot trigger, since it pierces a ply once a stroke; a fold carried under the needle along the seam shows as a hole off the line and fails seam_line, a tuck across the seam shows only as a gap > 3.5 mm (no_gaps), and gathering of both plies is not seen in material coordinates)	authored
ends_meet	the plies' near and far seam corners within 3 mm of each other	industry < 3 mm [read]

Check	Rule	Source of the threshold
pucker (recorded in `info`, not scored: below)	RMS height of the top ply's 20 mm seam band less the allowance's outer 4 mm, along the stitched length where it lies over the table, <= 0.5 mm; seam contraction <= 2 %	authored (no published numeric gate); the band excludes the work hanging over the table's edges and the cut edge's lift, a tick artifact (E17b); the seam's threads relax nearly slack (§8.8), so a tension pucker is not modelled and the check sees waves of the work and the feed's contraction
corner, leg1_line, leg2_line, continuous, leg2_reaches_left_edge	pivot: the turn within 1.6 mm of (10, 10) mm, both legs on their lines, no gap > 3.5 mm, the second leg to the left edge	authored from the instruction
curve: seam_line, plies_registered, no_gaps, stitch_density, no_long_threads, joins_secured, joined_far_end, reaches_left_edge	judged on the finished work over every stitch through both plies, forward or reverse, in any order or direction: every hole of every formed stitch (one-ply stitches included) within 1.6 mm of 10 mm from the edge path (right edge, the rounded corner, the near edge); the two plies' material points at each stitch within 3 mm; along the line 10 mm inside the edge, on each ply, no step between neighbouring stitches > 3.5 mm per stitch (a skipped stitch's span counts two), from within 6 mm of the far end to within 6 mm of the left edge; 9-11 stitches (12 allowed) in every inch of each sequence of sewing, on places (the seam's stitches in sewing order, a skipped stitch's holes included, collapse into one place while within 0.5 mm of a place's first stitch, so a pile, lock or bar tack is one place and a slow creep a place every 0.5 mm; a sequence is consecutive places of one seam piece and feed direction moving the same way along the line by up to 10 mm a step; sequences shorter than an inch are not counted), so overlaps, locks and restarts over the seam do not add and no stitch length escapes the count; no thread longer than 10 mm between consecutive stitches of one seam piece, in the work (a jump left without a trim, which the trimmer cannot cut; a skipped stitch's 5 mm and a restart up to four stitches back over the seam pass; a restart further back must be trimmed first); each end of a trimmed piece that lies within the seam (more than 6 mm from the far end and from the line's end) has the 1/4 in of seam inside it sewn at least twice, by its own backtack or lock or by another piece overlapping it. The seam may be made in several trimmed pieces, so secured. The mean allowance is recorded in `info`, not judged (an authored 0.8 mm mean would fail seams within the sourced 1/16 in everywhere)	training manual; ends meet < 3 mm (industry [read]); 9-11 SPI along the curve (Army 11 +- 1 [read]); ends of stitching backstitched or overstitched not less than 1/4 in (GL-PD-20023 3.5.2 [read]); the 3.5 mm step, 6 mm reach and 10 mm thread authored from the instruction
load: foot_down, ply_*single_layer,* *ply*_seam_allowance, ply*_from_end, edges_aligned, no_reflex	the foot down with load on the plies; exactly one layer of each ply under the needle, 10 +- 1.6 mm from its right edge and 3-8 mm from its far end; corners within 1.5 mm	instruction; tolerances authored

Check	Rule	Source of the threshold
common: needle_intact, no_reflex_while_sewing, threads_trimmed, released, no_thread_break	the needle 'ok': not damaged (it struck the plate or the hook: burred, replaced in practice), bent or broken; no robot protective stop while the machine ran: the arm hit something or exceeded a joint limit within reach of a running needle, a cell fault in any automated cell whether or not the seam was harmed (the G1 has no protective stop and cannot fail this); threads trimmed (the tails are recorded); no hand touching the cloth (< 0.05 N net on each hand; curve: < 0.05 N with each pad's contact force added as a magnitude, since a pinch's two fingers cancel in the net force: 1.15 N a finger read 0.007 N); no thread broken	authored (the tails are recorded against the Army major-defect boundary, 38 mm [read])

`success` requires every check; `score` is the fraction passed, set to 0 while the task's primary outcome is absent (no stitch through both plies; for load, the plies not under the needle), so that a policy that does nothing, which passes the common checks, scores 0 (measured in §13: the hands-off soak scores 0).

The curve's `score` is 1.0 on success and otherwise at most 0.95: the share of the seam's required span (6 mm from either end, along the line 10 mm inside the edge) sewn through both plies, each stitch covering 1.75 mm either side of its place with its credit, the best stitch at a place counting once. A stitch's credit is the least of: 1 within 1.6 mm of the line, falling linearly to 0 at 3.2 mm; 1 with the plies within 3 mm, falling to 0 at 6 mm apart; 1 with the mean step between places of its sequence over the inch about it (the step to its nearer neighbour where no other place is in that inch) inside the 9-12 stitches-an-inch band, falling to 0 at 1.5 and 3.5 mm; 0 where the thread to it broke. Times 1 - strays / the stitches the required span takes (172 at seed 0), counting a stitch that is the best stitch nowhere along the line as 1 - its place credit, a formed one-ply stitch as 1 - its line credit and a thread longer than 10 mm between consecutive stitches as 1: rework, which sewing over the seam on the line neither adds nor dilutes. The weight (one required stitch per stray, a 231 mm thread the same as one stray stitch), the narrow stitch-length ramp and the linear rework that can bring a complete seam with as many strays to 0 are authored calibrations. Times the needle's damage factor (damaged 0.75, bent or broken 0.5) and 0.75 for a broken thread: a burred needle is replaced and a bent one may have marked the work, so the seam they made is discounted, never erased. Times 0.90, + 0.05 once the threads are trimmed, + 0.05 for letting go once trimmed; times 0.95. Nothing sewn scores 0 without a gate. The order, direction, speed and method of sewing, pivots, presser lifts and backtacks earn nothing and cost nothing; a hand on the work before the trim costs nothing; a protective stop fails success and costs the partial score nothing. A seam made in trimmed pieces dips by up to 0.1 x 0.95 x coverage between a piece's trim and the next piece's first stitches, until about ten stitches recover it; its final score is unaffected. Every constant of this partial score beyond the checks' own tolerances is authored; none is measured. Not judged on the curve: the work's flatness and the seam's stretch or gathering (the plain seam's pucker measure is written for a straight band and is not applied); the seam's smoothness between holes inside the 1.6 mm band: the arc taken as three straight chords with 30 deg corners, the allowance swinging between 8.4 and 11.6 mm, passes (a heading rule was not adopted: no source gives a threshold, a rule per run of sewing is bypassed by trimming at each corner, and one over all places reads the zigzag of an overlapped restart as kinks). A faulted world is not judged: the backend reports its evaluation unavailable, with the fault and the last judged evaluation, marked interrupted, and `gar verify` refuses it.

Recorded, not scored (`info`): pucker, with its pass flag, because identical inputs can sew one of two seam shapes (§8.8); and the stitches' balance, the fraction balanced in the stitch-formation law's capstan band (FED-STD-751a's balanced stitch). Balance follows from the two set tensions, which no task lets the policy change, so it is the same for every stitch a policy can make and a check of it could not fail.

The evaluator is tested on fixtures with known verdicts (59 tests, no GPU; six of them hold the load task's fast ply-registration search to the exhaustive one): a perfect seam; a hole off the line; a fold carried under the needle; a ply pierced twice; the drape over the table's edge (not contraction); a gathered seam (contraction); missing backtacks; skipped stitches; a backtack-only seam; curves; plies registered or shifted; a damaged, bent or broken needle; the cut edge's lift (not pucker) against a wave of the seam band (pucker). The curve's: a seam sewn from the left edge, a reverse-first backtack, locks and a restart over the seam (pass); cycles past the edge (not skips); one skip and a run of two; an unsewn stretch; short ends; plies joined apart; each ply's own ends; a hand squeezing the work (not let go); reverse-only ledgers (a verdict, not an exception); pivots against a square corner; the episode's radius; and the score's order: nothing sewn 0, more seam more, finishing in proportion, further off the line or plies further apart never more, damage scaling the seam down; a seam within 1/16 in everywhere

whatever its mean (pass); 1.0, 1.25 and 3.4 mm stitches (fail density); an offset restart and a pile of stitches in one place (pass); thread strung across the work by sewing out of order (fail), the same trimmed, a short restart and a butted piece (pass); one-ply stitches off the line (fail); sewing over the seam again never changing the score; stitches crowded below 0.5 mm (fail) and longer than 3.5 mm (never more than 3.4 mm); one skip at a 2.55-2.7 mm realised stitch (pass); butted or one-sided trimmed joins (fail) against backtacked or overlapping ones (pass); the correct seam over an off-line one scoring less the further off it was.

11 Assisted demonstrations

All five tasks are attempted with labelled phantom loads in place of an operator's hands (the curve twice: on a short stack and on the task's own start state; every frame carries the ASSISTED badge; the phantom acts only on the cloth's vertices, mass-proportionally, and the machine is commanded through its control box). They show what the world does when the task's physical interactions are carried out; they are not robot solutions, and the unmodified evaluator judges each.

Every guiding policy here meets the feed's own turn: with nobody touching it the plain-seam start turns counterclockwise 0.55 deg per stitch at first (§12, the machine alone), so the guides and pivots below spend most of their effort holding that turn back, and each record says how much (`guide_sign_balance`, `turn_split`).

Load. The two 200 x 420 mm plies start stacked on the table left of the machine with the foot down on the bare plate. The control box lifts the foot (8.5 mm); a phantom carry pulls every vertex of both plies in proportion to its mass by a PD on the top ply's material point that must end under the needle (10 mm from the right edges, 5 mm from the far ends), along a 2.5 s minimum-jerk path, with a couple of zero net force that holds the stack square and, within 60 mm of the needle, the plies' leading 40 mm held ~2.5 mm above the plate. At the end it lets the leading part down, lowers the foot while still holding, and only then lets go. The foot came down on both plies, each a single layer under it at a 9.4 mm allowance, the plies 5.5 and 5.2 mm from their far ends, their edges aligned to 0.48 mm: the unmodified evaluator passes every check (score 1.0). Three other carries fail, each by the world's response to what the carry does: without the squaring couple the stack yaws ~25-30 deg on its way (the laminate grips harder than the steel bed) and ends 14 mm from the needle; with the couple but dragged flat, the bottom ply's leading edge catches on the raised feed dog's teeth and stops 22 mm short; and let go under the raised foot, the 420 mm stack's ~200 mm overhang beyond the table's front edge pulls it off the bed and the foot comes down on bare plate.

Plain seam: the whole seam, its line off at the end. The plies start under the lowered foot (the task's start state). The machine backtacks three stitches at the far end (touch-back held), sews forward at 600 sti/min until the stitches reach within 4 mm of the near end, backtacks three, stops needle-up and trims. While it sews forward, a phantom guide holds the seam line of the work still to be sewn on the needle's line: a lateral pull on the cloth ahead of the foot, PD on the line's measured offset (1e4 /s^2 per unit mass, its damping applied as its exact decay over the interval), clipped at 2 N, never along the seam and off during the backtacks; as the work ahead of the foot runs out it keeps its total stiffness and is applied more often. In the filmed run it sewed 171 stitches from backtack to backtack and trimmed, no fault, the needle intact, no reflex; the evaluator passes 16 of its 17 checks (seam mean 10.6 mm, stitch lengths 2.41-2.62 mm, both ends joined within 4.8 mm, both backtacks; pucker 0.30 mm RMS, recorded) and fails the seam line, 3.4 mm off at its worst against 1.6 mm: the top ply's holes stay inside the tolerance for the first ~395 mm and leave it over the last ~25 mm, where little cloth is left ahead of the foot for the guide to act on (figure below). The guide pulled the work the same way, away from the seam's edge, on all 2,208 of its applications (0.11 N on average): it spent the whole seam holding back the feed's counterclockwise turn.

Pivot. A 200 x 200 mm stack under the lowered foot. The machine sews the first leg 10 mm from the right edge at 600 sti/min with the plain seam's lateral guide; it stops needle-down with the corner hole ~10 mm from the near edge; the foot is lifted; a phantom couple turns the stack about the lowered needle (2 mN m per deg on a ring 40-90 mm out); a phantom finger presses the corner's tail flat (0.01 N in all on the cloth within 25 mm of the needle standing above the plate); the foot comes down and the second leg is sewn along the near edge with a guide that acts only within ~80 mm of the foot and takes hold over 0.2 s; then stop, trim. In the filmed run leg 1 reached the corner, the couple turned the stack 88.6 deg (the corner's tail standing up beside the foot until the finger pressed it, 1,878 points above the plate), and leg 2 ran to the left edge (its last hole 3.3 mm from it): 144 stitches, no fault, the needle intact. The evaluator passes 8 of 12 checks (0.67) and fails the corner (5.6 mm from its place), both lines (leg 1 5.5 mm, leg 2 9.8 mm off) and the continuity (an 8.0 mm span between two holes): leg 1 drifts from 10 to 15 mm from its edge as it nears the corner, and leg 2 runs from 13 mm in to the edge itself (holes at 0 mm about 250 mm along) and back out to 16 mm (figure below). The legs' guides pulled one way on 97 and 81 % of their applications, and at the start of leg 2 the line stood 5.8-9.0 mm off at the needle. The sequence is shaped by what fails in leg 2: a guide acting up to 200 mm ahead meets the turned line 6 deg off and crumples the stack, so the second guide acts only near the foot; a turn made with the foot lifted leaves

the corner's tail folded up against the foot, which catches it, so a finger presses the tail flat; and a holding field on the whole stack crumples it.

Curve (pivot-and-sew). A 150 x 150 mm stack (shorter than the task's 200 x 300 mm, to keep the run short) with its near-right corner rounded to 100 mm starts under the lowered foot. A three-stitch backtack (touch-back held) joins the seam at the far end; then the machine sews five stitches at a time at 600 sti/min, and whenever the stack's measured angle is more than 3 deg off the angle that lines the seam path up with the feed (corrected for the needle's distance from the edge) it stops needle-down, lifts the foot, and a phantom couple (a PD on the angle, 2e-4 N m/deg, clipped at 20 mN m, on a ring of the stack 40-90 mm from the needle) turns the work about the needle; the foot comes down and the machine sews on alone. 17 pivots took the work through the corner in 40.6 s: 99 stitches, every one formed, no fault, the needle intact, the threads trimmed, the work released; the film shows the stack flat throughout and the stitched curve along its edge. Of the 91 deg the work turned, the machine's stretches, nobody touching it, turned 52 deg and the couple's pivots 39 deg (`turn_split`). The unmodified evaluator fails it on accuracy (0.79): the seam runs 8.9 mm from the edge on average (10 +- 0.8 required), 3.3 mm off the 10 mm line at worst (+-1.6), and is joined 7.5 mm from the far end (<= 6): each pivot ends a few degrees short of its target and five stitches between pivots make a polygon, while the feed turns the work by itself in between. With the field guide pulling the cloth ahead of the foot between pivots (a diagnostic), the stretches turned the work 126 deg and the pivots turned it back 28 deg; 97 of 98 stitches formed and the hook struck the needle once (0.79). An operator steers continuously on this radius, which is not shown (§16); a steering field with the foot down crumples the work.

The curve on the task's own start state. The 200 x 300 mm plies with the 100 mm corner, as the task lays them, sewn the same way. It is not demonstrated (0.50). With nobody touching it, the work turned counterclockwise in every stretch (19.6 deg over 13), and the couple's 13 pivots, asking for 24-30 deg clockwise, turned it back only 16.6 deg in all: at those errors its proportional term is 4.7-6.0 mN m, under its 20 mN m cap, and the plain seam's draped stack needs ~40 mN m to turn clockwise. The seam left the edge (31 mm off its line at worst, 22 mm from the edge on average), a ridge rose beside the foot, and at 28.2 s the needle struck the plate and bent (67 stitches formed); the program kept the machine running to 62 s with no stitch formed after it (319 skips), and the work crumpled at the foot (filmed). The failure is the assistance's: a couple too weak to turn this stack against the feed's own turn. A stronger couple (the pivot demonstration's gain, its cap above 40 mN m) is the candidate and was not tried (§16).

Baste. The baste task's own start state (two plies under the lowered foot, a 6 mm allowance) sewn forward at its 5 mm stitch and 600 sti/min with the plain seam's lateral guide, no backtacks (the baste has none). It is not demonstrated (0.60). The guide held the line within 0.45-0.87 mm of the needle through 30 stitches, pulling the same way on 96 % of its 425 applications; at the 27th stitch the sewn work behind the foot lifted into a standing loop and heaped round the foot, and at the 32nd the needle struck the plate and bent (filmed). Sewn by the machine alone the same start state turns away from the toe and the needle fails at the 25th stitch instead (§12). On a 2.5 mm seam band the curl comes at the 2nd-5th stitch; why the sewn work lifts at the 27th stitch on the 1.25 mm band is not established (§16).

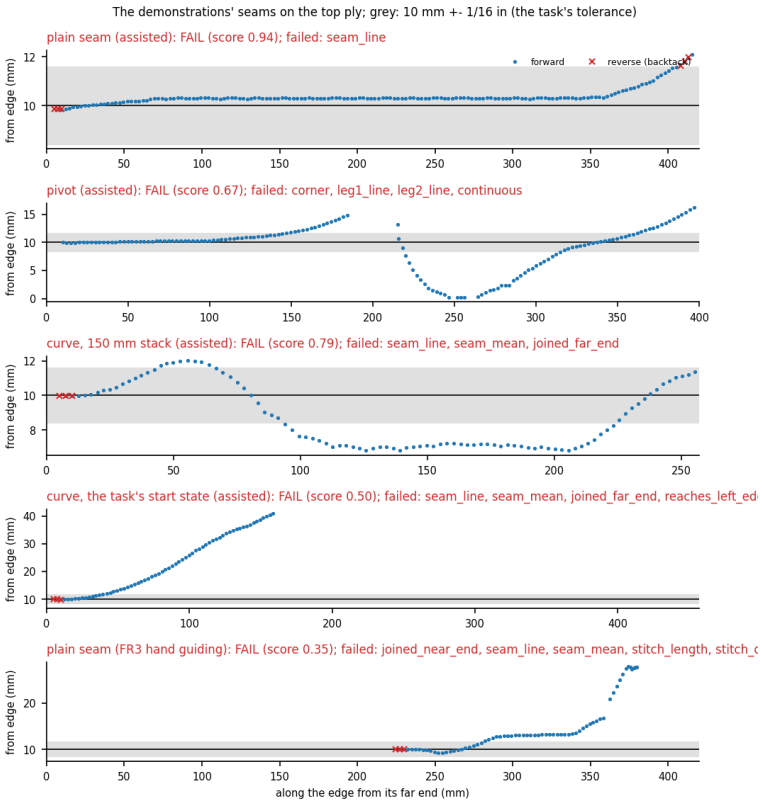


Figure 22: The assisted demonstrations' seams on the top ply: each hole's distance from the edge along the edge; grey: 10 mm +- 1/16 in

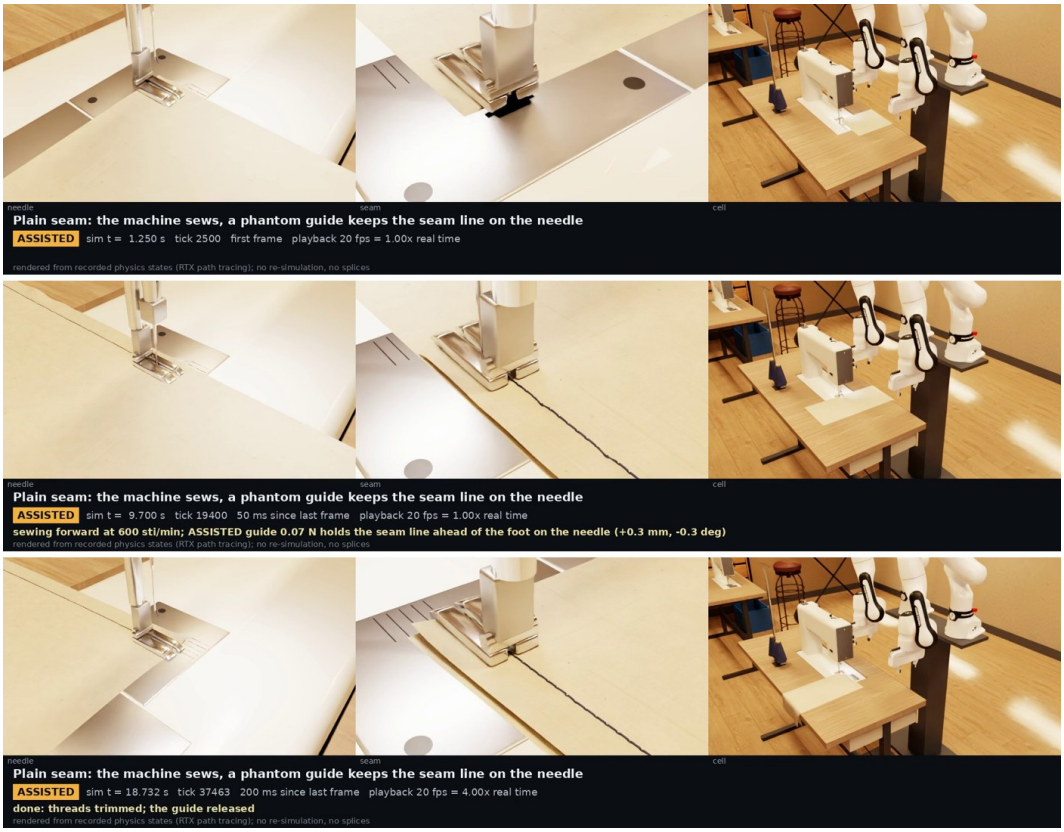


Figure 23: Plain seam (ASSISTED): first, middle and last frames of the film

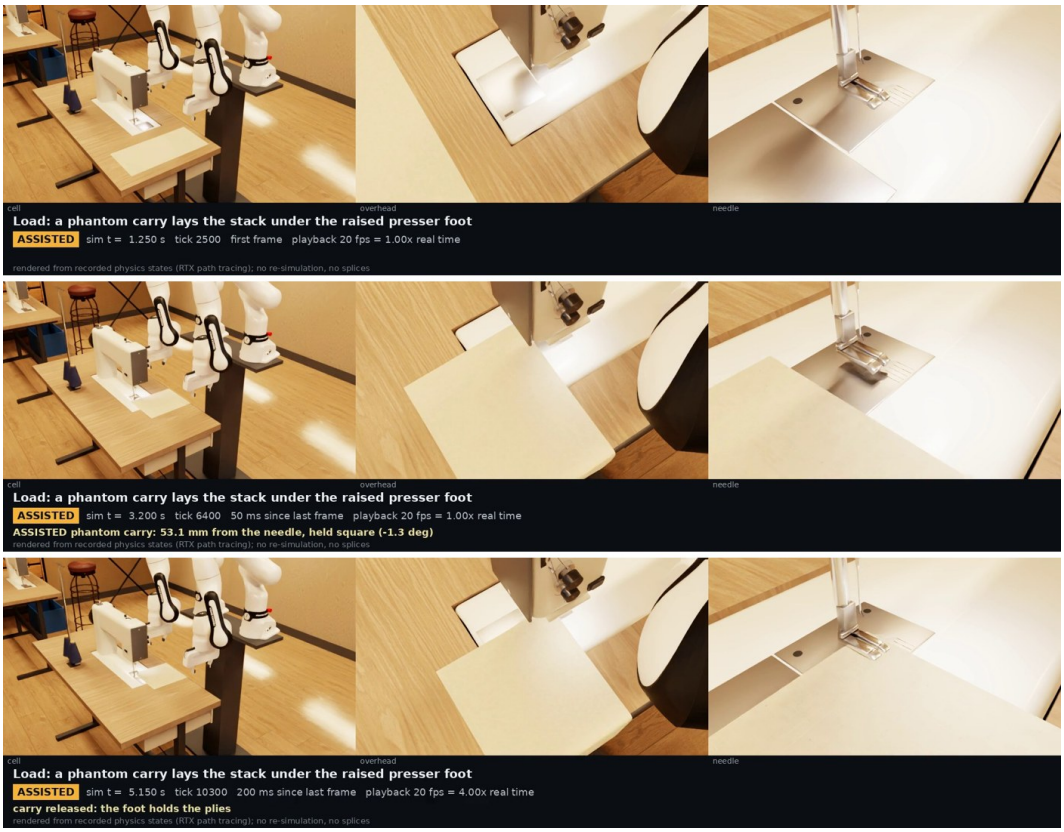


Figure 24: Load (ASSISTED): first, middle and last frames of the film

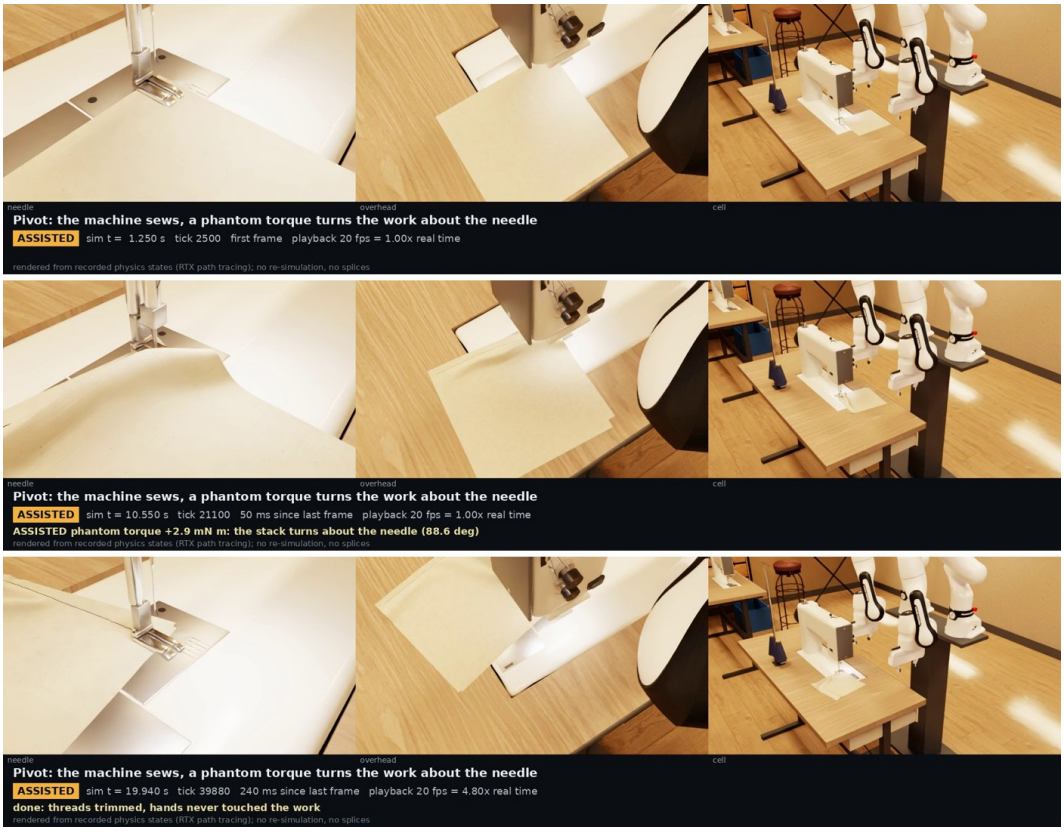


Figure 25: Pivot (ASSISTED): first, middle and last frames of the film



Figure 26: Curve on a 150 mm stack (ASSISTED): first, middle and last frames of the film



Figure 27: The curve on the task's own start state (ASSISTED): first, middle and last frames of the film (the work crumpled at the foot at the end)

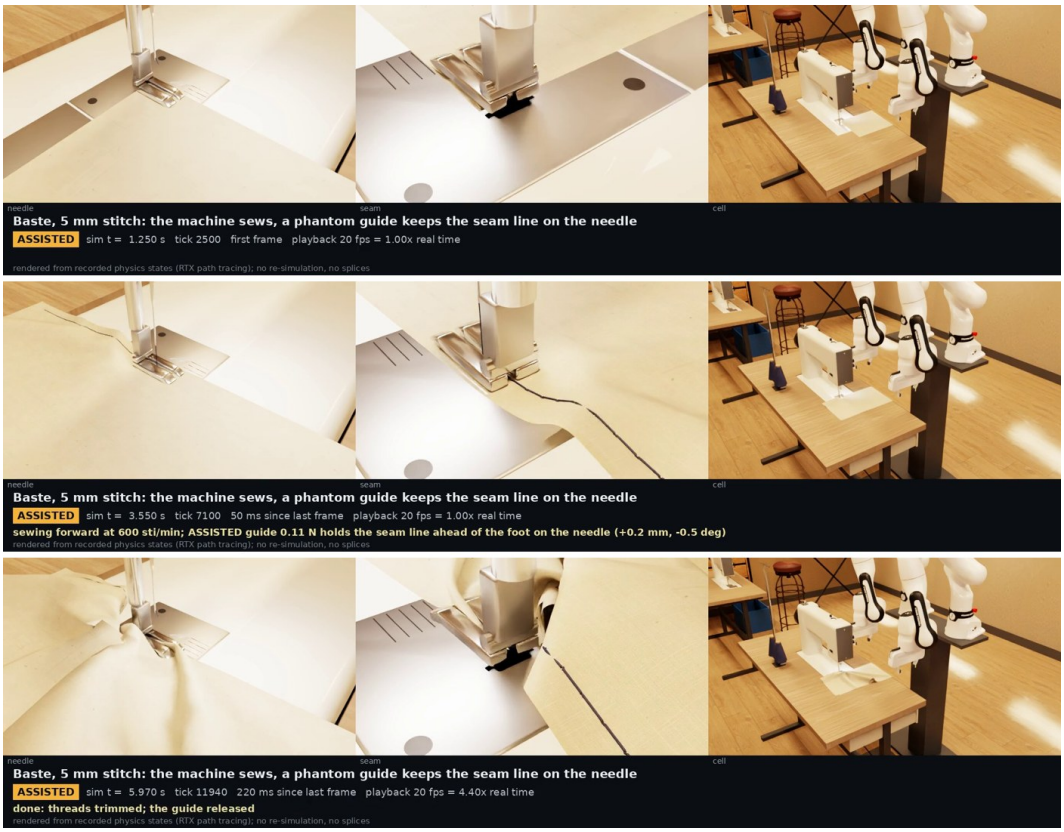


Figure 28: Baste (ASSISTED): first, middle and last frames of the film (the sewn work lifted behind the foot, the needle bent)

12 Scripted-robot demonstrations

Robot programs written with the plies' known start place, using only the policy interface (Cartesian targets through the environment's IK, the hand, the machine's control box); nothing assisted.

The machine alone (SCRIPTED MACHINE I/O). From the plain-seam start state the control box runs the machine at 600 sti/min for 6 s, stops it needle-up and trims; no robot moves and nothing is assisted (the task's own start, unguided). It forms 60 stitches, every one, 2.03-2.59 mm long, the threads trimmed, no fault, the needle intact. With nobody guiding, the work turns counterclockwise under the foot: 5.5 deg in the first 10 stitches, 13.0 in 30, 19.1 in 60, slowing as the turn carries the stack's edge in from the right toe; the stitch line runs from 9.9 mm to 38 mm in from the edge. The evaluator scores it 0.59 and fails it as a seam, as it should (60 stitches sew 150 mm of the 420 mm plies, no backtacks, the line 28 mm off). The baste's start at its 5 mm stitch (same 6 s): the work lies flat at the foot but turns faster, 10.1 deg in 10 stitches and 19.0 in 25; the needle, deflected 0.097 mm by the work against the hook's 0.07 mm clearance, is struck by the hook at the 25th stitch and bent at the 28th, after which the work crumples at the foot (28 of 60 stitches formed).

The guided seam (SCRIPTED ROBOT). A program for the FR3 pair through the policy's interface only (Cartesian targets through the environment's IK, the hands' position mode, the control box), written with the plies' known start place: both hands come over a 200 mm long stack along straight lines (E0d's checked plan), press lightly, the machine backtacks three stitches, sews forward at 600 sti/min while the hands travel with the work at the measured feed rate and regrip every 30 mm, backtacks at the near end, stops and trims. It does not sew a seam. In the filmed run the right hand never touched the work (the cloth's force on it read 0 N in all 200 samples: its pressing height, set from a static check of its tilted fingers against the bed, leaves it above the cloth in the running world); the left hand pressed in 58 of the samples (1.2 N median, at most 14 N) and its collision reflex tripped three times while the machine sewed; the needle struck the plate, both threads broke and the needle bent (4.5 N lateral; 72 stitches formed in all). The evaluator scores it 0.35 (the line 34 mm off, 32 skips, the near end 40 mm away, the needle bent). With one light, intermittent hand on it the work turns under the foot as it does with none; the program does not guide it. Its approach and places come from E0d: a one-shot joint-space move swings the right arm into the table, and the first two-hand places tried put the arms across each other.

The humanoid's hand on the work (SCRIPTED ROBOT). The G1 stands at the table in its ready

posture. Legal commands only (joint targets from the G1 hand IK, the control box), the program lowers the left Dex3 hand flat over the stack's left part, as an operator's left hand rests on the work, in 0.5 mm steps until the cloth pushes back with a press that lasts 50 ms; stopping instead if the palm lags its target by 8 mm or the IMU tilts past 1.5 deg (the robot has no balance controller). The press is read from the cloth's reaction on the hand's links, which stands in for the Dex3's fingertip tactile sensors (not simulated; not part of the policy's observation). The machine then sews 12 stitches at 600 sti/min while the hand travels with the work at the feed rate and its height is trimmed to hold ~3 N. Result: the press settled at 3.6 N with the palm 68 mm up; while sewing the hand held 3.4 N on average (2.7-4.1 N, sampled every 20 ms) and moved 22.9 mm while the cloth under it moved 22.0 mm; all 12 stitches formed, the needle intact, the robot standing (pelvis 0.784 m, tilt at most 0.44 deg). The press must last 50 ms: a program that takes the thumb's impact (8.3 N) for the press hovers over the work (0-0.2 N while sewing). With a 5 ms reaction filter the hand bounces on the work (13-86 N spikes) and stops on the palm's lag after 11 stitches (§8.9). This is the hand on the work while the machine sews, not a whole seam: no program guides a seam from end to end with the G1.

A robot hand moves the work by contact (SCRIPTED ROBOT). From the load task's start state (the plies on the table left of the machine), the left FR3's closed hand comes down on the stack (E18's approach), presses it with ~3 N and slides it 50 mm toward the machine in 1 mm Cartesian steps, then lifts. The hand travelled 50.36 mm and the cloth under it 50.16 mm, both plies together: it slipped 0.2 mm, the friction's 0.1 mm pre-slip and the sampling. The stack's corners moved 55.0 and 32.8 mm (it turned about 3 deg about a point beyond its far end, pushed off its centre of friction); the hand's force peaked at 4.1 N, no reflex (filmed: the `slide` camera sees the contact).

A robot hand holds the drape by pinching it (SCRIPTED ROBOT). From the plain-seam start state the left FR3 brings its open Franka Hand under the table's front edge (turned 30 deg about the vertical, so that its wrist clears the table top and its forearm the pedestal; every planned pose checked in the rigid cell with a 5 mm margin), straddles the hanging drape with its fingers 25 mm in from the drape's left edge, closes them on both plies in force mode at 2 N per finger, pulls the drape toward the operator and a little down in 0.5 mm steps (at most 15.5 mm, stopping early once the cloth pulls back 1.5 N), holds it 1 s, opens and withdraws. The grip is 2 N because at a usual grasp force the world's finger-cloth coupling fails (from one checkpoint, 20 N per finger throws the fingers open and 5 N chatters); this shows the hold the world supports, not a realistic grip. The fingers closed to 0.60 mm with 2.35 N on each, 7 and 4 of the two plies' material points between the pads. While the hand travelled 14.6 mm toward the operator and 2.7 mm down, the held cloth of both plies went with it, 14.3 and 14.6 mm (95 and 98 %), slipping 0.7 and 0.4 mm along the pull and ~2 mm down the pads; the squeeze, commanded at 2 N, read 1.8 N on average during the pull and 2.0 N during the hold (0.5-4.8 N). The pull never reached its 1.5 N stop: the hanging drape gave way, pulling back at most 0.23 N. As the fingers opened, the outer finger carried one ply 22 mm further toward the operator before it cleared, and the ply swung back. No reflex, no contact of the arm with the cell, the needle's zone untouched (filmed; the `pinch` camera sees the outer finger's pad on the drape's edge, the inner finger hidden behind the drape).

Steering the work with a hand while sewing (E20, SCRIPTED ROBOT). The curve task asks for continuous steering; body-force fields crumple limp cloth, so here the left FR3's hand presses the plain-seam stack 110 mm left of and 130 mm in front of the needle (~3 N) and travels with the work at the feed's advance while the machine sews at 600 sti/min: straight for 1 s, then moving 3 mm sideways over 1 s, then straight for 1 s. The experiment records the cloth's highest point within 30 mm of the needle and the turn of the cloth under the foot. The hand pressed 3.5 N and moved 3.35 mm sideways, the cloth under it 3.36 mm (the hand carries the work by friction, no slip); the holes' distance from the edge went from 9.86-9.98 mm on the straight to 11.1-11.6 mm at the end, a 1.46 mm shift of the seam line. The cloth within 30 mm of the needle, 1.2-2.5 mm high while the hand ran straight, rose to 5.9 mm as the steer went on and stayed at 5.5-7.1 mm after it (the film: a broad, shallow ridge in front of and left of the foot; the seam behind the foot flat), and the cloth under the foot turned 2.6 deg in all. All 30 stitches formed, the needle intact, no reflex. Under the foot the work turned 1.6 deg during the steer's second (it stood at 0.5-0.9 deg through the straight second before); a 100 mm-radius curve sewn at this feed turns it ~14 deg/s, about nine times faster, and no robot program sews a curve (§16).

A hand easing the sewn work through (E21, SCRIPTED ROBOT). A 0.5 N body force on the sewn work's free end puts 3.3-4.8 N on the needle and crumples the flap behind the foot. A hand pulls through a contact instead: after 2.4 s of sewing the left FR3's hand lands lightly (0.6 N commanded) on the sewn work 15 mm behind and 110 mm left of the needle and draws it at 1.15 times the feed while the machine sews, so that its friction pulls the work forward. The machine stops needle-down while the hand lands (from beside the machine's head along a statically clear route; E18's press). The hand pressed 1.0 N, travelled 37 mm with the work in 1.5 s and drew it forward by 0.21 N on average (0.41 N at most); the needle's peak lateral load was 0.24 N while the hand pulled against 0.36 N while the machine sewed unassisted, no fault, the needle intact, no reflex; the stitches came out 2.44 mm against 2.48 mm before (an expected ~0.2 mm per N predicts ~0.04 mm longer for this pull: the size is within the stitch-to-stitch spread, the sign is the other way; not explained).

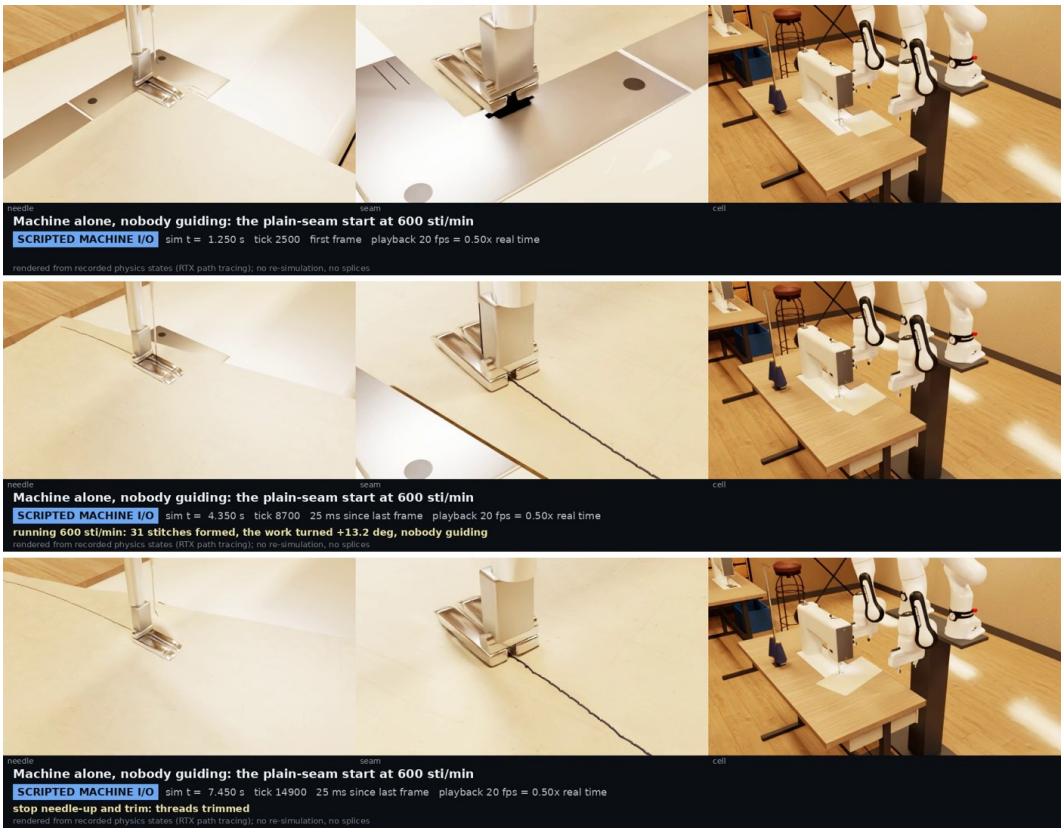


Figure 29: The machine alone (SCRIPTED MACHINE I/O): first, middle and last frames of the film

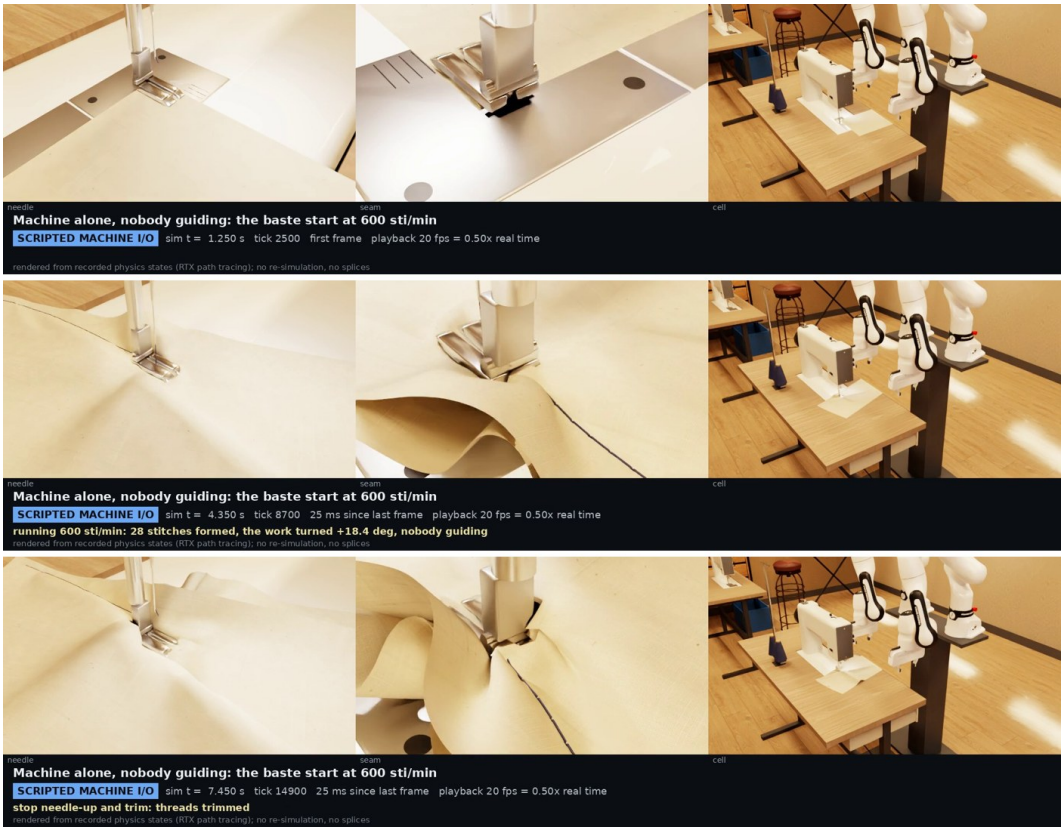


Figure 30: The baste start, the machine alone (SCRIPTED MACHINE I/O): first, middle and last frames (the needle bent at the 28th stitch)



Figure 31: The G1's hand on the work while the machine sews (SCRIPTED ROBOT): first, middle and last frames



Figure 32: The FR3 pair guiding a seam (SCRIPTED ROBOT): first, middle and last frames

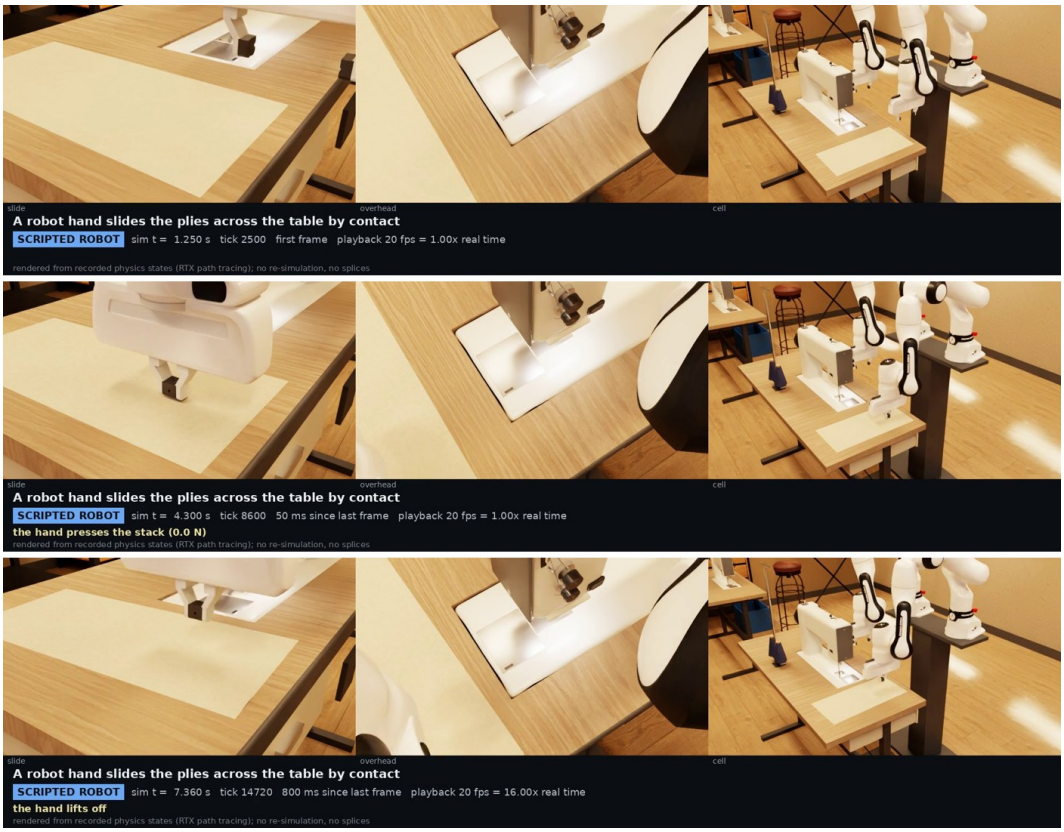


Figure 33: A robot hand slides the stack by contact (SCRIPTED ROBOT): first, middle and last frames



Figure 34: A robot hand pinches the drape at 2 N a finger (SCRIPTED ROBOT): first, middle and last frames

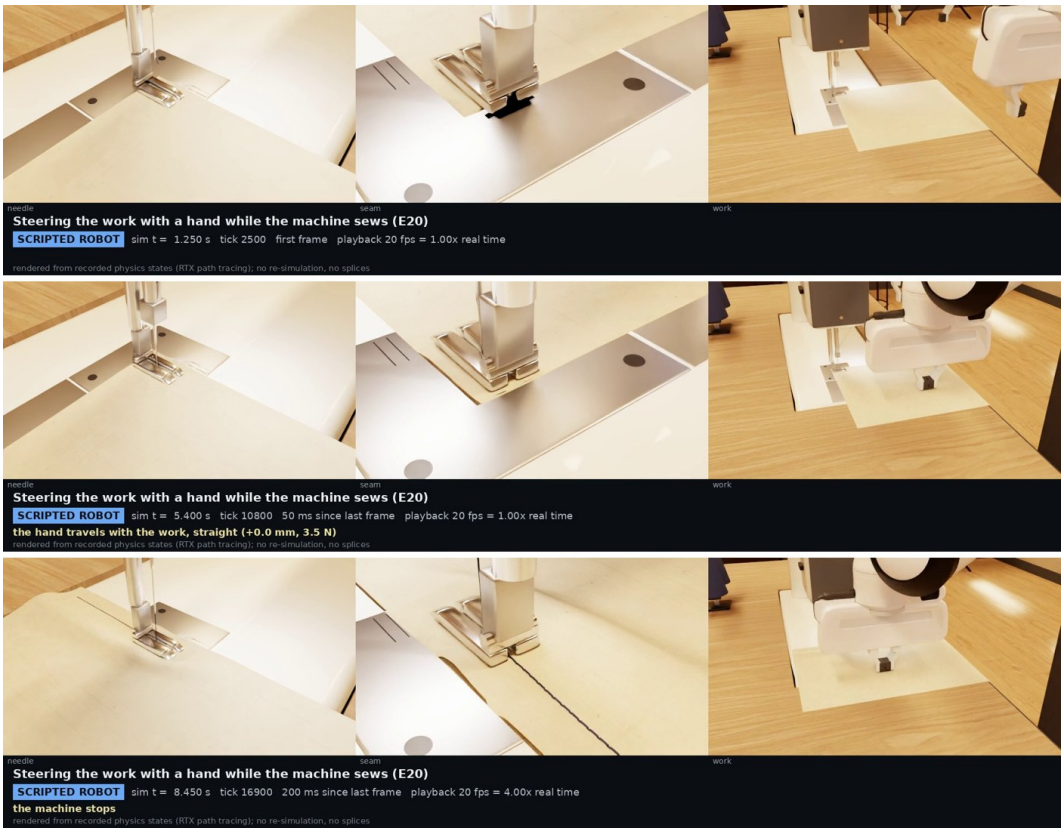


Figure 35: E20: a hand steering the work while the machine sews (SCRIPTED ROBOT): first, middle and last frames

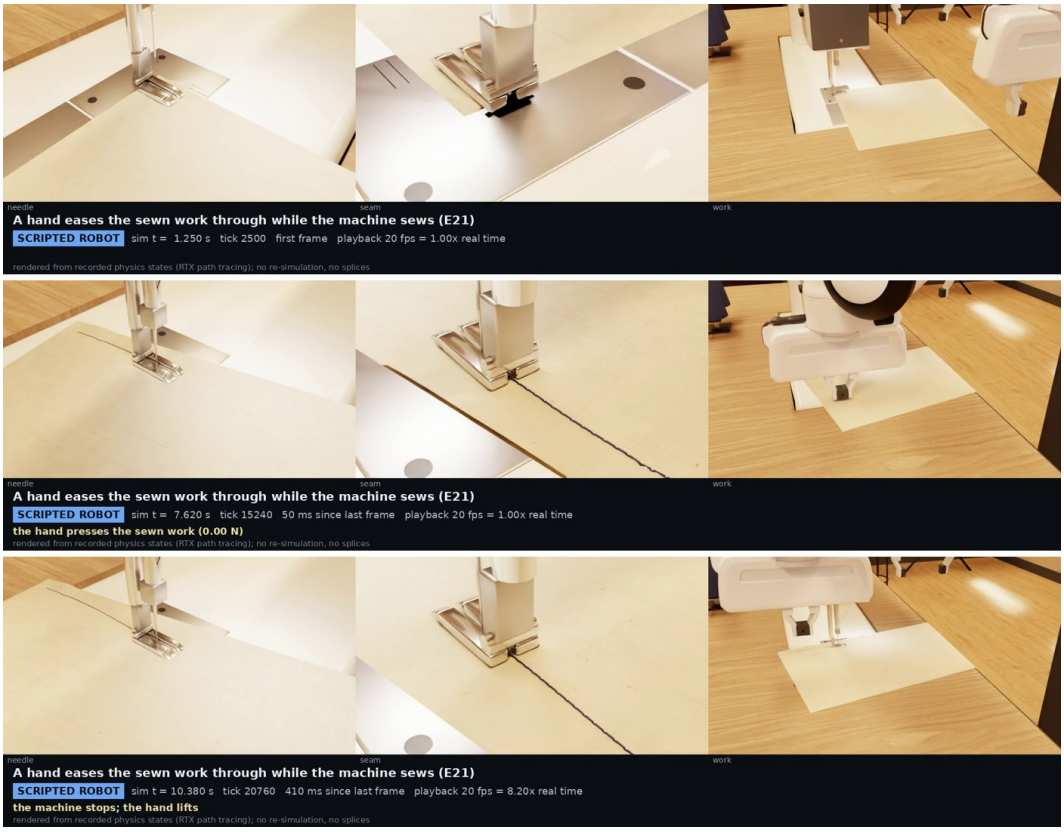


Figure 36: E21: a hand easing the sewn work through (SCRIPTED ROBOT): first, middle and last frames

13 Stability: rest and legal abuse

Rest (E14, UNASSISTED). Each start state stepped with no command, the arms (or the humanoid) holding their ready pose, the machine idle; per 0.25 s the largest step, the displacement from the start, the peak speed, the kinetic energy and a period-2 measure:

- plain-seam (dual-fr3): 30.00 s, acceptance passed; largest step per 0.25 s 0.0222 mm, largest displacement from the start 0.0254 mm (last 0.0157), drift over the second half: fitted rate beyond three standard errors 0.00 um/s (largest fitted rate 0.04 um/s, detection floor 0.11 um/s, largest scatter about the trend 1.7 um), peak speed 0.691 mm/s, kinetic energy <= 5.22e-11 J, period-2 <= 0.116 um, foot 41.18-41.18 N, arm joint drift 0.0e+00 rad, reflexes 0; do-nothing evaluation {'success': False, 'score': 0.0, 'primary': False, 'failed': ['stitches_formed']}
- plain-seam (g1): 15.00 s, acceptance passed; largest step per 0.25 s 0.2864 mm, largest displacement from the start 0.2544 mm (last 0.0931), drift over the second half: fitted rate beyond three standard errors 0.00 um/s (largest fitted rate 0.61 um/s, detection floor 7.08 um/s, largest scatter about the trend 30.8 um), peak speed 1.443 mm/s, kinetic energy <= 1.33e-09 J, period-2 <= 0.093 um, foot 41.18-41.18 N, arm joint drift 7.9e-04 rad, reflexes 0; do-nothing evaluation {'success': False, 'score': 0.0, 'primary': False, 'failed': ['stitches_formed']}
- plain-seam (dual-panda): 10.00 s, acceptance passed; largest step per 0.25 s 0.0350 mm, largest displacement from the start 0.0366 mm (last 0.0249), drift over the second half: fitted rate beyond three standard errors 0.00 um/s (largest fitted rate 0.28 um/s, detection floor 2.44 um/s, largest scatter about the trend 7.2 um), peak speed 0.567 mm/s, kinetic energy <= 6.15e-11 J, period-2 <= 0.086 um, foot 41.18-41.18 N, arm joint drift 0.0e+00 rad, reflexes 0; do-nothing evaluation {'success': False, 'score': 0.0, 'primary': False, 'failed': ['stitches_formed']}
- plain-seam (single-fr3): 10.00 s, acceptance passed; largest step per 0.25 s 0.0227 mm, largest displacement from the start 0.0264 mm (last 0.0177), drift over the second half: fitted rate beyond three standard errors 0.00 um/s (largest fitted rate 0.30 um/s, detection floor 1.07 um/s, largest scatter about the trend 3.6 um), peak speed 0.440 mm/s, kinetic energy <= 3.89e-11 J, period-2 <= 0.096 um, foot 41.18-41.18 N, arm joint drift 0.0e+00 rad, reflexes 0; do-nothing evaluation {'success': False, 'score': 0.0, 'primary': False, 'failed': ['stitches_formed']}
- load (dual-fr3): 15.00 s, acceptance passed; largest step per 0.25 s 0.0000 mm, largest displacement from the start 0.0000 mm (last 0.0000), drift over the second half: fitted rate beyond three standard errors 0.00 um/s (largest fitted rate 0.00 um/s, detection floor 0.00 um/s, largest scatter about the trend 0.0 um), peak speed 0.003 mm/s, kinetic energy <= 8.82e-16 J, period-2 <= 0.001 um, foot 0.00-0.00 N, arm joint drift 0.0e+00 rad, reflexes 0; do-nothing evaluation {'success': False, 'score': 0.0, 'primary': False, 'failed': ['foot_down', 'ply0_single_layer', 'ply1_single_layer']}

Every start state on every deployment: the default rig's five start states for 15-30 s, the other three deployments' five for 10-15 s:

Start state	Robot	Seconds	Every-tick checks	Fastest vertex after settling (mm/s)	Fitted drift beyond 3 SE / detection floor (um/s)	Lowest point dropped (um)	Foot load change	Robot
baste	FR3 pair	20	passed	0.39	0.00 / 0.07	0.13	0.000 %	joint drift 0.0e+00 rad
baste	Panda pair	10	passed	0.40	0.00 / 0.43	0.19	0.000 %	joint drift 0.0e+00 rad
baste	G1	10	passed	1.29	0.00 / 7.74	0.14	0.000 %	tilt 0.16 deg
baste	single FR3	10	passed	0.38	0.00 / 0.39	0.22	0.000 %	joint drift 0.0e+00 rad
curve	FR3 pair	20	passed	0.26	0.00 / 0.11	0.79	0.000 %	joint drift 0.0e+00 rad
curve	Panda pair	10	passed	0.26	0.00 / 0.29	0.69	0.000 %	joint drift 0.0e+00 rad
curve	G1	10	passed	0.45	0.00 / 0.45	1.53	0.000 %	tilt 0.16 deg
curve	single FR3	10	passed	0.21	0.00 / 0.26	0.40	0.000 %	joint drift 0.0e+00 rad
load	FR3 pair	15	passed	0.00	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
load	Panda pair	10	passed	0.01	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
load	G1	10	passed	0.00	0.00 / 0.00	0.00	0.000 %	tilt 0.16 deg

Start state	Robot	Seconds	Every-tick checks	Fastest vertex after settling (mm/s)	Fitted drift beyond 3 SE / detection floor (um/s)	Lowest point dropped (um)	Foot load change	Robot
load	single FR3	10	passed	0.01	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
pivot	FR3 pair	20	passed	0.01	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
pivot	Panda pair	10	passed	0.07	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
pivot	G1	10	passed	0.07	0.00 / 0.00	0.00	0.000 %	tilt 0.16 deg
pivot	single FR3	10	passed	0.07	0.00 / 0.00	0.00	0.000 %	joint drift 0.0e+00 rad
plain-seam	FR3 pair	30	passed	0.84	0.00 / 0.11	0.81	0.000 %	joint drift 0.0e+00 rad
plain-seam	Panda pair	10	passed	1.09	0.00 / 2.44	0.73	0.000 %	joint drift 0.0e+00 rad
plain-seam	G1	15	passed	1.88	0.00 / 7.08	0.52	0.008 %	tilt 0.16 deg
plain-seam	single FR3	10	passed	0.80	0.00 / 1.07	0.98	0.000 %	joint drift 0.0e+00 rad

The drift criterion is a trend, not a displacement: each cloth vertex's least-squares rate over the second half's samples, counted only beyond three standard errors (a bounded jitter at the implicit solve's velocity tolerance reads as zero; the smallest creep a run could have seen is its detection floor, recorded). A displacement measure (the largest displacement between the run's midpoint and its end, over half the run) would read that jitter as drift. All 20 soaks pass their every-tick checks. The draped plain-seam start state rests: over 30 s (filmed) its fastest vertex moves at most 0.84 mm/s after settling, the jitter at the implicit solve's velocity tolerance, and no vertex has a trend beyond three standard errors (the largest fitted rate 0.035 um/s against a detection floor of 0.11 um/s); the drape's lowest point moved 0.81 um. The G1's start rests too (fastest vertex 1.9 mm/s after settling, inside the 5 mm/s limit; no trend beyond the floor), and the humanoid stands unsupported through its soak (pelvis tilt 0.16 deg). No hands-off soak forms a stitch or scores.

Legal abuse (E15, SCRIPTED ROBOT). Commands a policy may legally send, each from the plain-seam start state; every case must stay finite, and each has a predeclared prediction:

Case	Passed	Finite	Needle	Stitches	Reflexes	Machine faults	Cloth peak speed (m/s)	Notes
(a) right fingertip's side across the needle's line above the foot, machine at 600 sti/min	True	True	broken	1	-	needle_struck_robot	0.044	struck before running: False
(b) hand rammed 30 mm into the table in 0.2 s	True	True	ok	0	fr3_joint4 405 N m	-	0.000	
(c) machine at 5,000 sti/min for 1 s, nobody guiding	True	True	ok	76	-	-	0.034	76 cycles, stack fed 146.8 mm
(d) presser, reverse and speed toggled every 20 ms for 1 s	True	True	ok	7	-	-	0.027	
(e) needle stopped down; hand drags the plies 30 mm toward the operator in 0.3 s	True	True	ok	1	fr3_joint2 25 N m	-	0.050	
(f) both hands press closed fingers onto the stack, full force mode (70 N per hand)	True	True	ok	0	fr3_joint3 20 N m; fr3_joint4 44 N m; fr3_joint3 83 N m; fr3_joint4 46 N m	-	0.000	finger force left 0.0 N, right 0.0 N

Every case stays finite; five meet their prediction and one, (e), does not. (a) A closed FR3 hand cannot stand on the needle's line: its fingers are 30 mm across and the presser bar stands 1 mm beside the needle, so the probe puts a fingertip's side 0.3 mm across the line, reached on a statically checked joint-space path; the running needle

meets the fingertip and breaks after one stitch. (b) The ram trips the FR3's collision reflex on the tick the hand meets the table; the external-torque estimate there reads 404.6 N m on joint 4 against the 20 N m threshold, a rigid impact resolved within one tick (repeated runs read the same: the rigid world is deterministic here). (c) Unguided at 5,000 sti/min the plain-seam stack fed 1.93 mm per stitch against the 2.5 mm setting and all 76 stitches formed, the needle intact; the outcome at top speed varies from run to run here as on the bench (§8.6). (d) Chattering control-box commands are accepted and change nothing physical beyond what they command. (e) Dragging the plies against the stopped needle did not go as predicted: the right arm's collision reflex stopped the drag (joint 2 at 25.0 N m against the 20 N m threshold of a moving arm) while the needle's lateral load was 1.6 N, below its 5.0 N bending load, and the needle stayed intact. (f) Both closed hands, fingers clamped at the hand's full force (35 N per finger), commanded 10 mm below the stack's top, into the table: both arms' collision reflexes fire on the way down (20-83 N m; runs with different hand-cloth couplings read the same, so a rigid contact trips them, not the cloth); the stack stays finite and at rest, and at the end neither hand presses it (0 N). A request for 70 N per finger is refused before anything runs (the catalog's FR3 hand is rated 35 N per finger).

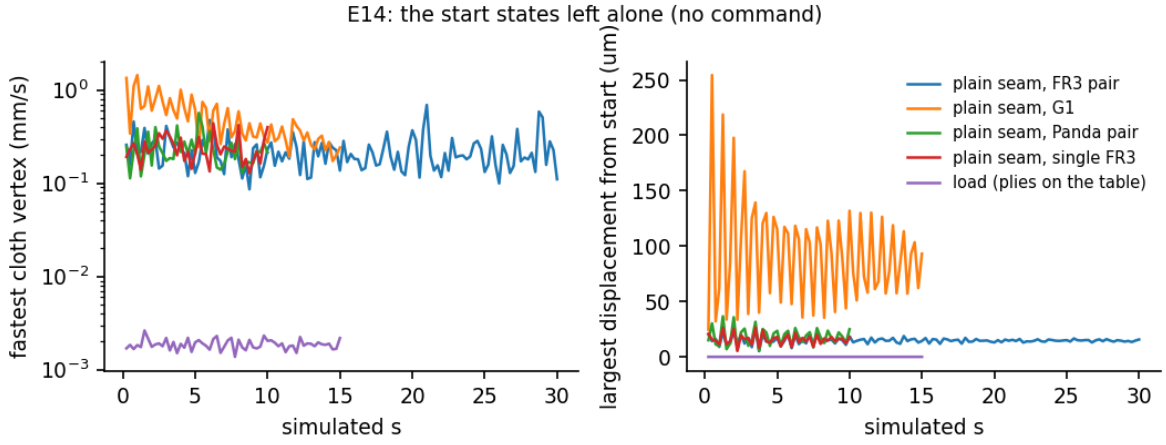


Figure 37: E14: the start states left alone: the fastest cloth vertex and the largest displacement over time

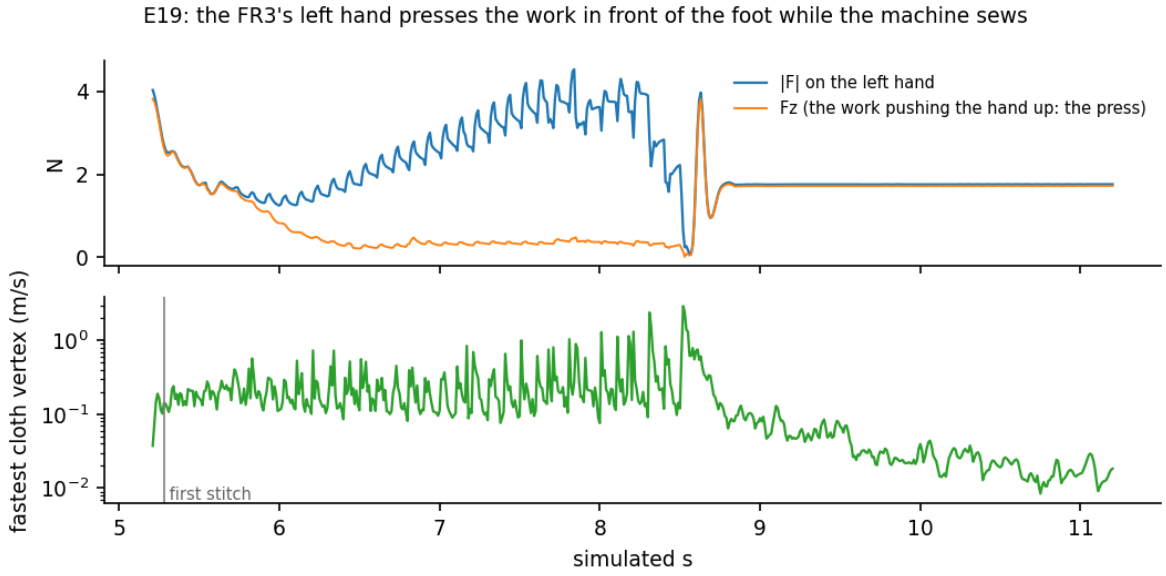


Figure 38: E19: a hand holding the work while the machine sews: the hand's force, the cloth's fastest vertex, the machine's faults

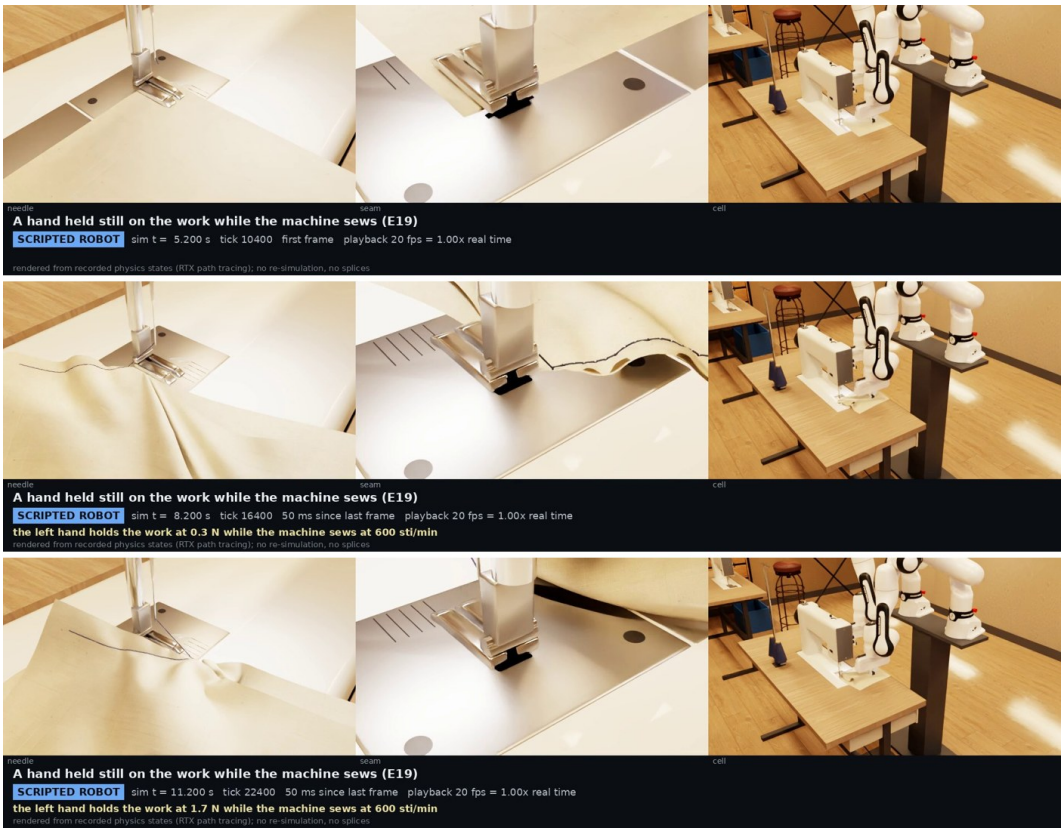


Figure 39: E19 (SCRIPTED ROBOT): first, middle and last frames of the film



Figure 40: E14, the plain seam's start state left alone, FR3 pair (UNASSISTED): first, middle and last frames of the 30 s film



Figure 41: E14, the plain seam's start state left alone, G1 (UNASSISTED): first, middle and last frames of the 15 s film

14 Runtime

The limit is 200 wall seconds per simulated second. E16 measures it through the public API as a user meets it: `gar run` with the policy key, every default policy camera rendered for every command (`operator`, `overhead` and `needle`; the G1's `head_rgb`, `overhead` and `needle`), the plain seam sewn at 600 sti/min through the control box while one hand travels with the work, one Cartesian target per command: eight 0.5 s actions, then three commands each of 0.2, 0.1, 0.05, 0.02 and 0.01 s, three one-tick `tool_io` commands and three observation-only reads, the host's load sampled after every command.

Where the limit holds. The criterion, set before these runs: on a quiet host every deployment's commands of 0.1 s and longer return inside 200x. Measured: every deployment's commands of 0.02 s and longer return inside it; 0.01 s commands do not (235-273x). Each deployment's quiet run started only after three minutes of 1-minute load under 4 and GPU utilisation under 30 % (sampled every 30 s), one deployment at a time; while each ran, the session's own render worker held the 1-minute load at a median of 3.5-4.5:

Run	Host load at start -> end (median after each command; 16-CPU quota); GPU busy	Launch (s)	Hand's approach, machine idle	Sewing	Slowest action	Inside 200x	Formed stitches; faults (the state keeps the last five); needle	Arm reflexes while sewing; peak cloth force on the fingers (sampled every 10 ms)
FR3 pair, quiet host	1.8 -> 3.1 (4.5); 0 -> 0 %	162	114x	121x	122x	yes	60; none; ok	0; 21 N
Panda pair, quiet host	1.6 -> 3.3 (3.8); 0 -> 0 %	162	113x	123x	124x	yes	60; none; ok	0; 21 N
single FR3, quiet host	31.3 -> 3.8 (3.7); 0 -> 5 %	155	107x	118x	122x	yes	59; hook struck needle; damaged	4; 20 N
G1, quiet host	1.3 -> 3.2 (3.5); 0 -> 0 %	160	111x	118x	121x	yes	60; none; ok	0; 5 N

Deployment	0.5 s	0.2 s	0.1 s	0.05 s	0.02 s	0.01 s	Shortest inside 200x	One-tick tool_io (s)	Observation only (s)	0.05 s command after a 5 s / 15 s pause (s); added by the 15 s pause (s)
FR3 pair, quiet host	121 / 122	127 / 127	132 / 133	144 / 146	196 / 199	235 / 239	0.02 s	1.01-1.11	0.71-0.71	7.1-7.4 / 7.3-7.7; +0.1
Panda pair, quiet host	123 / 124	128 / 129	135 / 136	150 / 155	177 / 189	239 / 241	0.02 s	1.01-1.16	0.72-0.73	7.2-7.6 / 7.3-7.6; -0.1
single FR3, quiet host	119 / 122	126 / 127	134 / 135	148 / 169	184 / 185	260 / 273	0.02 s	0.98-1.12	0.67-0.67	7.3-7.7 / 7.4-7.4; -0.0
G1, quiet host	118 / 121	123 / 125	129 / 132	143 / 161	173 / 180	246 / 256	0.02 s	1.21-1.44	0.91-0.92	7.0-7.8 / 7.0-7.5; -0.0

A request has a floor beyond its physics: an observation-only read takes 0.67-0.92 s and a one-tick `tool_io` 1.0-1.4 s (three images rendered, written and published; the server's handling), so a command needs ~0.02 s of physics or more for its time to fall inside 200x. The physics costs the rest: the seam band's 1.25 mm mesh costs 1.65 times the 2.5 mm mesh per tick while sewing (§6), and all three default cameras are rendered. On a host shared with other environments' jobs:

Run	Host load at start -> end (median after each command; 16-CPU quota); GPU busy	Launch (s)	Hand's approach, machine idle	Sewing	Slowest action	Inside 200x	Formed stitches; faults (the state keeps the last five); needle	Arm reflexes while sewing; peak cloth force on the fingers (sampled every 10 ms)
FR3 pair	9.0 -> 9.7 (9.5); 1 -> 1 %	26	115x	122x	125x	yes	60; none; ok	0; 21 N
Panda pair	11.8 -> 11.6 (13.0); 93 -> 2 %	27	153x	163x	176x	yes	60; none; ok	0; 21 N
single FR3 (right hand)	10.9 -> 13.5 (14.4); 2 -> 0 %	27	139x	121x	152x	yes	59; hook struck needle; damaged	4; 20 N
G1	9.4 -> 12.1 (11.0); 1 -> 100 %	28	112x	120x	150x	yes	60; none; ok	0; 5 N

By command length, wall seconds per simulated second (median / worst of the commands of that length), the shortest length at which every command stayed inside 200x, and the fixed cost of a request:

Deployment	0.5 s	0.2 s	0.1 s	0.05 s	0.02 s	0.01 s	Shortest inside 200x	One-tick tool_io (s)	Observation only (s)	0.05 s command after a 5 s / 15 s pause (s); added by the 15 s pause (s)
FR3 pair	123 / 125	129 / 130	133 / 138	152 / 153	190 / 197	234 / 241	0.02 s	1.06-1.76	0.72-0.73	7.1-7.6 / 7.2-7.9; -0.3
Panda pair	166 / 176	180 / 180	190 / 198	207 / 221	211 / 212	255 / 292	0.1 s	1.17-1.48	0.75-0.76	9.6-10.5 / 10.0-10.2; -0.2
single FR3 (right hand)	121 / 130	135 / 146	142 / 154	154 / 175	219 / 230	264 / 268	0.05 s	0.96-1.09	0.68-0.68	7.2-7.3 / 7.4-7.6; -0.3
G1	121 / 122	125 / 125	130 / 132	142 / 143	174 / 183	250 / 277	0.02 s	1.19-1.33	0.92-0.92	7.0-7.2 / 7.0-7.4; +0.3

What a pause between requests costs (E16b): with the machine idle and the robot at rest, 0.05 s commands sent back to back, after 5 s pauses and after 15 s pauses (three of each), every deployment:

Deployment	Back to back (s, three commands)	After a 5 s pause (s)	After a 15 s pause (s)	Added by the 15 s pause (median, s)	Host load (1 min) during the run
FR3 pair	6.8, 6.7, 6.7	6.7, 6.7, 6.7	6.7, 6.7, 6.8	+0.0	2.0-2.6

Deployment	Back to back (s, three commands)	After a 5 s pause (s)	After a 15 s pause (s)	Added by the 15 s pause (median, s)	Host load (1 min) during the run
Panda pair	6.8, 6.7, 6.8	7.0, 6.7, 6.8	6.7, 6.7, 6.8	-0.1	5.3-19.9
single FR3 (right hand)	6.5, 6.4, 6.4	6.4, 6.4, 6.8	6.3, 6.4, 7.0	-0.0	4.6-15.7
G1	6.7, 6.6, 6.9	6.7, 6.6, 6.9	7.3, 6.6, 6.6	-0.1	3.9-6.8

The render worker is kept 600 s after a request (`renderer_options.idle_release_s`), because a new worker adds 6.9-7.8 s to the next request: a 15 s pause costs -0.07 to +0.03 s against the same command sent back to back, on every deployment.

Of a short request's fixed cost, the evaluation, which runs on every request, takes 2.4 ms and the whole state 4 ms (timed in-process on the plain-seam state mid-seam); the rest is the render and the server's own handling (the request, its history, the images written and published).

The runtime note a policy reads (`capabilities.hardware.wall_time`) reports these measurements, and the command cap (2 s of simulated time) is set from the slowest rate they measured against the 900 s client timeout.

Other runs, wall seconds per simulated second as they ran (the films rendered inside the run where noted; the host shared as above):

Run	Simulated (s)	Wall (s)	Wall / simulated	Cloth step's share	Rigid step's share
machine feed (film)	6.2	4,173	673	83 %	0.3 %
load demonstration (film)	3.9	3,132	803	93 %	0.3 %
plain seam demonstration (film)	17.5	13,720	785	90 %	0.3 %
pivot demonstration (200 x 200 mm; film)	18.7	12,681	678	89 %	0.3 %
curve demonstration (150 x 150 mm; film)	40.6	22,887	564	87 %	0.4 %
curve demonstration, the task's start (200 x 300 mm; film)	62.0	39,614	639	91 %	0.3 %
humanoid's hand on the work (film)	3.5	2,478	710	91 %	0.4 %
FR3 pair guiding a seam (film)	14.8	9,812	663	86 %	0.3 %
hand slides the stack (film)	6.1	1,913	313	not recorded	not recorded
soak, plain seam, FR3 pair (film)	30.0	7,471	249	96 %	0.7 %
soak, plain seam, G1 (film)	15.0	3,359	224	96 %	1.1 %
soak, plain seam, Panda pair (no film)	10.0	6,964	696	99 %	0.3 %
soak, plain seam, single FR3 (no film)	10.0	6,144	614	99 %	0.3 %
soak, load (no film)	15.0	6,035	402	98 %	0.5 %

The cloth's implicit step takes most of every run's wall time, and a resting stack costs nearly what a sewing one does (on the 1.25 mm mesh, side by side on the shared host, 260-302 ms a tick at rest against 315-337 sewing; the 2.5 mm mesh 133-138 and 182-207). On the 2.5 mm mesh the draped soaks run 2.78-2.86 extra Newton iterations a tick at rest beyond the two every tick makes (E14, all four deployments; the flat load start 1.00). A run with the solver's step-by-step record (60 resting ticks of the plain-seam start) shows why: every tick's first two Newton steps move cloth lying on the table and over its edge by 2.6-3.4 um, the scale of the 2.5 um (g dt^2) that gravity's prediction puts into each tick and the contact must take out again, while the loop stops only when every step is below its velocity accuracy, 0.1 mm/s or 50 nm a tick; three more iterations take it there. A predictor that starts resting contacts where they ended the last tick (a warm start) might remove most of them; it is not tried.

The G1's rigid step costs 1.7 s per simulated second at rest: the scene zeroes Newton's default 100 mm detection gap on every shape (§5), leaving 8 contacts a tick, all of them the G1's two feet on the floor, the G1 standing (pelvis 0.784 m, tilt 0.17 deg). With the gap (MuJoCo 3.9 widens detection by it without generating force) the step carries ~1,075 contacts a tick, 1,014 of them pairs of its own links 0.1-199 mm apart with no force, and costs 25 wall seconds per simulated second on a shared host.

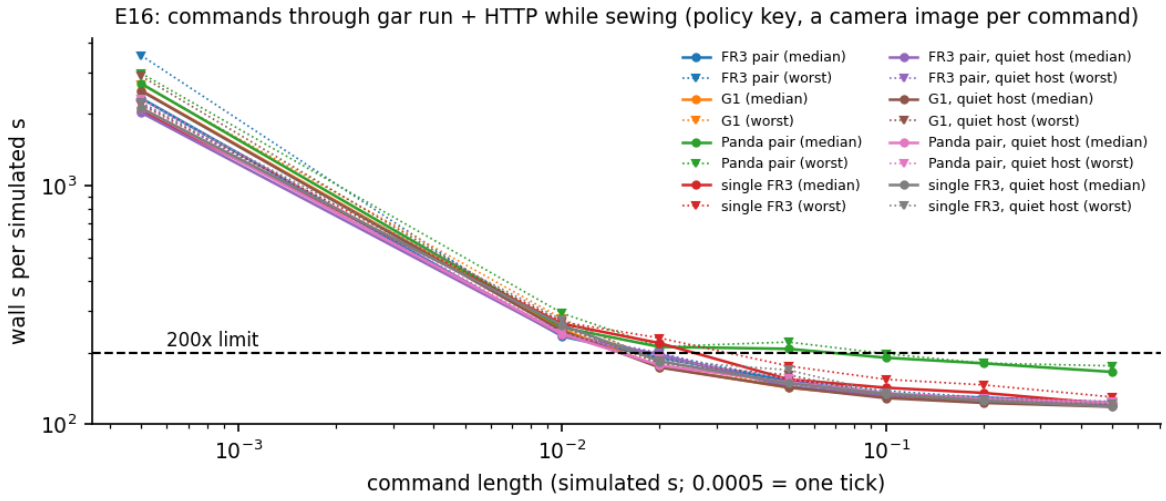


Figure 42: E16: wall seconds per simulated second by command length, every deployment on the shared host and the FR3 pair on a quiet host, against the 200x limit

15 Interfaces and visual review

GAR integration. The environment is registered in the shared runtime (`gym_anything_robotics/registry.py`: `sewing` with the aliases `sew`, `sewing-machine`, `lockstitch`, `sew-plain-seam`; `launch.py`; the task catalogue `sewing/benchmark.json` with the resolver `sewing.variants:task_resolve`) and served by `gym_anything_robotics/backends/newton_sewing.py`. The robot choices come from the catalog (`robots/catalog.py`, `ENVIRONMENTS["sewing"]`, validator `sewing.robot_setup:validate_setup`). `gar tasks inspect sewing` lists the five tasks and the selectors (preset, task, seed); `gar inspect sewing` the robot choices and the description. `gar doctor sewing` reports the coordinator's packages (newton, warp-lang, mujoco) and, through `sewing/doctor.py`, the environment's own needs without starting physics or a renderer: SciPy, Pillow and imageio-ffmpeg; the Isaac interpreter every RGB observation uses; the modules of other environments the renderer imports (`whiteboard.ipc`, `whiteboard.appearance`, `wiping`, `label_application.dressing`); and every offered deployment's locked robot assets, the default's required and the others reported with the option they serve.

Policy interface. The arms' observation carries each joint's estimated external torque (`joints.<arm>/<joint>.external_torque`, N m): the torque of the contacts on the arm and of the cloth's filtered reaction on the hand, the quantity libfranka reports as tau_ext_hat and the collision reflex judges. The G1 has no force or tactile channel (the Dex3's fingertip sensors are not simulated). A command may last at most 2 s of simulated time (`max_command_s`): at the slowest rate E16 measured it returns inside the declared 900 s client timeout, where an uncapped command could outlast it while the server kept stepping. The runtime note a policy reads (`capabilities.hardware.wall_time`) reports the E16 and E16b measurements.

Session features.

- *Variations* (`sewing/variations.py`): a catalogue of 50 entries (5 tasks x seeds 0-9, `sewing-sample-room-v1`, its digest in `benchmark.json`), served by `GET /v1/variations` and switched in a session by `POST /v1/variations/start` or a reset with `options ("variation_id": N)`. A switch builds the new world as a new session would (its start state from the cache: ~5 s; prepared afresh: 3.5-7 min measured), releases the render worker and the replaced world, and drops the checkpoints; a refused switch or reset changes nothing and leaves the session ready.
- *Per-action trace* (`sewing/trace.py`): `GET /v1/history/(id)/trace` returns, for every tick of the action, the joints (position, velocity, target, the motion generator's set point), the arms' external torque estimate, the cloth's filtered force on each hand, the machine (shaft angle, speed, needle tip and state, presser, foot height and load, cycle, holes, stitches, threads' highest tension) and the tick's wall time split; events (machine I/O, punctures, stitches, faults, protective stops). The cloth's vertices are not sampled (they stay on the GPU); the sampler costs 0.23-0.26 ms per tick against cloth ticks of 115-330 ms.
- *Checkpoints*: in memory, in the server process, restorable any number of times within the session and variation they were saved in; a server restart, a session close or a variation switch drops them (stated in the capabilities, the variation contract and every save/restore response).

- *Compliant arms*: the raw operation `joint_impedance` sets an FR3 or Panda arm's joint impedance, tau = K (q_target - q) - D dq, K per joint in [0, 14250] N m/rad as libfranka's Robot::setJointImpedance takes it, D by default at the ratios of libfranka's joint impedance example; null restores the position servo. The arm's gravity stays compensated as on the robot. Refused before any tick out of range; not offered to the G1, whose low-level gains have no sourced range here. In the rigid world 5 N down on link 7 deflects joint 2 by 2.6 mrad on the servo and 24.9 at K 50.
- *Observation*: every policy camera is rendered by default, `operator` (the G1: `head_rgb`), `overhead` (a D435-class cell camera on a boom above the work) and `needle`; a launcher may ask for fewer.
- *The G1's task text* says that the machine is driven through its control box (tool_io), not the pedal, knee lifter and touch-back switch a standing person would use.
- *Acceleration limits by robot model*: the FR3's 10 rad/s^2 (libfranka main), the Panda's 15/7.5/10/12.5/15/20/20 (libfranka master), the G1's ramp an environment value (no motion generator on the robot).
- *Doctor*: the default robot's (FR3) assets are resolved, verified and required.

Through the session (E16, dashboard). E16 starts `gar run sewing --task plain-seam --policy-key` for each of the four deployments, and an HTTP client with the policy key lowers one hand onto the stack (the left; the single FR3's right, from 0.15 m above), starts the machine through `tool_io` and sews eight 0.5 s actions while the hand travels with the work, then the short commands of §14, then stops and trims; every action returns the policy cameras' images. The hand program is open-loop, and the sessions' seams show it: the hand is lowered in one 0.5 s command and then travels at the dial's feed, 25 mm/s, 90-190 mm left of the needle, wherever the work actually goes; the Franka fingers pressed up to 19-21 N (E18's press, lowered until the cloth pushes back, holds 3.5 N; at that press the same hand travelling with the work for 3 s sews 30 stitches clean). Every session's seam formed 60 of 60 stitches with the needle intact, on the shared host and on the quiet one, except the single FR3's, whose hook struck the needle once in each (59 of 60, the needle damaged). Every session reported its faults in the observation and kept answering. A dashboard check (headless Chromium) connects to the session's `/dashboard` with the policy key (connecting does not step), sends the machine at 600 sti/min for 0.25 s, a 20 mm move of the right hand and a stop with the foot lifted from the action editor, and an out-of-range speed: each committed exactly its ticks, the machine sewed (the counter rose to 7) and lifted its foot, the out-of-range command was rejected without stepping, and the page reported no script errors (passed).

Visual review. Every film was looked at through its review frames (five per film) and its first, middle and last frames in this report, and the films whose records named an event were opened frame by frame around it; every still was looked at on labelled contact sheets. In short: nothing floats, interpenetrates or moves without a cause in any frame looked at; the cloth lies on the bed, the table and the plate and hangs over the table's front edge as a drape; the stitch line grows behind the foot; the robots hold their ready poses where a film says they do, and the humanoid stands, with no arm pedestal or pedal drawn beside it. The frames show what the result files say: the machine alone turning the work under the foot; the plain seam sewn end to end and lying flat; the pivot's corner tail standing up during the turn until the phantom finger presses it; the baste's sewn work lifting into a loop behind the foot at the 27th stitch (assisted) and lying flat until the needle fails (unguided); the task's own curve crumpled at the foot by the end; E19's held work buckled into folds in front of the foot; E20's shallow ridge beside the foot during the steer; the pinch camera seeing the outer finger's pad on the drape's edge (the inner finger hidden behind the drape). Two visual defects remain: the seam allowance's raw edge stands up as a dark line along the seam (the feed dog's impact resolved within one tick, §9), and the seam camera shows dark notches cutting into the raw edge while the machine sews, absent at rest: probably the edge's lifted segments, not verified. No stills of the baste's start state were rendered.

16 Limitations and work not built

Stated plainly; each is a gap in the world, not a property of a policy.

Scope against the design. Built: preset A (sample room) on the A head with five tasks (A0 load, A1 plain seam, A2 baste, A5 pivot, A6 curve) and the `dual-fr3`, `dual-panda`, `single-fr3` and `g1` robots. Not built:

- preset A's tasks A3 (pinned seam: pins and their strikes), A4 (taper on denim along a chalk line), A7 (strap: a pre-pressed fold and hand-over-hand regrasping), A8 (ease between notches), A9 (tension adjustment: the needle tension is not a control-box output, and balance is a law, not an emergent interlock);
- presets B (domestic machine at a dining table: buttons, handwheel), C (bindery: paper on the A head, perforation and ligament tearing) and D (upholstery: the walking-foot head, vinyl, welt cord).

- for A6 (curve) an operator steers continuously at a slower speed; the assisted demonstrations turn the work in pivot-and-sew steps instead (a steering field crumples the work). A robot hand does steer the work by contact (E20, §12). The world does not over-resist turning under the foot; it turns the work by itself (below), and on the curve most of the turning is the feed's own: of the assisted curve's 91 deg, the machine's stretches, with nobody touching the work, turned it 52 deg and the phantom pivots 39 deg (`turn_split`).

The humanoid. The G1 stands on its servos with no balance controller (as in the repository's other G1 environments): pushed hard (a hand against the machine) it can fall, which it does in the simulation as it would on the real robot without its balance controller. Its Dex3 hands meet the cloth as boxes around the links' meshes; the thumb of a palm-down Dex3 points at the work, so the hand presses the cloth with its folded thumb before the palm reaches it. From its square stance the machine's face plate hides the needle from the head camera. No program guides a whole seam with the G1: its demonstration rests the left hand on the work (a 3.6 N press, 3.4 N on average while sewing) while the machine sews 12 stitches (§12).

Physics that is approximate or missing.

- The needle's puncture, deflection and the stitch's formation are laws on the simulated state: the hook, the loop and the thread's path through the cloth are not simulated; balance is the tension ratio inside the capstan band, not where the interlock sits; thread elements are straight springs between interlocks.
- Cloth: no yarn structure; no damage except holes (no tearing, no fraying). The reference fabric's four KES hysteresis measures are modelled (bending 2HB, compression RC, tension RT, shear 2HG and 2HG5), each as friction with a micro-slip span that is an estimate; the tensile friction is taken proportional to the elastic tension and the shear friction linear in the shear angle (two-point fits), and KES measures shear under 10 gf/cm while the model's shear friction does not depend on the tension; the compression law takes KES T0 (quoted at 0.5 gf/cm2) as the zero-pressure thickness, so between 1 and 49 Pa a ply is up to 0.13 mm thinner than KES's curve (Tm, WC, LC and RC hold); the set's recovery time (1 s) is an estimate.
- Air: the squeeze film and drag are modelled, the added mass of the air is not (for the 200 mm overhang about 1.5 times the cloth's mass): free-hanging parts swing faster and ring longer than real cloth.
- **The work turns under the running foot by itself**: unguided, the plain-seam start turns counter-clockwise 0.55 deg per stitch over the first 10 stitches, slowing as the turn carries the stack's edge out from under the right toe: 19.1 deg after 60 stitches (6 s at 600 sti/min), the stitch line 38 mm in from the edge at the end. The baste's start at the 5 mm stitch turns twice as fast per stitch (10.1 deg after 10 stitches, 19.0 after 25); there the needle, deflected 0.097 mm by the work (the hook's clearance is 0.07 mm), was struck by the hook at the 25th stitch and bent at the 28th, the work lying flat at the foot until then (the film). What deflected it, beyond the turn, was not separated. The feed dog's two rows slip at each hand-off between plate and dog (the teeth move forward while their load builds and falls), and the foot's unequal toes (6.0 and 4.5 mm, an estimate from photographs) put more of its load on the right row, which slips less: the right row feeds 0.04-0.08 mm more per stroke. With equal toes, or the needle on the stack's centre line, the turn vanishes. Feed drift of this kind is a known defect of drop feeds (Singer, US 4,554,878), but no measured drift and no drawing of the foot was found: the turn's rate rests on the unmeasured toe widths. Every guiding policy meets it (the assisted seam's guide pulled the work the same way on all 2,208 of its applications, 0.11 N on average; the pivot's two legs 97 and 81 %).
- Friction coefficients for cloth on the laminate table (0.30) and on the feed dog's teeth (1.0, effective: the teeth are below mesh scale) are open; cloth-steel 0.15 and cloth-cloth 0.65 are read ranges' midpoints. The teeth are smooth boxes: raised at a needle-up stop they hold an unloaded bottom ply against a weak couple through their sides (section 8.7); how strongly real teeth do is not measured.
- The cloth sees the table's front edge rounded (6 mm, an estimate within 2-12 mm), and the render draws it so; the rigid table the robots touch keeps a square edge.
- A sewn seam settles in one of two shapes from run to run on the 2.5 mm mesh: flat (0.12-0.32 mm RMS) or lifted (0.58-1.75 mm, above the evaluator's 0.5 mm), once with its start curled 16.5 mm; runs are not bit-reproducible. On the 1.25 mm mesh E7's bench seam (16 stitches, settled 0.4, 1.0 and 2.0 s), run six times, gave one shape: all six flat, 0.106-0.107 mm RMS in the evaluator's band and 0.17-0.18 % contraction after 2.0 s (0.092 after 0.4 s), the highest point 1.43 mm at the cut edge, every stitch formed. The task's own seams were not repeated, so pucker is recorded, not scored.
- Seams relax: thread elements end nearly slack after a seam settles. Tension pucker shows only as in-plane gathering (the contraction rises from 0.19 to 0.45 % at three times the needle tension, section 8.8), not as waves: there is no yarn jamming in the in-plane compression and the seam band's 1.25 mm mesh barely resolves a 2.5 mm stitch's wave.

- Shear locks past 50 deg (derived from a comparable plain cotton's thread count, 45-62 deg) with the fabric's tensile toe stiffness (an estimate); between the KES test's 8 deg and the locking angle the shear law is the KES law extrapolated, which softens slightly where a real fabric stiffens.
- The needle's measured puncture peak reaches the penetration law at low speed (82 % at 600 sti/min) but not at speed (57 % at 3,000 sti/min): the plies yield into the plate's needle hole (section 8.8).
- The rule that keeps a stored stick force when a contact is pressed harder acts on the cloth's contacts with the machine, the table and the hands; contacts between the two plies keep the plain elastic-plastic law, whose stored displacement scales with the load. With the rule on the ply-on-ply contacts too, the top ply's free end flipped up and heaped 13-20 mm behind the foot at the 4 mm feed in 10 of 11 runs; why is not established (open). The presser foot landing on the stack locks less force between the plies than without the rule (0.5-0.8 N per contact against 1.9-2.2), and what remains is the ply-on-ply law's.
- Static friction is elastic-plastic with a pre-slip of 0.1 mm (an estimate for fibrous contacts; no measured value for cotton on steel was found): a contact below its limit deforms up to that before it slides. The pre-slip costs the feed dog part of each stroke, and the dog's stroke is calibrated to the dial at this value (derived at 600 sti/min, 40 N): a different pre-slip would move the feed (§3), uncalibrated, and at other speeds and presser forces the fed length is what E2 measures, not the dial. The coefficient pairs are load-independent.
- A ply folded under the needle is pierced once (the machine makes at most one hole per ply per stroke, at the layer the needle meets first); a real needle pierces every layer of the fold. A fold carried under the needle is still caught by the evaluator (the hole lies far from the seam line), but the stitch through it binds one layer.
- The feed dog's impact on the cloth is resolved within one 0.5 ms tick: the seam allowance's free edge keeps a lift that halves at half the tick (section 9); the pucker check leaves the allowance's outer 4 mm out of its band so that it reads the work, not this artifact.
- A finished seam is slack: the thread elements, drawn to the needle thread's 0.28 N at the interlock, relax to 0.005-0.022 N within 0.4 s (E7, §4), so a seam holds no residual thread tension, a tension pucker cannot form, and the pucker check measures waves of the work and the feed's contraction only.
- The stitches' balance follows from the two set tensions through the capstan law; no task lets the policy change a tension, so balance is recorded in the evaluation's info, not scored.
- Runs of one configuration are not bit-reproducible (the GPU's atomic reductions; section 8.8 measured the spread of one quantity).
- The machine's shaft follows its speed command (the servo's torque far exceeds the needle's load); the hand lifter lever and the handwheel are not operable by the robot.
- Robot-cloth contact is limited to the hands: the cloth sees the Franka Hand's URDF finger boxes and housing and the Dex3's links and palm (boxes around their meshes); every other link of every robot passes through the cloth, which a program reaching across the work will meet. Each hand's boxes ride on one compliant body, free along the vertical only (3e4 N/m, zeta 0.3, 10 g: estimates; no measured FR3 or Dex3 stiffness at the tool was found); across the vertical and in rotation they follow the arm rigidly, so a hand dragged sideways into the work has no compliance in that direction, and the pads themselves are rigid boxes (Franka's own finger mesh has the same square-edged tip). The FR3 and Panda arms take a joint impedance (raw op `joint_impedance`, libfranka's range); there is no Cartesian impedance, and the G1 has no compliant mode here. The coupling to the arm is explicit: the spring's force and the cloth's contact and friction forces reach the arm one tick later, filtered (1 ms, a numerical damper; with 5 ms a light hand pressing the work chatters, with none it alternates tick to tick: E18).
- **A Franka Hand squeezing the plies between its fingers holds them only lightly**: along the closing axis the pads are rigid for the cloth and the cloth's reaction reaches the fingers a tick late, so the plies, with nowhere to go, are crushed into their contact stop within a tick. From one checkpoint, 2 N per finger in force mode and the position mode (~1.2 N) hold steadily, 5 N chatters (0-142 N) and 20 N, a usual grasp, throws the fingers open and the arm into the pedestal with no fault raised. The fix (each finger a body of the cloth's implicit solve along its closing axis, as the hands' vertical compliance is) is a solver change, not made; the gripper's description tells a policy.
- The dual-FR3 rig (both arms on one pedestal, bases 0.4 m apart) cannot put both hands in front of the needle at once: the arms cross (E0d). Its two hands can work together with the right hand 170 mm in front of the needle, turned and tilted, and the left 150 mm to the left, regripping every 30 mm of travel with the work. Moving the right arm's base (five places from 0.10 to 0.40 m right of the needle line) does not help: with the right hand at the seam edge 50-120 mm in front of the needle, the right arm's links 5-7 meet the machine's head and overarm in every placement tried. A rig with an arm behind or beside the machine is not built.

- Robot motion: the IK computes targets, not collision-free paths. A joint-space move near the machine can swing a wrist through the overarm (the FR3's reflex then stops it), and a Cartesian turn of the hand can cross a wrist singularity; the scripted programs move along short Cartesian steps or joint-space moves between configurations checked statically for contacts (E15 case a).
- At the machine's top speed the presser foot leaves the stack for about a fifth of the time (it is thrown by the rising feed dog); its moving mass (0.12 kg) is an estimate. Unguided at that speed a short stack on the bed can run away (the stretched strip springs back when the grip at the needle is lost, and the cloth drags the needle into the hook): on the 1.25 mm band no run of six ran away at either tick (the far end never overran the cloth at the needle, which stayed 1.2-8.1 mm ahead of it) and the hook struck the needle in one of six at each (E2b); with a 2.5 mm band the outcome depends on the tick. Six runs a tick cannot show that the dependence is gone. A missing loss is suspected and not identified (the presser bar's bushing friction and damping are not modelled); what real cloth does here is not measured (open). The control box's speed output says so to a policy: the model is converged in time at 600 sti/min (E17), and the speeds between that and 5,000 are not qualified. The presser bar's position, 1 mm beside the needle, is authored: a closed Franka hand cannot reach the needle's line except with a fingertip's side (E15 case a).
- The bending hinges: where contact holds one wing of a hinge flat (the cloth leaving the presser foot, the table's edge), the hinge's curvature should live in its free wing alone; the solver cannot know which wing is held, so there the cloth bends about half a mesh row (~1 mm in the seam band) further in than the continuum (a first-order discretisation error; the Peirce test's clamp carries the correction). The hinge energy also couples the two bending directions with a ratio of -1/3, which the cylinder calibration does not see: a strip free to curl sideways would bend at 0.83-0.89 of its B; in the tests the membrane suppresses the curl (0.819 to 0.823 B with a correction term), and its effect on free, doubly curved drape is not measured (open).
- **The baste is not sewn**: at its 5 mm stitch with the seam guided along the edge, the sewn work lifts behind the foot at the 27th stitch and the needle strikes the plate at the 32nd; unguided, the work turns away and the hook strikes the needle at the 25th. On the 2.5 mm band the edge curled at the 2nd-5th stitch; the 1.25 mm band delays it; what lifts the work then is not established.
- **On the 1.25 mm seam band at the default tick**, sliding friction reads 2 % low (a coupon on a 12 deg incline slides 5.2 % fast; exact at half the tick or on a 2.5 mm mesh), and the Peirce cantilever on a mesh half the band's (0.625 mm) bends 4.5-13.9 % short (B 13-36 % low); both fail their acceptance checks and neither is explained (§8.5).
- The mesh is 1.25 mm in a 22 mm band along each seam the task sews (the right edge; for the pivot also the near edge; for the curve the whole rounded path) and 7.5 mm elsewhere: folds and bends tighter than ~5 mm radius outside the band are under-resolved (the virtual KES loop on the 7.5 mm mesh: B 0.80 of the input across the weft at 1-2.5 /cm).
- Several machine dimensions (feed-dog footprint, the foot's sole and toes, rod/crank ratio) and needle-law thresholds (missed-loop and flagging limits, hole stiffness, yield) are estimates listed in the parameter ledger (§7).
- A hand held still on the work while the machine sews (E19) stays finite with the needle intact, but the shear lock does not bound the plies' shear there: as the held work is fed, the 99th percentile climbs to 38-47 deg and single triangles reach 82-89 deg beside the left toe, then the sheared region lets go with the cloth moving at up to 2.9-4.2 m/s (two runs; without the lock, 72-87 deg). The lock's stiffness is an estimate, too soft against the feed dog's drive; no measured post-locking shear stiffness of a plain cotton was found (open).
- The force a policy can sense: the FR3 and Panda arms report each joint's estimated external torque (as libfranka's tau_ext_hat); the G1 reports none (the Dex3's fingertip tactile sensors are not simulated), and the scripted G1 program finds the work by the privileged cloth force on its hand.
- The machine is driven through its control box (`tool_io`: speed, needle stop, presser lift, reverse, trim), as an automated cell drives a head; for the standing G1 this same channel stands in for the pedal, the knee lifter and the touch-back switch a person would use, and the handwheel and the hand lifter are not robot-operable; the G1's task text says so. Whether the humanoid should work the pedal and knee lifter with its body is an open design decision.

17 Reproduction

The environment ships as the `sewing` package of RoboWorld (Newton 1.5.0 / Warp 1.16; images from Isaac Sim 5.1 as a render-only worker, `ISAAC_RENDER_PYTHON`). The experiments of this report were run by scripts that are

not part of the release. What the release runs:

What	Command
Discovery / dependencies	`gar inspect sewing`; `gar doctor sewing`; `gar tasks inspect sewing` (the five tasks and their selectors)
Managed session	`gar run sewing --task plain-seam` (`--robot dual-panda`, `single-fr3` or `g1` for the other deployments)
Parameter ledger	`python -m sewing.params` (writes `PARAMETERS.md`)
Visual assets	`python -m sewing.visual_assets` (fetches the pinned CC0 assets and checks their hashes)

The start states are prepared on first use and cached by the physics hash (`env.physics_hash()`; it covers the sewing modules that shape the physics, the robot catalog, the robot models and the sewing rigs). Experiment inputs are deterministic (seeded layouts, fixed ticks); runs on the GPU are not bit-reproducible (CUDA atomics in the contact accumulation; E2b's repeats measure the spread at top speed), and the runtime figures depend on what else the host is doing (§14, where each action's host load is recorded). The films are rendered from recorded states, and their frame counts are decoded back from the files.

18 The acceptance record

Every result in this report comes from one evidence campaign, in which the physics did not change. Jobs run again after a probe or a demonstration was corrected (never the physics) are recorded with their reason. An acceptance step then judged every collected result against a stated criterion, taken from the design's experiment table where it has one (E1, E2, E7), from the fabric's measured KES values the model was built from (E1b-d), or from the experiment's own every-tick checks (E14), and ran the CPU test suites. It records a result it does not judge (an open question, a scripted robot's attempt, whose outcome is the script's) separately, never as passed, and lists what is not claimed.

Overall: **not passed** (26 of 32 checks; failed: e1_incline_friction, e1_peirce_bending_length, demo_seam_assisted, demo_pivot_assisted, demo_curve_assisted, demo_curve_task).

Check	Verdict	Criterion	Source
`campaign_frozen_and_complete`	pass	the same physics source from the first job to the last; every job exited 0	the evidence campaign
`e1_tension_law`	pass	strip stiffness within 5 % of the prescribed n(e), warp, weft and bias	E1 criterion
`e1_incline_friction`	**fail**	sticks below the slip angle (no acceleration, < 0.2 mm of pre-slip); slides above it at g(sin - mu cos) to 1 %	Coulomb friction
`e1_peirce_bending_length`	**fail**	bending length within 5 % of the B-derived value (elastic cantilever)	E1 criterion
`e1b_bending_loop`	pass	B and 2HB of the seam-band mesh within 5 % of the KES-FB2 inputs, warp and weft	KES-FB2 of Ramaiah fabric C; 5 %: the E1 tolerance
`e1c_compression`	pass	Tm, WC and RC within 5 % of KES-FB3 (T0 and LC differ by construction: the law's zero pressure is at T0, §3)	KES-FB3 of Ramaiah fabric C; 5 %: the E1 tolerance on the cloth's laws
`e1d_membrane_loops`	pass	RT, LT, WT, EMT, G, 2HG and 2HG5 within 5 % of KES-FB1	KES-FB1 of Ramaiah fabric C; 5 %: the E1 tolerance on the cloth's laws
`e2_feed`	pass	at standard settings (40 N, 600 sti/min) the fed length within ~3 % of the dial, every setting; the other speeds and forces recorded	E2 criterion
`e7_stitch`	pass	every attempted stitch forms; >= 95 % balanced at the default tensions, top-tight at x3 the needle tension, bottom-tight at x0.3; no thread breaks	E7 criterion
`e14_rest_plain_fr3`	pass	every tick finite; after settling: cloth speed, the drift over the second half (a fitted trend beyond three standard errors), kinetic energy, sag, foot load, joint drift, reflexes, stitches, faults and the G1's tilt within their limits	E14 (the whiteboard environment's soak bar)

Check	Verdict	Criterion	Source
e14_rest_plain_g1	pass	every tick finite; after settling: cloth speed, the drift over the second half (a fitted trend beyond three standard errors), kinetic energy, sag, foot load, joint drift, reflexes, stitches, faults and the G1's tilt within their limits	E14 (the whiteboard environment's soak bar)
e14_rest_load_fr3	pass	every tick finite; after settling: cloth speed, the drift over the second half (a fitted trend beyond three standard errors), kinetic energy, sag, foot load, joint drift, reflexes, stitches, faults and the G1's tilt within their limits	E14 (the whiteboard environment's soak bar)
e14_rest_plain_dual_panda	pass	every tick finite; after settling: cloth speed, the drift over the second half (a fitted trend beyond three standard errors), kinetic energy, sag, foot load, joint drift, reflexes, stitches, faults and the G1's tilt within their limits	E14 (the whiteboard environment's soak bar)
e14_rest_plain_single_fr3	pass	every tick finite; after settling: cloth speed, the drift over the second half (a fitted trend beyond three standard errors), kinetic energy, sag, foot load, joint drift, reflexes, stitches, faults and the G1's tilt within their limits	E14 (the whiteboard environment's soak bar)
e15_legal_abuse	pass	every legal abuse case stays finite and ends in the physical outcome the case names	E15
e18_press_hold_dual-fr3	pass	a hand held still on the work keeps its press: never below 0.2 N and no fluctuation peak above 20 Hz (the stitch rate is 10 Hz), idle and sewing; no reflex	E18 (§8.9)
e18_press_hold_g1	pass	a hand held still on the work keeps its press: never below 0.2 N and no fluctuation peak above 20 Hz (the stitch rate is 10 Hz), idle and sewing; no reflex	E18 (§8.9)
e18_press_hold_dual-panda	pass	a hand held still on the work keeps its press: never below 0.2 N and no fluctuation peak above 20 Hz (the stitch rate is 10 Hz), idle and sewing; no reflex	E18 (§8.9)
e18_press_hold_single-fr3	pass	a hand held still on the work keeps its press: never below 0.2 N and no fluctuation peak above 20 Hz (the stitch rate is 10 Hz), idle and sewing; no reflex	E18 (§8.9)
e19_finite	pass	finite throughout	E19
demo_seam_assisted	**fail**	the unmodified task evaluator passes the assisted demonstration (assistance labelled on every frame)	the task evaluator (`sewing/task.py`)
demo_load_assisted	pass	the unmodified task evaluator passes the assisted demonstration (assistance labelled on every frame)	the task evaluator (`sewing/task.py`)
demo_pivot_assisted	**fail**	the unmodified task evaluator passes the assisted demonstration (assistance labelled on every frame)	the task evaluator (`sewing/task.py`)
demo_curve_assisted	**fail**	the unmodified task evaluator passes the assisted demonstration (assistance labelled on every frame)	the task evaluator (`sewing/task.py`)
demo_curve_task	**fail**	the unmodified task evaluator passes the assisted demonstration (assistance labelled on every frame)	the task evaluator (`sewing/task.py`)
hand_slide_contact	pass	a robot hand pressing the stack moves it by contact: the cloth under the hand follows the hand to within 0.5 mm (the friction's 0.1 mm preslip and the sampling), no reflex	§12
e16_every_deployment_measured	pass	runtime through gar run + HTTP measured for every offered deployment	E16

Check	Verdict	Criterion	Source
e16_quiet_inside_200x_from_0p1s	pass	on a quiet host, every deployment's commands of 0.1 s and longer return inside 200x (wall per simulated second, worst of each length) with every default policy camera; the shortest length inside is reported	E16
e16b_pause_costs_no_new_worker	pass	every deployment: a 0.05 s command after a 15 s pause within 1 s of one sent back to back (a new render worker costs 6.9-7.8 s)	E16b
films_encoded_and_decoded	pass	every film written, decoded back frame for frame, published	the film writer
stills	pass	every start state rendered from every camera (non-trivial images)	the still renderer
tests	pass	the CPU test suites pass	unit and contract tests

Recorded, not judged:

- e1b_bias_and_bulk_mesh: the bias is not calibrated; the 7.5 mm bulk mesh is coarser than the loop's curvature range
- e14_drift_plain_fr3: m/s and m: the trend beyond three standard errors (checked), the smallest creep the run could see
- e14_drift_plain_g1: m/s and m: the trend beyond three standard errors (checked), the smallest creep the run could see
- e14_drift_load_fr3: m/s and m: the trend beyond three standard errors (checked), the smallest creep the run could see
- e14_drift_plain_dual_panda: m/s and m: the trend beyond three standard errors (checked), the smallest creep the run could see
- e14_drift_plain_single_fr3: m/s and m: the trend beyond three standard errors (checked), the smallest creep the run could see
- e19_hold_against_feed: a hand holding the work against the feed: the outcome (needle, thread, reflex) is the operator's; the world must stay finite
- e19_energy: the energy ledger: R2 > 0 is energy from nowhere, R2 < 0 a dissipation the ledger does not book; e19_energy_check is the ledger's own closure on one stitch with nobody touching the work
- e19_energy_check: the energy ledger: R2 > 0 is energy from nowhere, R2 < 0 a dissipation the ledger does not book; e19_energy_check is the ledger's own closure on one stitch with nobody touching the work
- e20_hand_steer: a scripted robot's contact on the work while the machine sews: recorded (the steer's line shift and the pull's needle load are the world's response to the program)
- e21_hand_pull: a scripted robot's contact on the work while the machine sews: recorded (the steer's line shift and the pull's needle load are the world's response to the program)
- e16_runtime: wall seconds per simulated second on the shared host as it was (load recorded per command in the records); the 200x limit is judged per command length (shortest_command_inside_200x_s), not passed or failed here
- e16_runtime_quiet: the same runs on a quiet host (each started after three minutes of 1-minute load under 4 and GPU use under 30 %, §14); every deployment
- e16b_pause: what a pause between requests costs, every deployment
- demo_g1_guide: a scripted robot's attempt: its outcome is the script's, recorded
- demo_guided_seam: a scripted robot's attempt: its outcome is the script's, recorded

Not claimed:

- Autonomous task solution: the assisted demonstrations use labelled phantom guides; no policy is shipped
- Measured hand compliance: HAND_STIFFNESS, HAND_DAMPING_RATIO and HAND_MASS are estimates
- Measured pre-slip of cotton on steel: FRICTION_PRESLIP is an estimate; the feed is calibrated to the dial at that value

- The short stack at 5,000 sti/min
- Real-time simulation